

Ansible Simple Q&A 1–500 for AWS DevOps:

1. What is Ansible?

- Ansible is an open-source automation tool for configuration management, application deployment, and orchestration.

2. What are the key features of Ansible?

- Agentless, simple YAML syntax, idempotent operations, modular architecture.

3. What is an Ansible playbook?

- A YAML file containing a set of instructions (tasks) to automate configurations or deployments.

4. What is an Ansible role?

- A structured way to organize playbooks, tasks, variables, templates, and files.

5. What is an Ansible inventory?

- A list of hosts and groups that Ansible manages, defined in **INI** or **YAML** format.

6. What is the difference between static and dynamic inventory in Ansible?

- Static: manually defined hosts.
- Dynamic: generated from a script or cloud source (e.g., AWS EC2).

7. How do you install Ansible on a Linux machine?

- Using `sudo apt install ansible` (Debian/Ubuntu) or `sudo yum install ansible` (RHEL/CentOS).

8. What is an Ansible module?

- A reusable unit of code in Ansible that performs a specific task (e.g., `yum`, `apt`, `copy`).

9. Name some common Ansible modules.

- `file`, `copy`, `yum`, `apt`, `service`, `command`, `shell`, `user`.

10. How do you run an Ansible playbook?

- `ansible-playbook <playbook.yml>`

11. How do you run an Ansible command on a single host?

- `ansible <host> -m <module> -a "<args>"`

12. What is idempotency in Ansible?

- Ensuring a task can run multiple times without changing the result after the first run.

13. How do you define variables in Ansible?

- Variables can be defined in playbooks, inventory, or external files (YAML).

14. What is an Ansible template?

- A Jinja2 file that can dynamically generate configuration files using variables.

15. How do you use loops in Ansible?

- Use loop or with_items to iterate over a list of values in a task.

16. What is an Ansible fact?

- Predefined information gathered from managed hosts, like OS, IP address, memory, etc.

17. How do you gather facts in Ansible?

- Use the setup module: ansible <host> -m setup

18. How do you skip fact gathering in a playbook?

- Set gather_facts: no at the start of the playbook.

19. What is a handler in Ansible?

- A task triggered by notify when another task changes, often used to restart services.

20. How do you notify a handler in Ansible?

- Add notify: <handler_name> under a task that should trigger it.

21. How do you include one playbook inside another?

- Use import_playbook or include directives in the main playbook.

22. How do you include tasks from another file?

- Use include_tasks or import_tasks in a playbook.

23. How do you define groups in Ansible inventory?

- Using [group_name] in INI format or groups: in YAML inventory.

24. How do you run Ansible on multiple hosts?

- Use -I <host_pattern> or define multiple hosts in inventory.

25. How do you pass variables at runtime?

- Use -e "var_name=value" when running the playbook.

26. How do you secure sensitive data in Ansible?

- Use **Ansible Vault** to encrypt variables or files.

27. How do you create a vault file?

- ansible-vault create <filename>

28. How do you edit a vault file?

- ansible-vault edit <filename>

29. How do you run a playbook with encrypted vault variables?

- ansible-playbook <playbook.yml> --ask-vault-pass

30. How do you check syntax of a playbook without executing it?

- `ansible-playbook <playbook.yml> --syntax-check`

31. How do you check tasks that will run without executing them?

- `ansible-playbook <playbook.yml> --check (dry run)`

32. How do you list all hosts in an inventory?

- `ansible all --list-hosts`

33. How do you ping all hosts using Ansible?

- `ansible all -m ping`

34. How do you update packages on multiple Linux hosts using Ansible?

- Use `yum` or `apt` modules in a playbook with `loop` if needed.

35. How do you copy a file to multiple hosts?

- Use the `copy` module with `hosts: all`.

36. How do you create a user using Ansible?

- Use the `user` module: `- name: Create user name: deploy state: present`

37. How do you restart a service in Ansible?

- Use `service` module: `- name: Restart nginx service: name=nginx state=restarted`

38. How do you manage files and directories in Ansible?

- Use the `file` module to create directories or set permissions.

39. How do you run shell commands in Ansible?

- Use `shell` or `command` modules:
 - `shell: "uptime"`
 - `command: ls -l /tmp`

40. How do you manage AWS resources using Ansible?

- Use **AWS modules** like `ec2`, `s3`, `iam`, `cloudformation`.
- Provide AWS credentials via environment variables, IAM roles, or profiles.

41. How do you launch an EC2 instance using Ansible?

- Use the `ec2` module with parameters like `image_id`, `instance_type`, `key_name`.

42. How do you create an S3 bucket using Ansible?

- Use the `aws_s3` module with `mode=create` and specify bucket name.

43. How do you attach an IAM role to EC2 instance using Ansible?

- Use `iam_role` module to create or attach a role.

- Specify the instance and role details.

44. How do you deploy an application to multiple EC2 instances using Ansible?

- Define hosts in inventory.
- Use tasks to copy files, install packages, and start services.

45. How do you handle errors in Ansible tasks?

- Use `ignore_errors: yes` to continue on failure.
- Use `failed_when` for custom failure conditions.

46. How do you run tasks only on certain hosts?

- Use `hosts: <host_pattern>` or `when` conditions.

47. How do you use conditionals in Ansible tasks?

- Use `when`: with variables or facts.
 - Example: `when: ansible_os_family == "RedHat"`

48. How do you register output of a task in Ansible?

- Use `register: variable_name` and reference it with `{{ variable_name.stdout }}`

49. How do you debug variables in Ansible?

- Use the `debug` module: `- debug: var=variable_name`

50. How do you create a scheduled job (cron) using Ansible?

- Use the `cron` module:
 - `- name: Add cron job`
 - `cron:`
 - `name: "Backup"`
 - `minute: "0"`
 - `hour: "2"`
 - `job: "/usr/local/bin/backup.sh"`
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51. How do you install a package on multiple hosts using Ansible?

- Use `yum` or `apt` modules in a playbook:
- `- name: Install nginx`
- `yum:`
- `name: nginx`
- `state: present`

52. How do you remove a package using Ansible?

- Use state: absent with package module:
- - name: Remove httpd
- yum:
- name: httpd
- state: absent

53. How do you ensure a service is running on multiple hosts?

- Use service module:
- - name: Ensure nginx is running
- service:
- name: nginx
- state: started
- enabled: yes

54. How do you check the OS version of hosts using Ansible?

- Use setup module and check ansible_distribution or ansible_distribution_version.

55. How do you add a repository in Linux using Ansible?

- Use yum_repository for RHEL/CentOS or apt_repository for Debian/Ubuntu.

56. How do you create a directory with specific permissions using Ansible?

- Use file module:
- - name: Create directory
- file:
- path: /opt/mydir
- state: directory
- mode: '0755'

57. How do you copy multiple files to multiple hosts?

- Use copy module with loop:
- - name: Copy files
- copy:
- src: "{{ item }}"
- dest: /tmp/
- loop:

- - file1.txt
- - file2.txt

58. How do you move or rename files using Ansible?

- Use file module with state: moved:
- - name: Move file
- command: mv /tmp/file1.txt /tmp/file2.txt

59. How do you set environment variables using Ansible?

- Use environment parameter in tasks or playbooks:
- - name: Run command with env
- shell: echo \$MYVAR
- environment:
- MYVAR: "value"

60. How do you reboot a server using Ansible?

- Use reboot module:
- - name: Reboot server
- reboot:
- reboot_timeout: 300

61. How do you run tasks as a different user?

- Use become: yes and become_user: <username>:
- - name: Run as root
- shell: whoami
- become: yes
- become_user: root

62. How do you install Python dependencies using Ansible?

- Use pip module:
- - name: Install requests
- pip:
- name: requests
- state: present

63. How do you check if a file exists before performing a task?

- Use stat module with when condition:

- - stat:
- path: /tmp/file.txt
- register: file_check
-
- - debug: msg="File exists"
- when: file_check.stat.exists

64. How do you download a file from a URL using Ansible?

- Use get_url module:
- - name: Download file
- get_url:
- url: https://example.com/file.txt
- dest: /tmp/file.txt

65. How do you upload a file to S3 using Ansible?

- Use aws_s3 module:
- - name: Upload to S3
- aws_s3:
- bucket: mybucket
- object: file.txt
- src: /tmp/file.txt
- mode: put

66. How do you create an IAM user using Ansible?

- Use iam_user module:
- - name: Create IAM user
- iam_user:
- name: deploy_user
- state: present

67. How do you attach an IAM policy to a user using Ansible?

- Use iam_policy module with iam_user:
- - iam_policy:
- iam_type: user
- iam_name: deploy_user

- policy_name: AdministratorAccess
- state: present

68. How do you stop/start an EC2 instance using Ansible?

- Use ec2 module with state: stopped or running:
- - name: Stop EC2
- ec2:
- instance_ids: i-123456
- state: stopped

69. How do you tag AWS resources using Ansible?

- Use ec2_tag module:
- - name: Tag EC2
- ec2_tag:
- resource: i-123456
- state: present
- tags:
- Name: WebServer

70. How do you install AWS CLI using Ansible?

- Use pip module: - name: Install AWS CLI pip: name: awscli state: present

71. How do you manage CloudFormation stacks using Ansible?

- Use cloudformation module:
- - name: Create stack
- cloudformation:
- stack_name: my-stack
- state: present
- template: /tmp/template.yaml

72. How do you install a specific version of a package using Ansible?

- Use version parameter in package module:
- - yum:
- name: nginx-1.18
- state: present

73. How do you copy templates to multiple hosts?

- Use template module:
- - name: Copy nginx.conf template
- template:
- src: nginx.conf.j2
- dest: /etc/nginx/nginx.conf

74. How do you enable/disable services on boot?

- Use enabled parameter in service module:
- - service:
- name: nginx
- enabled: yes

75. How do you add users to groups using Ansible?

- Use user module with groups:
- - user:
- name: deploy
- groups: sudo
- state: present

76. How do you remove users from a system using Ansible?

- Use state: absent in user module:
- - user:
- name: tempuser
- state: absent

77. How do you manage file permissions using Ansible?

- Use file module with mode:
- - file:
- path: /tmp/file.txt
- mode: '0644'

78. How do you check a service status using Ansible?

- Use service_facts module and debug:
- - service_facts:
- - debug:
- var: ansible_facts.services

79. How do you check disk space using Ansible?

- Use ansible_mounts fact: - debug: var=ansible_mounts

80. How do you reboot AWS EC2 instances and wait until they are up?

- Use reboot module with connect_timeout and pre_reboot_delay parameters.

81. How do you create security groups using Ansible?

- Use ec2_group module:
- - name: Create SG
- ec2_group:
- name: web-sg
- description: Web servers SG
- vpc_id: vpc-123456

82. How do you add ingress rules to security groups?

- Use ec2_group with rules:
- - ec2_group:
- name: web-sg
- rules:
- - proto: tcp
- from_port: 80
- to_port: 80
- cidr_ip: 0.0.0.0/0

83. How do you create multiple EC2 instances with Ansible?

- Use count parameter in ec2 module:
- - ec2:
- image: ami-123456
- instance_type: t2.micro
- count: 3

84. How do you manage AWS Lambda functions using Ansible?

- Use lambda module to create/update functions, handlers, and environment variables.

85. How do you manage S3 bucket policies using Ansible?

- Use aws_s3 module with policy parameter specifying JSON policy.

86. How do you restart multiple services sequentially using Ansible?

- Use serial keyword in playbook:
- - hosts: all
- serial: 1

87. How do you schedule tasks (cron jobs) on multiple AWS EC2 instances?

- Use cron module in a playbook targeting EC2 hosts.

88. How do you install multiple packages using Ansible?

- Use loop:
- - name: Install packages
- yum:
- name: "{{ item }}"
- state: present
- loop:
- - git
- - vim
- - curl

89. How do you download files from S3 to EC2 instances?

- Use aws_s3 module with mode=get.

90. How do you deploy Docker containers using Ansible on AWS EC2?

- Use docker_container module after installing Docker with docker_image module.

91. How do you ensure idempotency in Ansible tasks?

- Use modules properly instead of raw shell commands.

92. How do you check the uptime of servers using Ansible?

- Use command: uptime with register to store output.

93. How do you run tasks on a subset of hosts?

- Use -I option or host patterns in playbook.

94. How do you include variables from a file?

- Use vars_files in playbook:
- vars_files:
- - vars/main.yml

95. How do you define default variables in a role?

- Use defaults/main.yml in the role directory.

96. How do you override role defaults?

- Define variables in playbook under vars: section.

97. How do you set facts dynamically during a playbook run?

- Use set_fact module:
- - set_fact:
- my_var: "value"

98. How do you include multiple tasks conditionally?

- Use include_tasks with when: condition.

99. How do you install a specific version of a Python package using Ansible?

- Use pip module with version parameter:
- - pip:
- name: boto3==1.26.0

100. How do you gather AWS EC2 instance facts in Ansible?

- Use ec2_instance_facts module to collect metadata like instance type, ID, IPs, etc.
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101. How do you define a play in Ansible?

- A play maps a group of hosts to tasks, defining what should run on which hosts.

102. How do you run multiple plays in a single playbook?

- Define multiple - hosts: sections sequentially in the YAML file.

103. How do you use variables in Jinja2 templates?

- Use {{ variable_name }} to reference variables inside templates.

104. How do you set default values for variables in a template?

- Use the default filter: {{ variable_name | default('value') }}

105. How do you debug a template output?

- Use the template module to copy to a temp location, then debug module to inspect.

106. How do you manage different environments (dev, prod) in Ansible?

- Use separate inventory files or group_vars directories for each environment.

107. How do you check connectivity of hosts before running tasks?

- Use ansible all -m ping to ensure SSH access and host availability.

108. How do you run Ansible with a specific inventory file?

- Use -i <inventory_file> option:

- `ansible-playbook -i inventory.yml playbook.yml`

109. How do you install a specific version of Ansible?

- Use pip: `pip install ansible==<version>`

110. How do you upgrade Ansible to the latest version?

- Use `pip install --upgrade ansible`

111. How do you check Ansible version installed?

- Run `ansible --version`

112. How do you include a role in a playbook?

- Use roles: section in the play:
- roles:
- - role_name

113. How do you define handlers in a role?

- Place tasks in `handlers/main.yml` and call them via `notify`.

114. How do you use loops with dictionaries in Ansible?

- Use `dict2items` filter or `with_dict` for looping through key-value pairs.

115. How do you install multiple AWS CLI tools or SDKs?

- Use pip module with loop:
- - pip:
- name: "{{ item }}"
- loop:
- - awscli
- - boto3

116. How do you wait for a port to become available using Ansible?

- Use `wait_for` module:
- - wait_for:
- port: 22
- host: "{{ inventory_hostname }}"
- timeout: 300

117. How do you copy files recursively in Ansible?

- Use `copy` module with `remote_src: yes` or `synchronize` module for rsync-like behavior.

118. How do you install a package only if it is not already installed?

- Package modules like yum or apt are idempotent and only install if absent.

119. How do you stop tasks if a condition is not met?

- Use failed_when or when conditions.

120. How do you create and manage directories recursively?

- Use file module with state: directory and desired permissions.

121. How do you set up AWS credentials for Ansible?

- Use environment variables (AWS_ACCESS_KEY_ID, AWS_SECRET_ACCESS_KEY), IAM roles, or profiles.

122. How do you run playbooks on EC2 instances in a specific region?

- Use region parameter in AWS modules or set environment variable AWS_REGION.

123. How do you check the existence of an S3 bucket using Ansible?

- Use aws_s3 module with mode=list or aws_s3_bucket_info module.

124. How do you create multiple security group rules in Ansible?

- Use ec2_group module with rules list.

125. How do you manage AWS CloudWatch alarms using Ansible?

- Use cloudwatch_metric_alarm module with appropriate thresholds and actions.

126. How do you set a task to run only on the first host of a group?

- Use run_once: yes in the task.

127. How do you skip tasks for certain hosts?

- Use when conditions referencing hostnames or variables.

128. How do you include a task file multiple times?

- Use include_tasks with loop or repeat inclusion as needed.

129. How do you install Docker using Ansible?

- Use docker role or yum/apt modules for Docker packages.

130. How do you ensure a Docker service is running using Ansible?

- Use docker_container module with state: started and restart_policy.

131. How do you pull Docker images using Ansible?

- Use docker_image module:
- - docker_image:
- name: nginx
- source: pull

132. How do you remove Docker images using Ansible?

- Use `docker_image` module with `state: absent`.

133. How do you manage Docker volumes using Ansible?

- Use `docker_volume` module to create, remove, or inspect volumes.

134. How do you manage Docker networks using Ansible?

- Use `docker_network` module to create or remove networks.

135. How do you check EC2 instance status using Ansible?

- Use `ec2_instance_facts` or `ec2_metadata_facts` module.

136. How do you tag multiple EC2 instances at once?

- Use `ec2_tag` module with a list of instance IDs.

137. How do you stop all instances in a specific AWS region?

- Use `ec2` module with filters for region and `state: stopped`.

138. How do you create multiple IAM users in Ansible?

- Loop through a list of usernames with `iam_user` module.

139. How do you delete multiple IAM users in Ansible?

- Loop through usernames with `state: absent` in `iam_user` module.

140. How do you create EC2 key pairs using Ansible?

- Use `ec2_key` module specifying name and `key_material` or `generate new`.

141. How do you create a VPC using Ansible?

- Use `ec2_vpc_net` module with CIDR and `state: present`.

142. How do you create subnets in a VPC?

- Use `ec2_vpc_subnet` module specifying VPC ID and CIDR.

143. How do you attach an internet gateway to a VPC?

- Use `ec2_vpc_igw` module to create and attach.

144. How do you create route tables using Ansible?

- Use `ec2_vpc_route_table` module and associate with subnets.

145. How do you create security group rules with specific protocols?

- Define `proto`, `from_port`, `to_port`, and `cidr_ip` in `ec2_group` module.

146. How do you check AWS resource existence before creating?

- Use `register` and `when conditions`, or `state: present/absent`.

147. How do you install Node.js using Ansible?

- Use yum, apt, or community.general.nvm module for version management.

148. How do you manage cron jobs in AWS EC2 using Ansible?

- Use cron module specifying minute, hour, and command.

149. How do you upload files to multiple S3 buckets?

- Loop through a list of bucket names with aws_s3 module.

150. How do you set AWS region dynamically in Ansible playbook?

- Use a variable aws_region and pass it to AWS modules:
 - - ec2:
 - region: "{{ aws_region }}"
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151. How do you install Nginx on multiple Linux hosts using Ansible?

- Use yum or apt module depending on OS:
- - name: Install Nginx
- yum:
- name: nginx
- state: present

152. How do you start and enable a service using Ansible?

- Use service module:
- - service:
- name: nginx
- state: started
- enabled: yes

153. How do you stop and disable a service?

- Use service module with state: stopped and enabled: no.

154. How do you copy a file to multiple hosts?

- Use copy module:
- - copy:
- src: /local/path/file.txt
- dest: /remote/path/file.txt

155. How do you create a directory on remote hosts?

- Use file module:

- - file:
- path: /opt/mydir
- state: directory

156. How do you remove a directory or file?

- Use file module with state: absent.

157. How do you create multiple users on hosts?

- Use user module with loop:
- - user:
- name: "{{ item }}"
- state: present
- loop:
- - user1
- - user2

158. How do you set user passwords in Ansible?

- Use user module with password parameter (encrypted with mkpasswd or openssl passwd).

159. How do you add users to groups?

- Use user module with groups and append: yes.

160. How do you remove users from the system?

- Use user module with state: absent.

161. How do you install packages conditionally?

- Use when: with yum or apt module based on OS facts.

162. How do you upgrade all packages on a host?

- Use yum: name=* state=latest or apt: upgrade=dist.

163. How do you gather host facts like CPU and memory?

- Use setup module: ansible all -m setup.

164. How do you check if a service is running?

- Use service_facts module and inspect ansible_facts.services.

165. How do you restart a service only if its configuration file changed?

- Use notify and handlers in the playbook.

166. How do you deploy templates to multiple hosts?

- Use template module with variables:

- - template:
- src: myconfig.j2
- dest: /etc/myconfig.conf

167. How do you manage files with different content per host?

- Use host_vars or group_vars and templates with Jinja2 variables.

168. How do you copy directories recursively?

- Use copy module with recursive: yes.

169. How do you check for the existence of a file before creating it?

- Use stat module with register: and when:.

170. How do you install Python packages on remote hosts?

- Use pip module:
- - pip:
- name: boto3
- state: present

171. How do you manage cron jobs for multiple users?

- Use cron module with user: parameter.

172. How do you deploy applications from a Git repository?

- Use git module with repo: and dest:.

173. How do you ensure idempotency when using Git?

- Specify version;; Ansible pulls updates only if the repository changed.

174. How do you fetch files from remote hosts?

- Use fetch module.

175. How do you execute shell commands on remote hosts?

- Use shell module; prefer command when no shell features are needed.

176. How do you manage environment variables for a task?

- Use environment: in the task:
- - shell: echo \$MY_VAR
- environment:
- MY_VAR: "value"

177. How do you create symbolic links on remote hosts?

- Use file module with state: link.

178. How do you reboot a host and wait until it comes back?

- Use reboot module.

179. How do you check uptime of hosts?

- Use command: uptime with register: to capture output.

180. How do you manage firewall rules on multiple hosts?

- Use firewalld or ufw modules depending on OS.

181. How do you manage SELinux status?

- Use selinux module or edit /etc/selinux/config via template.

182. How do you check open ports on remote hosts?

- Use command: ss -tuln or ansible_facts for network info.

183. How do you copy SSH keys to users on multiple hosts?

- Use authorized_key module with user: and key:.

184. How do you create temporary files on remote hosts?

- Use tempfile module.

185. How do you remove old log files older than X days?

- Use find module with age and file_type and remove with file.

186. How do you install multiple packages in one task?

- Use list in name: parameter for yum, apt, or package modules.

187. How do you install a specific package version?

- Specify version: in the package module.

188. How do you check disk usage before creating files?

- Use ansible_mounts fact or command: df -h.

189. How do you configure timezone on remote hosts?

- Use timezone module or link /etc/localtime via file.

190. How do you manage swap memory?

- Use command module to create swapfile and add to /etc/fstab.

191. How do you enable services to start at boot?

- Use service module with enabled: yes.

192. How do you check the distribution and version of Linux hosts?

- Use ansible_distribution and ansible_distribution_version facts.

193. How do you copy multiple files conditionally?

- Use with_items loop and when: conditions.

194. How do you remove duplicate lines from files?

- Use lineinfile with regexp:.

195. How do you monitor file changes during playbook execution?

- Run playbook with --diff option.

196. How do you ensure a task runs only once on all hosts?

- Use run_once: yes.

197. How do you create users with a specific shell?

- Use user module with shell: parameter.

198. How do you set file ownership and permissions?

- Use file or copy module with owner, group, and mode.

199. How do you check free memory on a host?

- Use ansible_memtotal_mb fact.

200. How do you manage services with different names on different OS distributions?

- Use when: conditions based on ansible_distribution.
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201. How do you create a new group on Linux hosts using Ansible?

- Use group module:
- - name: Create group
- group:
- name: devops
- state: present

202. How do you add a user to a group?

- Use user module:
- - user:
- name: deploy
- groups: devops
- append: yes

203. How do you remove a user from a group?

- Use user module with groups excluding the group or state: absent to remove user.

204. How do you check if a directory exists before creating it?

- Use stat module:
- - stat:
- path: /opt/mydir
- register: dir_check
-
- - file:
- path: /opt/mydir
- state: directory
- when: not dir_check.stat.exists

205. How do you create a file with content using Ansible?

- Use copy module with content parameter:
- - copy:
- dest: /tmp/test.txt
- content: "Hello World"

206. How do you ensure a file is absent?

- Use file module:
- - file:
- path: /tmp/test.txt
- state: absent

207. How do you execute multiple shell commands sequentially?

- Use multiple shell or command tasks in the playbook.

208. How do you manage crontab entries using Ansible?

- Use cron module:
- - cron:
- name: "Backup"
- minute: "0"
- hour: "2"
- job: "/usr/bin/backup.sh"

209. How do you copy files from remote hosts to the control node?

- Use fetch module:
- - fetch:

- src: /var/log/messages
- dest: /tmp/logs/

210. How do you create a symbolic link on multiple hosts?

- Use file module:
- - file:
- src: /opt/source
- dest: /opt/link
- state: link

211. How do you install services only if a certain condition is true?

- Use when: condition:
- - yum:
- name: nginx
- state: present
- when: ansible_distribution == "CentOS"

212. How do you ensure services are enabled at boot?

- Use service module with enabled: yes.

213. How do you remove multiple packages in one task?

- Use yum or apt with a list of packages.

214. How do you check the status of a package?

- Use package_facts module and inspect ansible_facts.packages.

215. How do you reboot a Linux host and wait until it's back?

- Use reboot module:
- - reboot:
- reboot_timeout: 300

216. How do you run a task only once across all hosts?

- Use run_once: yes.

217. How do you ensure a service is restarted only if a configuration file changes?

- Use notify with handlers.

218. How do you copy templates to multiple hosts?

- Use template module with Jinja2 variables.

219. How do you gather host facts like memory and disk space?

- Use setup module: `ansible all -m setup`

220. How do you check the uptime of a host?

- Use command: `uptime` with register: to save output.

221. How do you install a specific version of a package?

- Specify version: in the package module:
- - yum:
- name: `nginx-1.18`
- state: `present`

222. How do you copy directories recursively?

- Use copy module with `recursive: yes`.

223. How do you create temporary files on remote hosts?

- Use `tempfile` module.

224. How do you delete files older than X days?

- Use `find` module with `age` and `file_type`, then loop with `file` module.

225. How do you disable SELinux on multiple hosts?

- Use `selinux` module or edit `/etc/selinux/config` via template.

226. How do you check if a service is running before restarting it?

- Use `service_facts` module and `when: condition`.

227. How do you check free disk space on hosts?

- Use `ansible_mounts` fact: - debug: `var=ansible_mounts`

228. How do you set hostname on remote hosts?

- Use `hostname` module.

229. How do you install software from a URL?

- Use `get_url` to download, then `shell` to install.

230. How do you manage firewall rules?

- Use `firewalld` or `ufw` modules depending on the OS.

231. How do you manage Python packages on remote hosts?

- Use `pip` module.

232. How do you deploy SSL certificates to hosts?

- Use `copy` or `template` module to deploy `.crt` and `.key` files.

233. How do you change file ownership?

- Use file module with owner: and group:.

234. How do you schedule repeated tasks using Ansible?

- Use cron module with desired frequency.

235. How do you create users with a specific shell and home directory?

- Use user module with shell: and home:.

236. How do you manage system limits (ulimits)?

- Use lineinfile module to edit /etc/security/limits.conf.

237. How do you copy multiple files using a loop?

- Use with_items or loop in copy module.

238. How do you remove duplicate lines from a file?

- Use lineinfile module with state: present and regexp.

239. How do you install Java on Linux hosts?

- Use yum or apt module with package name java-1.8.0-openjdk.

240. How do you restart multiple services sequentially?

- Define multiple service tasks in order.

241. How do you monitor changes to files during playbook runs?

- Use --diff option when running ansible-playbook.

242. How do you upload files to remote hosts conditionally?

- Use when: based on facts or variables.

243. How do you check the last login of users?

- Use command: lastlog or shell: last.

244. How do you ensure a task runs only if a file exists?

- Use stat module with when: condition.

245. How do you enable swap memory on multiple hosts?

- Use command module to create swapfile and shell to activate.

246. How do you configure timezone on hosts?

- Use timezone module or link /etc/localtime via file module.

247. How do you check open ports on a host?

- Use command: ss -tln or ansible_facts network modules.

248. How do you manage host-specific SSH keys?

- Use authorized_key module.

249. How do you set environment variables permanently?

- Use lineinfile to edit /etc/profile or .bashrc.

250. How do you check running processes on remote hosts?

- Use shell: ps -ef and register: for output.
-

251. How do you copy files from one directory to another on multiple hosts?

- Use copy module with remote_src: yes:
- - copy:
- src: /source/path/
- dest: /destination/path/
- remote_src: yes
- recursive: yes

252. How do you manage file permissions recursively?

- Use file module with state: directory and mode: 0755 with recurse: yes.

253. How do you check if a package is installed before installing it?

- Use package_facts module and conditionally install with when:.

254. How do you remove multiple users from Linux hosts?

- Loop over a list with user module:
- - user:
- name: "{{ item }}"
- state: absent
- loop:
- - user1
- - user2

255. How do you create multiple directories in one task?

- Use loop with file module:
- - file:
- path: "{{ item }}"
- state: directory
- loop:

- - /opt/app1
- - /opt/app2

256. How do you check if a service is enabled and running?

- Use service_facts and check ansible_facts.services['service_name'].state and .enabled.

257. How do you install multiple Python packages with different versions?

- Use pip module with loop:
- - pip:
- name: "{{ item.name }}" == "{{ item.version }}"
- loop:
- - { name: boto3, version: 1.28.0 }
- - { name: requests, version: 2.31.0 }

258. How do you remove a directory only if it exists?

- Use stat module with when: condition on stat.exists.

259. How do you manage multiple cron jobs for different users?

- Use cron module with user: parameter for each task.

260. How do you ensure a file is created with specific ownership and permissions?

- Use file or copy module with owner, group, and mode.

261. How do you check mounted filesystems on remote hosts?

- Use ansible_mounts fact: - debug: var=ansible_mounts.

262. How do you restart multiple services only if configuration changes?

- Use notify and handlers in the playbook.

263. How do you deploy multiple templates with variables?

- Use template module with loop or with_items referencing different variables.

264. How do you fetch log files from multiple hosts into a single directory?

- Use fetch module with flat: yes and loop through hosts.

265. How do you install software from a downloaded RPM or DEB file?

- Use yum or apt module with remote_src: yes.

266. How do you check the last modified time of a file?

- Use stat module and inspect stat.mtime.

267. How do you disable a service on multiple hosts?

- Use service module with enabled: no and state: stopped.

268. How do you install a package only if another package is not installed?

- Use package_facts with a conditional when: statement.

269. How do you copy only if the source file is newer?

- Use copy module with force: yes and remote_src: yes.

270. How do you create a file with content from a variable?

- Use copy module with content: "{{ my_variable }}".

271. How do you manage environment variables for all users?

- Use lineinfile to edit /etc/profile or /etc/environment.

272. How do you ensure swap is enabled after reboot?

- Use command module to create swap and add entry in /etc/fstab.

273. How do you run tasks only if a certain package is installed?

- Use package_facts and a when: condition.

274. How do you manage sysctl parameters?

- Use sysctl module specifying name and value.

275. How do you check for available memory on hosts?

- Use ansible_memtotal_mb fact.

276. How do you remove old log files older than 7 days?

- Use find module with age: 7d and file_type: file then remove using file module.

277. How do you install multiple AWS CLI tools at once?

- Use pip module with a loop:
- - pip:
- name: "{{ item }}"
- loop:
- - awscli
- - boto3

278. How do you wait for a TCP port to open on multiple hosts?

- Use wait_for module with port: and host: "{{ inventory_hostname }}".

279. How do you ensure idempotency when creating directories?

- Use file module with state: directory; it won't recreate if already exists.

280. How do you deploy multiple configuration files in one task?

- Use template module with with_items or loop.

281. How do you register output of a command for later use?

- Use register: parameter:
- - command: cat /etc/os-release
- register: os_info

282. How do you debug registered variables?

- Use debug module:
- - debug:
- var: os_info.stdout

283. How do you stop a service only if it is running?

- Use service_facts and conditional when:.

284. How do you create multiple user accounts with specific UID and shell?

- Use user module with loop and variables.

285. How do you copy SSH keys to multiple hosts?

- Use authorized_key module with user: and key:.

286. How do you manage hosts with different Linux distributions?

- Use ansible_distribution fact with when: conditions.

287. How do you ensure a service is reloaded if config changes?

- Use notify and handlers in combination with template.

288. How do you manage disk quotas on Linux?

- Use command or shell module to configure quota commands.

289. How do you check disk usage before creating new files?

- Use ansible_mounts or shell: df -h with register:.

290. How do you schedule a task to run as root?

- Use cron module with user: root.

291. How do you create multiple symbolic links in one task?

- Use loop with file module.

292. How do you deploy application files only if new version exists?

- Use checksum or force: yes in copy or template module.

293. How do you check connectivity of all hosts before running tasks?

- Use ping module: ansible all -m ping.

294. How do you deploy software from Git repository?

- Use git module with repo: and dest: parameters.

295. How do you ensure idempotency when deploying from Git?

- Use version: parameter in git module; Ansible only pulls updates if needed.

296. How do you manage configuration files with different content per host?

- Use host_vars and template module.

297. How do you remove duplicate lines from configuration files?

- Use lineinfile with state: present and regexp for uniqueness.

298. How do you ensure logs are rotated automatically?

- Use logrotate module or deploy /etc/logrotate.d templates.

299. How do you check if a kernel module is loaded?

- Use ansible_facts.modules or command: lsmod.

300. How do you load a kernel module if not already loaded?

- Use modprobe with command module and conditional check with stat or shell: lsmod.
-

301. How do you create multiple directories with different permissions?

- Use file module with loop and specify mode: for each directory.

302. How do you check if a user exists before creating it?

- Use getent command or ansible_facts and a when: condition.

303. How do you copy files only if they don't already exist?

- Use copy module with force: no.

304. How do you install packages only if they are not already installed?

- Use package_facts with a conditional when: check.

305. How do you restart multiple services sequentially?

- Define multiple service tasks in order in the playbook.

306. How do you check for existing SSH keys before deploying new ones?

- Use stat or authorized_key module with key: check.

307. How do you manage multiple cron jobs per user?

- Use cron module with user: parameter for each job.

308. How do you fetch multiple log files from hosts into a single directory?

- Use fetch module with flat: yes and loop through hosts or files.

309. How do you check if a port is open before starting a service?

- Use `wait_for` module with `port:` and `host:`.

310. How do you install software from a local RPM or DEB file?

- Use `yum` or `apt` module with `remote_src: yes`.

311. How do you deploy multiple templates with host-specific variables?

- Use `template` module with `host_vars` or `group_vars`.

312. How do you create symbolic links only if the target exists?

- Use `file` module with `state: link` and `conditional when:`.

313. How do you remove files older than a certain number of days?

- Use `find` module with `age:` and `file_type: file`, then `file` module to remove.

314. How do you check if a kernel module is loaded?

- Use `command: lsmod` with `register:` or `ansible_facts.modules`.

315. How do you load a kernel module only if it's not loaded?

- Use `modprobe` with a conditional check using `shell` or `stat`.

316. How do you deploy configuration files only if changes occur?

- Use `template` module with `notify` to trigger handlers.

317. How do you gather disk usage statistics on remote hosts?

- Use `ansible_mounts` fact or `command: df -h` with `register:`.

318. How do you create users with a specific UID and group?

- Use `user` module with `uid:` and `group:` parameters.

319. How do you check service status across multiple hosts?

- Use `service_facts` and loop through `ansible_facts.services`.

320. How do you disable swap memory on remote hosts?

- Use `command` module with `swapoff -a` and remove swap entry from `/etc/fstab`.

321. How do you check free memory before running resource-intensive tasks?

- Use `ansible_memfree_mb` fact and `when: condition`.

322. How do you copy a directory structure without files?

- Use `find` module with `file_type: directory` and `with_items` to recreate directories.

323. How do you check the Linux distribution version before installing packages?

- Use `ansible_distribution` and `ansible_distribution_version` facts in `when:`.

324. How do you install a package with dependencies?

- Use `yum`, `apt`, or `package` modules; they handle dependencies automatically.

325. How do you ensure that only one instance of a task runs at a time?

- Use serial: 1 in the playbook or run_once: yes.

326. How do you copy files from remote hosts to control node with host-specific names?

- Use fetch module with dest: "/tmp/{{ inventory_hostname }}/"

327. How do you manage environment variables per user?

- Use lineinfile to edit .bashrc or /etc/profile.

328. How do you install multiple Python packages with pip?

- Use loop or list in name: parameter of pip module.

329. How do you deploy SSL certificates on multiple hosts?

- Use copy or template module with .crt and .key files.

330. How do you ensure idempotency when creating files?

- Use file or copy modules with proper state;; Ansible will skip if already correct.

331. How do you restart a service if a file exists?

- Use stat to check file and when: condition with service module.

332. How do you remove users only if they exist?

- Use getent passwd username check with command or conditional user module.

333. How do you check running processes on remote hosts?

- Use shell: ps -ef and register: to capture output.

334. How do you create temporary directories safely?

- Use tempfile module with state: directory.

335. How do you copy files only if checksum differs?

- Use checksum: parameter in copy module.

336. How do you deploy configuration files from templates with dynamic values?

- Use template module with Jinja2 variables.

337. How do you schedule root cron jobs?

- Use cron module with user: root.

338. How do you remove duplicate lines from a file?

- Use lineinfile module with regexp: and state: present.

339. How do you configure hostnames on remote machines?

- Use hostname module.

340. How do you check if a directory is empty before removing it?

- Use find module with file_type: file and length check in when:.

341. How do you ensure services start on boot across multiple OS types?

- Use service module with enabled: yes and when: based on OS.

342. How do you create multiple files from a list of content?

- Use with_items loop with copy module and content: parameter.

343. How do you monitor changes in files during playbook execution?

- Run playbook with --diff to see differences.

344. How do you remove old log files safely?

- Use find module to list files older than X days, then remove with file module.

345. How do you configure firewall rules conditionally?

- Use firewallld or ufw module with when: conditions based on host facts.

346. How do you deploy software only if the host has enough free memory?

- Use ansible_memfree_mb fact with when: condition.

347. How do you check kernel version on remote hosts?

- Use ansible_kernel fact or command: uname -r.

348. How do you install multiple packages from a list in one task?

- Use loop with yum or apt module:
- - yum:
- name: "{{ item }}"
- state: present
- loop:
- - git
- - wget
- - curl

349. How do you ensure a command runs only if a file exists?

- Use stat module to check file and when: condition.

350. How do you copy files from a local machine to multiple hosts in parallel?

- Use copy module in a play targeting multiple hosts; Ansible executes tasks in parallel by default.
-

351. How do you ensure a task runs only on a specific host?

- Use hosts: in the play or when: inventory_hostname == "hostname".

352. How do you create multiple users with different shells and home directories?

- Use user module with loop and name, shell, home parameters.

353. How do you remove multiple files only if they exist?

- Use file module with state: absent and when: stat.exists after stat check.

354. How do you copy templates with variables for each host?

- Use template module and define host-specific variables in host_vars.

355. How do you deploy a package only on CentOS hosts?

- Use yum module with when: ansible_distribution == "CentOS".

356. How do you gather only selected facts to improve performance?

- Use setup module with gather_subset: option.

357. How do you check if a service is enabled before starting it?

- Use service_facts module and when: condition.

358. How do you ensure idempotency when copying files?

- Use copy module; Ansible only copies if content differs.

359. How do you remove old files in a directory recursively?

- Use find module with recurse: yes and age: filter, then remove with file module.

360. How do you run a command on all hosts but register results separately?

- Use register: in the task; Ansible stores output per host.

361. How do you deploy configuration files with dynamic content from variables?

- Use template module with Jinja2 placeholders.

362. How do you copy directories while preserving permissions?

- Use copy module with recursive: yes and mode: parameters.

363. How do you ensure a service is restarted only if a configuration changes?

- Use notify with a handler that restarts the service.

364. How do you check if a process is running before starting another task?

- Use shell: pgrep process_name with register: and when: condition.

365. How do you add multiple users to the same group?

- Use user module in a loop with groups: and append: yes.

366. How do you remove multiple users from a group?

- Use user module with groups: (excluding group) and append: yes.

367. How do you deploy software only if enough disk space is available?

- Use ansible_mounts facts to check free space with when: condition.

368. How do you install multiple packages from a list using one task?

- Use loop with yum or apt module:
- - yum:
- name: "{{ item }}"
- state: present
- loop:
- - git
- - wget
- - curl

369. How do you create symbolic links with checks to avoid duplicates?

- Use file module with state: link and when: based on stat module.

370. How do you gather facts about network interfaces only?

- Use setup module with gather_subset: network.

371. How do you copy a file only if the source has changed?

- Use checksum: in copy module.

372. How do you fetch multiple files from multiple hosts efficiently?

- Use fetch module with flat: yes and loops over hosts/files.

373. How do you check CPU usage before running resource-heavy tasks?

- Use ansible_processor_cores or ansible_processor_vcpus facts with when:.

374. How do you remove users only if they exist?

- Use getent command or check /etc/passwd with stat and when: condition.

375. How do you ensure a command is executed only once across all hosts?

- Use run_once: yes.

376. How do you deploy a configuration file only if it differs from the existing one?

- Use template module with notify and diff: yes.

377. How do you restart multiple services in order?

- Define multiple service tasks sequentially in the playbook.

378. How do you manage firewall rules only if certain ports are closed?

- Use firewalld or ufw modules with when: conditions checking ansible_facts.

379. How do you deploy SSL certificates to multiple hosts?

- Use copy or template module to place .crt and .key files.

380. How do you ensure Python packages are installed only if missing?

- Use pip module with state: present and when: based on package_facts.

381. How do you check if a directory is empty before removing it?

- Use find module to count files and conditional when:.

382. How do you ensure a service is started only on hosts with a specific OS?

- Use when: ansible_distribution == "Ubuntu" with service module.

383. How do you copy files to remote hosts using variables for destination paths?

- Use copy module with dest: "{{ var_path }}".

384. How do you remove duplicate lines from configuration files?

- Use lineinfile with regexp:.

385. How do you monitor file changes during playbook execution?

- Run with --diff or use stat module to check differences.

386. How do you deploy applications from a Git repository safely?

- Use git module with version: parameter to ensure idempotency.

387. How do you check connectivity of all hosts before executing tasks?

- Use ping module: ansible all -m ping.

388. How do you manage multiple crontab entries conditionally?

- Use cron module with when: and user: parameters.

389. How do you copy files from local machine to hosts only if changed?

- Use copy module with checksum: or default idempotency.

390. How do you restart a service only if a file exists?

- Use stat module to check file and when: condition with service.

391. How do you ensure a command runs only if a package is installed?

- Use package_facts with when:.

392. How do you configure timezone on multiple hosts?

- Use timezone module or file module to link /etc/localtime.

393. How do you manage SELinux policies with Ansible?

- Use selinux module or edit /etc/selinux/config via template.

394. How do you gather only memory and CPU facts?

- Use setup module with gather_subset: hardware.

395. How do you deploy software from a local archive file?

- Use unarchive module with src: and dest:.

396. How do you ensure swap is enabled after reboot?

- Use command module to create swapfile and add entry in /etc/fstab.

397. How do you manage multiple users' SSH keys efficiently?

- Use authorized_key module with loop over users.

398. How do you remove old files safely without affecting new ones?

- Use find module with age: and file_type: file then remove with file.

399. How do you deploy configuration files with host-specific variables?

- Use template module with host_vars or group_vars.

400. How do you check and restart services only if they are not running?

- Use service_facts to check state and when: with service module.
-

401. How do you ensure a task runs only on hosts in a specific group?

- Use hosts: group_name in the playbook.

402. How do you create users only if they do not exist?

- Use user module with state: present; Ansible is idempotent by default.

403. How do you remove users only if they exist?

- Use user module with state: absent; combine with getent check if needed.

404. How do you copy files from a local directory to multiple hosts?

- Use copy module with src: and dest:; use loop for multiple files.

405. How do you copy a directory recursively?

- Use copy module with recursive: yes.

406. How do you ensure a service is running on multiple hosts?

- Use service module with state: started and enabled: yes.

407. How do you stop a service on multiple hosts?

- Use service module with state: stopped.

408. How do you restart a service only if its configuration changed?

- Use notify and a handler in the playbook.

409. How do you deploy multiple configuration files using templates?

- Use template module with `with_items` or `loop`.

410. How do you fetch files from remote hosts to the control node?

- Use `fetch` module with `flat: yes` and `dest:.`

411. How do you remove files older than a certain number of days?

- Use `find` module with `age:` and `file_type: file`, then remove with `file` module.

412. How do you check disk usage on remote hosts?

- Use `ansible_mounts` fact or `shell: df -h` with `register:.`

413. How do you manage multiple cron jobs for different users?

- Use `cron` module with `user:` parameter.

414. How do you create multiple directories with different owners and permissions?

- Use `file` module with `loop` and `owner`, `group`, `mode`.

415. How do you remove directories only if they exist?

- Use `file` module with `state: absent` and `stat check with when:.`

416. How do you install packages only if they are not already installed?

- Use `package_facts` with `conditional when:.`

417. How do you install a specific version of a package?

- Use `version:` parameter in `yum`, `apt`, or `package` module.

418. How do you deploy software from a Git repository?

- Use `git` module with `repo:` and `dest:.`

419. How do you ensure idempotency when using Git?

- Specify `version:;` Ansible pulls updates only if changed.

420. How do you check host connectivity before running tasks?

- Use `ping` module: `ansible all -m ping`.

421. How do you manage firewall rules on multiple hosts?

- Use `firewalld` or `ufw` module depending on OS.

422. How do you set environment variables for tasks?

- Use `environment:` parameter in the task.

423. How do you restart services sequentially across hosts?

- Use `serial:` in the play or define tasks in order.

424. How do you copy files only if they changed?

- Use checksum: in copy module; idempotent by default.

425. How do you ensure a service is started only on certain OS distributions?

- Use when: `ansible_distribution == "Ubuntu"` with service module.

426. How do you create symbolic links on remote hosts?

- Use file module with state: link and src:.

427. How do you remove duplicate lines from configuration files?

- Use lineinfile module with regexp: and state: present.

428. How do you check CPU and memory facts?

- Use setup module or facts like `ansible_memtotal_mb` and `ansible_processor_cores`.

429. How do you deploy configuration files with host-specific variables?

- Use template module with `host_vars` or `group_vars`.

430. How do you install multiple Python packages with pip?

- Use pip module with loop or list of packages in name:.

431. How do you ensure swap memory is enabled after reboot?

- Use command module to create swapfile and update `/etc/fstab`.

432. How do you manage SELinux settings on multiple hosts?

- Use selinux module or edit `/etc/selinux/config` via template.

433. How do you fetch multiple log files efficiently?

- Use fetch module with loops and flat: yes.

434. How do you check if a process is running before executing a task?

- Use shell: `pgrep process_name` with register: and when:.

435. How do you manage multiple crontab entries for the same user?

- Use cron module with name: parameter for each job.

436. How do you deploy SSL certificates to multiple hosts?

- Use copy or template module for .crt and .key files.

437. How do you copy files with dynamic destination paths?

- Use copy module with dest: `"{{ var_path }}"`.

438. How do you monitor file changes during playbook execution?

- Use `--diff` option with `ansible-playbook`.

439. How do you install software only if the host has enough disk space?

- Use `ansible_mounts` facts with conditional when:.

440. How do you check kernel version on remote hosts?

- Use `ansible_kernel` fact or command: `uname -r`.

441. How do you create temporary files safely?

- Use `tempfile` module with `state: file`.

442. How do you remove users only if they exist and are not logged in?

- Use `getent` or `who` commands with `when: condition`.

443. How do you copy files from local to remote hosts in parallel?

- Use `copy` module; Ansible executes tasks on multiple hosts in parallel by default.

444. How do you check if a service is enabled before starting it?

- Use `service_facts` module and `when: with ansible_facts.services`.

445. How do you ensure a command runs only once across all hosts?

- Use `run_once: yes`.

446. How do you remove old files without affecting new ones?

- Use `find` module with `age: filter` and `remove: with file`.

447. How do you configure hostnames on multiple hosts?

- Use `hostname` module.

448. How do you ensure services are restarted only if needed?

- Use `notify` with a handler.

449. How do you manage multiple directories with different owners and permissions?

- Use `file` module with `loop`, `owner`, `group`, and `mode:.`

450. How do you ensure idempotency when creating files or directories?

- Use `file` or `copy` module; Ansible skips if the desired state is already met.

451. How do you run a task only if a host has a specific fact value?

- Use `when: with the relevant fact`, e.g., `when: ansible_distribution == "Ubuntu"`.

452. How do you create a file with specific content?

- Use `copy` module with `content: parameter:`
- - `copy:`
- `dest: /tmp/myfile.txt`
- `content: "Hello World"`

453. How do you append a line to a file only if it doesn't exist?

- Use lineinfile module with state: present.

454. How do you remove a line from a file?

- Use lineinfile with state: absent and regexp:.

455. How do you copy files from one remote host to another?

- Use synchronize module with mode: push or pull.

456. How do you manage package updates automatically?

- Use yum or apt module with state: latest.

457. How do you check the existence of a package before installing?

- Use package_facts and conditional when:.

458. How do you deploy a configuration file only if it differs from the existing one?

- Use template module; Ansible handles idempotency automatically.

459. How do you run a shell command only if a file exists?

- Use stat module to check file, then when: in the task.

460. How do you restart a service if a certain package was installed?

- Use register: on package task and notify: in a conditional handler.

461. How do you manage multiple users' SSH authorized keys?

- Use authorized_key module with a loop over users.

462. How do you create multiple groups at once?

- Use group module with loop.

463. How do you remove multiple groups only if they exist?

- Use group module with state: absent and a conditional check.

464. How do you check disk space before creating files?

- Use ansible_mounts fact or command: df -h with register:.

465. How do you deploy multiple templates with host-specific variables?

- Use template module with host_vars or group_vars.

466. How do you install multiple packages from a list in one task?

- Use loop with yum or apt module:
- - yum:
- name: "{{ item }}"
- state: present
- loop:

- - git
- - wget
- - curl

467. How do you ensure idempotency when creating users or groups?

- Ansible's user and group modules are idempotent by default.

468. How do you manage firewall rules only if a port is not open?

- Use firewalld or ufw module with when: based on ansible_facts.

469. How do you copy directories while preserving permissions and ownership?

- Use copy module with recursive: yes and mode:.

470. How do you fetch logs from multiple hosts into host-specific directories?

- Use fetch module with dest: "/tmp/{{ inventory_hostname }}/\" and flat: yes.

471. How do you check if a process is running before restarting it?

- Use shell: pgrep process_name with register: and conditional when:.

472. How do you remove files older than X days?

- Use find module with age: and remove with file module.

473. How do you create temporary directories for tasks?

- Use tempfile module with state: directory.

474. How do you check free memory before executing a resource-heavy task?

- Use ansible_memfree_mb fact with when:.

475. How do you ensure a task runs only on a specific host?

- Use when: inventory_hostname == "hostname".

476. How do you deploy configuration files dynamically with Jinja2 variables?

- Use template module with variables from host_vars or group_vars.

477. How do you manage multiple crontab entries for different users?

- Use cron module with name: and user: for each entry.

478. How do you ensure a command runs only once across all hosts?

- Use run_once: yes.

479. How do you monitor changes in files during playbook execution?

- Use --diff option when running the playbook.

480. How do you restart a service only if a configuration file changes?

- Use notify with a handler linked to the task.

481. How do you install Python packages only if missing?

- Use pip module with state: present and conditional when:.

482. How do you deploy SSL certificates on multiple hosts?

- Use copy or template module for .crt and .key files.

483. How do you configure hostnames on multiple hosts?

- Use hostname module.

484. How do you remove users only if they exist and are not logged in?

- Use getent or who with when: condition before using user module.

485. How do you copy files from local to remote hosts efficiently?

- Use copy module; Ansible executes tasks in parallel on multiple hosts by default.

486. How do you check CPU cores and memory facts?

- Use ansible_processor_cores and ansible_memtotal_mb facts.

487. How do you remove directories only if empty?

- Use find module with file_type: file to check contents before removing with file.

488. How do you ensure services start on boot?

- Use service module with enabled: yes.

489. How do you deploy software from a local archive file?

- Use unarchive module with src: and dest:.

490. How do you restart multiple services sequentially?

- Define multiple service tasks in order in the playbook.

491. How do you copy files with dynamic paths?

- Use copy module with dest: "{{ variable_path }}".

492. How do you remove old log files safely?

- Use find module with age: filter and remove with file.

493. How do you gather only specific facts to improve performance?

- Use setup module with gather_subset:.

494. How do you ensure idempotency when managing files or directories?

- Use file or copy module; Ansible skips if desired state is already met.

495. How do you check service status before restarting it?

- Use service_facts and when: condition.

496. How do you deploy applications from a Git repository safely?

- Use git module with version: to ensure idempotency.

497. How do you manage swap memory across hosts?

- Use command module to create swapfile and update /etc/fstab.

498. How do you ensure firewall rules are applied only if needed?

- Use when: conditions based on ansible_facts with firewalld or ufw modules.

499. How do you run tasks conditionally based on OS type?

- Use when: ansible_distribution == "Ubuntu" or similar condition.

500. How do you ensure a configuration file is only updated if content changes?

- Use template module with idempotency; Ansible updates only if content differs.

Medium Ansible Q&A 501–550 for AWS DevOps:

501. How do you use include_tasks to run a set of tasks conditionally?

- Use include_tasks: tasks.yml with when: to include tasks only under certain conditions.

502. How do you use import_tasks vs include_tasks?

- import_tasks is static (evaluated at parse time); include_tasks is dynamic (evaluated at runtime).

503. How do you use block to group tasks with error handling?

- Use block: with rescue: and always: to manage errors for multiple tasks together.

504. How do you use delegate_to to run a task on a different host?

- Add delegate_to: hostname in the task to execute it on another host.

505. How do you use run_once with delegate_to?

- Combines execution on a single host while delegating the task to another host.

506. How do you use when with multiple conditions?

- Use and/or operators:
- when: ansible_os_family == "RedHat" and ansible_distribution_major_version == "8"

507. How do you gather only selected facts to optimize performance?

- Use setup module with gather_subset: network, hardware.

508. How do you use vars_files to manage multiple variable files?

- Include variable files in a playbook:
- vars_files:

- - vars/common.yml
- - vars/production.yml

509. How do you loop through a list of dictionaries with `with_items`?

- Example:
- - name: Create users
- user:
- name: "{{ item.name }}"
- shell: "{{ item.shell }}"
- with_items:
- - { name: "alice", shell: "/bin/bash" }
- - { name: "bob", shell: "/bin/zsh" }

510. How do you use `register` to store task output for later use?

- Add register: result and reference `{{ result.stdout }}` in later tasks.

511. How do you use `failed_when` to customize failure conditions?

- Example:
- failed_when: "'error' in result.stdout"

512. How do you use `changed_when` to control change detection?

- Example:
- changed_when: result.stdout != "

513. How do you use `with_fileglob` to loop through files in a directory?

- Example:
- - copy:
- src: "{{ item }}"
- dest: /tmp/
- with_fileglob: "/local/path/*.conf"

514. How do you use `with_nested` to loop through multiple lists?

- Example:
- - debug:
- msg: "{{ item }}"
- with_nested:
- - [1,2]

- - [a,b]

515. How do you use `ansible.builtin.include_role` dynamically?

- Include roles conditionally:
- - `include_role:`
- `name: common`
- `when: ansible_os_family == "Debian"`

516. How do you use `set_fact` to create dynamic variables during playbook execution?

- Example:
- - `set_fact:`
- `my_var: "{{ ansible_hostname }}_data"`

517. How do you use `with_dict` to loop through a dictionary?

- Example:
- - `debug:`
- `msg: "{{ item.key }}={{ item.value }}"`
- `with_dict: "{{ my_dict }}"`

518. How do you combine `loop` and `include_tasks`?

- Example:
- - `include_tasks: create_user.yml`
- `loop: "{{ users_list }}"`
- `loop_control:`
- `loop_var: user`

519. How do you use `until` and `retries` for task retries?

- Example:
- - `name: Wait for service`
- `command: curl http://localhost:8080`
- `register: result`
- `retries: 5`
- `delay: 10`
- `until: result.rc == 0`

520. How do you use `block` with `rescue` to handle failures gracefully?

- Example:

- block:
- - command: /bin/false
- rescue:
- - debug: msg="Task failed but handled"

521. How do you use `ansible.builtin.pause` to wait for user input?

- Example:
- - pause:
- prompt: "Press Enter to continue"

522. How do you run tasks only on hosts with a specific group membership?

- Use when: "'webservers' in group_names".

523. How do you use `ansible_facts` to make decisions?

- Example:
- when: `ansible_facts['os_family'] == 'RedHat'`

524. How do you use `with_sequence` to loop over a sequence of numbers?

- Example:
- - debug:
- msg: "{{ item }}"
- with_sequence: start=1 end=5

525. How do you conditionally include roles based on facts?

- Use `include_role` with when: statement.

526. How do you use `with_together` to iterate over multiple lists simultaneously?

- Example:
- - debug: msg="{{ item.0 }}" and "{{ item.1 }}"
- with_together:
- - [a,b]
- - [1,2]

527. How do you use `ansible.builtin.wait_for` to wait for a port to become available?

- Example:
- - wait_for:
- host: "{{ inventory_hostname }}"
- port: 22

- timeout: 300

528. How do you use `ansible.builtin.copy` with `backup: yes`?

- Example:
- - copy:
- src: /local/file
- dest: /remote/file
- backup: yes

529. How do you use `ansible.builtin.slurp` to read remote file contents?

- Example:
- - slurp:
- src: /etc/hosts
- register: slurped_file

530. How do you use `ansible.builtin.fetch` to download remote files efficiently?

- Example:
- - fetch:
- src: /var/log/syslog
- dest: ./logs/{{ inventory_hostname }}/
- flat: yes

531. How do you use `ansible.builtin.file` with `state: touch`?

- Creates an empty file if it does not exist.

532. How do you use `ansible.builtin.replace` to update file content?

- Example:
- - replace:
- path: /etc/config
- regexp: 'old_value'
- replace: 'new_value'

533. How do you use `ansible.builtin.lineinfile` to ensure a line exists at the end of a file?

- Example:
- - lineinfile:
- path: /etc/config
- line: "export PATH=\$PATH:/my/bin"

- state: present

534. How do you use ansible.builtin.template with host-specific variables?

- Place variables in host_vars or group_vars and reference in template using {{ variable }}.

535. How do you use ansible.builtin.yum or apt modules with update_cache: yes?

- Ensures the package manager cache is updated before installing.

536. How do you use ansible.builtin.service with enabled: yes and state: started?

- Ensures service starts on boot and is currently running.

537. How do you use ansible.builtin.systemd for systemd-specific tasks?

- Example:
- - systemd:
- name: nginx
- state: restarted
- enabled: yes

538. How do you use ansible.builtin.user to manage SSH key access?

- Example:
- - user:
- name: alice
- ssh_key: "{{ lookup('file', '/home/alice/.ssh/id_rsa.pub') }}"

539. How do you use ansible.builtin.group to manage group memberships dynamically?

- Example:
- - group:
- name: devops
- state: present

540. How do you use ansible.builtin.debug with var: to inspect variables?

- Example:
- - debug:
- var: ansible_facts

541. How do you use ansible.builtin.assert to validate conditions?

- Example:
- - assert:
- that:

- - ansible_os_family == "RedHat"

542. How do you use ansible.builtin.meta: end_play to stop a play dynamically?

- Stops play execution when certain conditions are met.

543. How do you use ansible.builtin.meta: flush_handlers to run handlers immediately?

- Forces handlers to execute at a specific point in the playbook.

544. How do you use ansible.builtin.set_fact with conditional logic?

- Example:
- - set_fact:
- my_var: "{{ 'value1' if ansible_os_family == 'RedHat' else 'value2' }}"

545. How do you use ansible.builtin.command vs ansible.builtin.shell safely?

- Use command when no shell features are needed; shell when using pipes, redirection, or variables.

546. How do you use ansible.builtin.raw to run commands on hosts without Python?

- Example:
- - raw: cat /etc/hosts

547. How do you use ansible.builtin.include_vars to load variables dynamically?

- Example:
- - include_vars: "{{ inventory_hostname }}.yaml"

548. How do you use ansible.builtin.set_fact to compute values based on host facts?

- Example:
- - set_fact:
- free_memory_gb: "{{ ansible_memfree_mb / 1024 }}"

549. How do you use ansible.builtin.pause with a timeout for automation?

- Example:
- - pause:
- seconds: 10
- prompt: "Waiting before next task"

550. How do you use ansible.builtin.wait_for_connection to ensure hosts are reachable?

- Waits for SSH connectivity before running tasks.
-

551. How do you use serial to control batch execution of hosts?

- Example:
- - hosts: webservers
- serial: 2

This runs tasks on 2 hosts at a time.

552. How do you use strategy: free to allow hosts to run tasks independently?

- Example:
- - hosts: all
- strategy: free

Tasks on faster hosts proceed without waiting for slower ones.

553. How do you use async and poll to run tasks asynchronously?

- Example:
- - command: /long/running/script.sh
- async: 600
- poll: 0

Runs task in background without waiting.

554. How do you use wait_for with a path instead of a port?

- Example:
- - wait_for:
- path: /tmp/file_ready
- state: present
- timeout: 300

555. How do you use with_inventory_hostnames to loop over hosts in a group?

- Example:
- - debug:
- msg: "{{ item }}"
- with_inventory_hostnames: webservers

556. How do you combine loop_control with loop_var for clarity?

- Example:
- - debug:
- msg: "{{ user.name }}"

- loop: "{{ users }}"
- loop_control:
- loop_var: user

557. How do you use import_playbook to modularize playbooks?

- Example:
- - import_playbook: setup_webservers.yml

Static import at parse time.

558. How do you use include_role dynamically with variables?

- Example:
- - include_role:
- name: "{{ role_name }}"

559. How do you use tags to run only specific tasks?

- Example:
- ansible-playbook site.yml --tags "deploy,configure"

560. How do you skip tasks with tags?

- Example:
- ansible-playbook site.yml --skip-tags "cleanup"

561. How do you use force_handlers: true in a playbook?

- Ensures all handlers run even if no tasks report changes.

562. How do you use meta: clear_facts to remove all host facts?

- Clears previously gathered facts during playbook execution.

563. How do you use pre_tasks and post_tasks in a playbook?

- pre_tasks run before main roles/tasks, post_tasks after all tasks.

564. How do you use environment to set environment variables for tasks?

- Example:
- - command: echo \$MY_VAR
- environment:
- MY_VAR: "Hello"

565. How do you use with_items for nested dictionaries?

- Example:
- - debug:

- msg: "{{ item.key }}={{ item.value }}"
- with_items: "{{ my_dict | dict2items }}"

566. How do you use failed_when with multiple conditions?

- Example:
- failed_when: result.rc != 0 and "'ignore' not in result.stdout"

567. How do you use changed_when to mark tasks as changed conditionally?

- Example:
- changed_when: "'updated' in result.stdout"

568. How do you use when with hostvars for cross-host conditions?

- Example:
- when: hostvars['host1'].ansible_os_family == 'Debian'

569. How do you use with_indexed_items to get index and value?

- Example:
- - debug:
- msg: "{{ item.0 }}: {{ item.1 }}"
- with_indexed_items:
- - a
- - b

570. How do you use assert to validate variable values dynamically?

- Example:
- - assert:
- that:
- - my_var is defined
- - my_var != ""

571. How do you use include_vars conditionally?

- Example:
- - include_vars: "{{ item }}"
- with_items:
- - dev.yml
- - prod.yml
- when: ansible_env == 'production'

572. How do you use `with_random_choice` to pick random items from a list?

- Example:
- - debug:
- `msg: "{{ item }}"`
- `with_random_choice:`
- - a
- - b
- - c
- - d

573. How do you use `register with loop` to collect results per item?

- Example:
- - command: `echo {{ item }}`
- `loop: [1,2,3]`
- `register: results`

574. How do you use `loop_control.label` to customize output in loops?

- Example:
- - debug:
- `msg: "Processing {{ item.name }}"`
- `loop: "{{ users }}"`
- `loop_control:`
- `label: "{{ item.name }}"`

575. How do you use `until with retries and delay` to wait for a service?

- Example:
- - shell: `curl http://localhost:8080`
- `register: result`
- `until: result.rc == 0`
- `retries: 5`
- `delay: 10`

576. How do you use `block with rescue and always` for advanced error handling?

- Example:
- `block:`

- - command: /bin/false
- rescue:
- - debug: msg="Failed but handled"
- always:
- - debug: msg="Always runs"

577. How do you use `ansible.builtin.uri` to check HTTP status of a service?

- Example:
- - uri:
- url: http://localhost:8080
- return_content: yes
- register: result

578. How do you use `ansible.builtin.wait_for_connection` in large environments?

- Ensures hosts are reachable before executing tasks.

579. How do you use `with_inventory_groups` to loop over groups?

- Example:
- - debug:
- msg: "{{ item }}"
- with_inventory_groups:
- - webserver

580. How do you use tags with roles to run selective parts of a playbook?

- Assign tags in role tasks and run:
- ansible-playbook site.yml --tags "install,configure"

581. How do you use `vars_prompt` to ask for sensitive input at runtime?

- Example:
- vars_prompt:
- - name: "db_password"
- prompt: "Enter DB password"
- private: yes

582. How do you use `with_subelements` to loop through nested lists?

- Example:
- - debug:

- msg: "{{ item.0.name }}" uses "{{ item.1 }}"
- with_subelements:
- - users
- - groups

583. How do you use `ansible.builtin.debug` with `msg` to inspect complex variables?

- Example:
- - debug:
- msg: "{{ my_complex_var | to_nice_json }}"

584. How do you use `ansible.builtin.include_role` with `tasks_from` to select tasks?

- Example:
- - include_role:
- name: common
- tasks_from: setup.yml

585. How do you use `with_together` to iterate over multiple lists simultaneously?

- Already shown in Medium 501–550; useful for mapping users to groups.

586. How do you use `ansible.builtin.fail` to stop playbook on certain conditions?

- Example:
- - fail:
- msg: "Required variable missing!"
- when: my_var is not defined

587. How do you use `ansible.builtin.meta: noop`?

- No-op command to satisfy syntax; can be used in dynamic task inclusion.

588. How do you use `ansible.builtin.debug` with verbosity levels?

- Example:
- - debug:
- msg: "Detailed info"
- verbosity: 2

589. How do you use `ansible.builtin.template` with conditional blocks inside the template?

- Example Jinja2:
- {% if ansible_os_family == 'RedHat' %}
- value=1

- {% else %}
- value=2
- {% endif %}

590. How do you use `ansible.builtin.include_tasks` with `loop` and `loop_control`?

- Dynamically include tasks for each item in a list.

591. How do you use `ansible.builtin.meta: reset_connection` to refresh SSH connections?

- Useful for reloading facts or reconnecting after network changes.

592. How do you use `ansible.builtin.meta: noop` for conditional task execution?

- Acts as a placeholder; can prevent errors in dynamic includes.

593. How do you use `ansible.builtin.wait_for` with multiple conditions?

- Combine host, port, path, state, and timeout.

594. How do you use `ansible.builtin.assert` with multiple conditions?

- Example:
- - assert:
- that:
- - ansible_os_family == "RedHat"
- - ansible_processor_cores > 2

595. How do you use `ansible.builtin.set_fact` with loops to build lists?

- Example:
- - set_fact:
- my_list: "{{ my_list | default([]) + [item] }}"
- loop: [1,2,3]

596. How do you use `ansible.builtin.copy` with `remote_src: yes`?

- Copies file already present on remote host to another remote location.

****597. How do you use `ansible.builtin.get_url` to download files from internet?**

- Example:
- - get_url:
- url: http://example.com/file.tar.gz
- dest: /tmp/file.tar.gz

598. How do you use `ansible.builtin.unarchive` with remote files?

- Example:
- - unarchive:
- src: /tmp/file.tar.gz
- dest: /opt/
- remote_src: yes

599. How do you use `ansible.builtin.set_fact` to calculate host-specific values?

- Example:
- - set_fact:
- log_dir: "/var/log/{{ inventory_hostname }}"

600. How do you use `ansible.builtin.include_tasks` with `when` for conditional task execution?

- Example:
- - include_tasks: install.yml
- when: ansible_os_family == "Debian"

601. How do you use `ansible.builtin.git` to deploy code from a Git repository?

- Example:
- - git:
- repo: 'https://github.com/user/repo.git'
- dest: /opt/myapp
- version: main

602. How do you use `ansible.builtin.synchronize` to sync directories between hosts?

- Example:
- - synchronize:
- src: /src/path/
- dest: /dest/path/
- mode: push

603. How do you use `ansible.builtin.yum` with `disable_gpg_check`?

- Example:
- - yum:
- name: httpd
- state: present

- `disable_gpg_check: yes`

604. How do you use `ansible.builtin.apt` with `update_cache: yes`?

- Example:
- `- apt:`
- `name: nginx`
- `state: latest`
- `update_cache: yes`

605. How do you use `ansible.builtin.package` module to install packages in a generic way?

- Example:
- `- package:`
- `name: "{{ item }}"`
- `state: present`
- `loop:`
- `- git`
- `- curl`

606. How do you use `ansible.builtin.service` with `enabled: no`?

- Disables service from starting on boot:
- `- service:`
- `name: httpd`
- `state: stopped`
- `enabled: no`

607. How do you use `ansible.builtin.systemd` to mask a service?

- Example:
- `- systemd:`
- `name: apache2`
- `masked: yes`

608. How do you use `ansible.builtin.user` to add a user with a specific UID?

- Example:
- `- user:`
- `name: deploy`
- `uid: 1500`

- state: present

609. How do you use `ansible.builtin.group` to add a group with GID?

- Example:
- - group:
- name: devops
- gid: 2000
- state: present

610. How do you use `ansible.builtin.file` with mode and owner?

- Example:
- - file:
- path: /opt/myapp
- state: directory
- owner: deploy
- group: devops
- mode: '0755'

611. How do you use `ansible.builtin.copy` with `force: no`?

- Avoids copying if destination file is unchanged:
- - copy:
- src: myconfig.conf
- dest: /etc/myconfig.conf
- force: no

612. How do you use `ansible.builtin.lineinfile` with `regexp` to update config lines?

- Example:
- - lineinfile:
- path: /etc/myapp.conf
- regexp: '^PORT='
- line: 'PORT=8080'

613. How do you use `ansible.builtin.replace` for complex pattern replacement?

- Example:
- - replace:
- path: /etc/myapp.conf

- regexp: 'DEBUG=(True|False)'
- replace: 'DEBUG=False'

614. How do you use `ansible.builtin.template` with a Jinja2 expression?

- Example:
- - template:
- src: app.conf.j2
- dest: /etc/myapp/app.conf

615. How do you use `ansible.builtin.get_url` with checksum validation?

- Example:
- - get_url:
- url: http://example.com/file.tar.gz
- dest: /tmp/file.tar.gz
- checksum: sha256:abcd1234...

616. How do you use `ansible.builtin.unarchive` with `remote_src: yes`?

- Example:
- - unarchive:
- src: /tmp/file.tar.gz
- dest: /opt/myapp
- remote_src: yes

617. How do you use `ansible.builtin.fetch` to collect log files from remote hosts?

- Example:
- - fetch:
- src: /var/log/myapp.log
- dest: ./logs/{{ inventory_hostname }}/
- flat: yes

618. How do you use `ansible.builtin.slurp` to read file content into a variable?

- Example:
- - slurp:
- src: /etc/hosts
- register: hosts_content

619. How do you use `ansible.builtin.stat` to check file properties?

- Example:
- - stat:
- path: /etc/myapp.conf
- register: file_stat

620. How do you use `ansible.builtin.debug` to print registered variable values?

- Example:
- - debug:
- var: file_stat

621. How do you use `ansible.builtin.wait_for` to check for a TCP port?

- Example:
- - wait_for:
- host: "{{ inventory_hostname }}"
- port: 80
- timeout: 120

622. How do you use `ansible.builtin.wait_for` to wait for a file to appear?

- Example:
- - wait_for:
- path: /tmp/myfile_ready
- state: present
- timeout: 300

623. How do you use `ansible.builtin.shell` with `creates` to ensure idempotency?

- Example:
- - shell: tar -xzf file.tar.gz -C /opt/myapp
- args:
- creates: /opt/myapp

624. How do you use `ansible.builtin.command` with `chdir`?

- Example:
- - command: ./deploy.sh
- args:
- chdir: /opt/myapp

625. How do you use `ansible.builtin.user` with `groups` and `append`?

- Example:
- - user:
- name: alice
- groups: "devops,sudo"
- append: yes

626. How do you use `ansible.builtin.lineinfile` with `insertbefore` or `insertafter`?

- Example:
- - lineinfile:
- path: /etc/myapp.conf
- line: "NEW_SETTING=1"
- insertafter: '^# Settings'

627. How do you use `ansible.builtin.replace` with multiline patterns?

- Example:
- - replace:
- path: /etc/myapp.conf
- regexp: 'start=.*\nend=.*'
- replace: 'start=1\nend=2'

628. How do you use `ansible.builtin.service` with `state: restarted`?

- Example:
- - service:
- name: nginx
- state: restarted

629. How do you use `ansible.builtin.systemd` to reload a service?

- Example:
- - systemd:
- name: nginx
- state: reloaded

630. How do you use `ansible.builtin.systemd` with `enabled: yes` to ensure startup?

- Example:
- - systemd:
- name: nginx

- enabled: yes

631. How do you use ansible.builtin.pip to install Python packages?

- Example:
- - pip:
- name: requests
- state: present

632. How do you use ansible.builtin.pip with virtualenv?

- Example:
- - pip:
- name: flask
- virtualenv: /opt/venv
- state: present

633. How do you use ansible.builtin.assert to validate host configuration?

- Example:
- - assert:
- that:
- - ansible_os_family == "RedHat"
- - ansible_processor_cores > 2

634. How do you use ansible.builtin.meta: flush_handlers to trigger handlers immediately?

- Forces handlers to execute without waiting for playbook end.

635. How do you use ansible.builtin.meta: end_play to stop execution on certain conditions?

- Example:
- - meta: end_play
- when: some_condition

636. How do you use ansible.builtin.set_fact with conditional assignment?

- Example:
- - set_fact:
- my_var: "{{ 'value1' if ansible_os_family == 'RedHat' else 'value2' }}"

637. How do you use ansible.builtin.include_tasks with loop for dynamic task execution?

- Include a task file multiple times with different loop items.

638. How do you use ansible.builtin.include_vars to load environment-specific variables?

- Example:
- - include_vars: "{{ env }}.yml"

639. How do you use ansible.builtin.raw for hosts without Python?

- Example:
- - raw: cat /etc/hosts

640. How do you use ansible.builtin.debug with verbosity to control output?

- Example:
- - debug:
- msg: "Detailed info"
- verbosity: 2

641. How do you use ansible.builtin.pause with seconds to delay playbook execution?

- Example:
- - pause:
- seconds: 10

642. How do you use ansible.builtin.wait_for_connection for hosts behind dynamic IPs?

- Ensures SSH is reachable before tasks run.

643. How do you use ansible.builtin.include_role with tasks_from to select specific role tasks?

- Example:
- - include_role:
- name: common
- tasks_from: setup.yml

644. How do you use ansible.builtin.set_fact to build a list dynamically during playbook run?

- Example:
- - set_fact:
- my_list: "{{ my_list | default([]) + [item] }}"
- loop: [1,2,3]

645. How do you use ansible.builtin.lineinfile with backrefs: yes?

- Allows using regex capture groups in replacement lines.

646. How do you use ansible.builtin.get_url with headers and authentication?

- Example:
- - get_url:

- url: http://example.com/file
- dest: /tmp/file
- headers:
- Authorization: "Bearer token"

647. How do you use `ansible.builtin.uri` to perform REST API calls?

- Example:
- - uri:
- url: http://api.example.com/status
- method: GET
- return_content: yes

648. How do you use `ansible.builtin.expect` to handle interactive prompts?

- Example:
- - expect:
- command: passwd user
- responses:
- 'New password:': 'MyPassword'
- 'Retype new password:': 'MyPassword'

649. How do you use `ansible.builtin.file` with `recurse: yes` to delete directories?

- Example:
- - file:
- path: /tmp/old_files
- state: absent
- recurse: yes

650. How do you use `ansible.builtin.group_by` to create dynamic host groups based on facts?

- Example:
 - - group_by:
 - key: "{{ ansible_os_family }}"
-

651. How do you use `ansible.builtin.apt_key` to add repository keys?

- Example:
- - apt_key:
- url: `https://packages.example.com/key.gpg`
- state: present

652. How do you use `ansible.builtin.apt_repository` to add a PPA or repo?

- Example:
- - apt_repository:
- repo: `ppa:deadsnakes/ppa`
- state: present

653. How do you use `ansible.builtin.yum_repository` to add a YUM repo?

- Example:
- - yum_repository:
- name: epel
- description: EPEL repo
- baseurl: `http://download.fedoraproject.org/pub/epel/$releasever/$basearch/`
- gpgcheck: yes
- state: present

654. How do you use `ansible.builtin.ping` module to check host availability?

- Example:
- - ping:

655. How do you use `ansible.builtin.debug` to inspect nested dictionary values?

- Example:
- - debug:
- var: `my_dict['nested']['key']`

656. How do you use `ansible.builtin.assert` to ensure a package is installed?

- Example:
- - assert:
- that:
- - `"'nginx' in ansible_facts.packages"`

657. How do you use `ansible.builtin.service_facts` to get service status?

- Example:
- - service_facts:
- - debug:
- var: ansible_facts.services

658. How do you use ansible.builtin.group_by to dynamically categorize hosts?

- Example:
- - group_by:
- key: "{{ ansible_distribution }}_servers"

659. How do you use ansible.builtin.set_fact with hostvars for cross-host calculations?

- Example:
- - set_fact:
- total_memory: "{{ hostvars['host1'].ansible_memtotal_mb + hostvars['host2'].ansible_memtotal_mb }}"

660. How do you use ansible.builtin.include_tasks with with_items?

- Dynamically include tasks for each item in a list.

661. How do you use ansible.builtin.meta: reset_connection after changing network settings?

- Refreshes SSH connection to ensure tasks continue.

662. How do you use ansible.builtin.set_fact with conditional loops?

- Example:
- - set_fact:
- active_users: "{{ active_users | default([]) + [item.name] }}"
- loop: "{{ users }}"
- when: item.active

663. How do you use ansible.builtin.slurp with b64decode to read file contents?

- Example:
- - slurp:
- src: /etc/hosts
- register: content
- - set_fact:
- hosts_content: "{{ content.content | b64decode }}"

664. How do you use ansible.builtin.fetch with flat: yes to avoid nested directories?

- Example:
- - fetch:
- src: /var/log/syslog
- dest: ./logs/
- flat: yes

665. How do you use `ansible.builtin.stat` to check if a file is a directory?

- Example:
- - stat:
- path: /opt/myapp
- register: dir_stat
- - debug:
- msg: "{{ dir_stat.stat.isdir }}"

666. How do you use `ansible.builtin.wait_for` with `search_regex`?

- Example:
- - wait_for:
- path: /var/log/app.log
- search_regex: "Service started"
- timeout: 120

667. How do you use `ansible.builtin.shell` with `creates` for idempotency?

- Example:
- - shell: tar -xzf app.tar.gz -C /opt/app
- args:
- creates: /opt/app/app_binary

668. How do you use `ansible.builtin.command` with `removes`?

- Example:
- - command: rm /tmp/tempfile
- args:
- removes: /tmp/tempfile

669. How do you use `ansible.builtin.user` to remove users and their home directories?

- Example:
- - user:

- name: olduser
- state: absent
- remove: yes

670. How do you use `ansible.builtin.group` to remove groups?

- Example:
- - group:
- name: oldgroup
- state: absent

671. How do you use `ansible.builtin.file` to set symbolic links?

- Example:
- - file:
- src: /opt/myapp/current
- dest: /opt/myapp/latest
- state: link

672. How do you use `ansible.builtin.copy` with `remote_src: yes`?

- Copies a file that already exists on remote host.

673. How do you use `ansible.builtin.template` to configure multiple hosts dynamically?

- Use `host_vars` or `group_vars` to provide host-specific variables.

674. How do you use `ansible.builtin.get_url` with `timeout` and `retries`?

- Example:
- - get_url:
- url: http://example.com/file.tar.gz
- dest: /tmp/file.tar.gz
- timeout: 30
- retries: 3

675. How do you use `ansible.builtin.unarchive` with `remote_src: yes` and `creates`?

- Avoids re-extraction if files already exist.

676. How do you use `ansible.builtin.pip` to upgrade a package?

- Example:
- - pip:
- name: requests

- state: latest

677. How do you use `ansible.builtin.service` with state: restarted safely?

- Ensures service restarts if configuration changed.

678. How do you use `ansible.builtin.systemd` to reload a unit without restart?

- Example:
- - systemd:
- name: nginx
- state: reloaded

679. How do you use `ansible.builtin.assert` to validate multiple facts?

- Example:
- - assert:
- that:
- - ansible_os_family == 'RedHat'
- - ansible_memtotal_mb > 2048

680. How do you use `ansible.builtin.meta: flush_handlers` for immediate execution?

- Forces all handlers to run at that point in the playbook.

681. How do you use `ansible.builtin.meta: end_play` to stop playbook conditionally?

- Example:
- - meta: end_play
- when: not my_condition

682. How do you use `ansible.builtin.include_tasks` with conditional variables?

- Include different task files depending on OS or environment.

683. How do you use `ansible.builtin.set_fact` to calculate host-specific directories?

- Example:
- - set_fact:
- log_dir: "/var/log/{{ inventory_hostname }}"

684. How do you use `ansible.builtin.slurp` to read sensitive files securely?

- Use b64decode to decode content after slurping.

685. How do you use `ansible.builtin.fetch` to collect files from all hosts in a group?

- Use a loop with inventory_hostname for organization.

686. How do you use `ansible.builtin.stat` to check file ownership and permissions?

- Example:
- - stat:
- path: /etc/myapp.conf
- register: file_info
- - debug:
- msg: "Owner: {{ file_info.stat.pw_name }}, Mode: {{ file_info.stat.mode }}"

687. How do you use `ansible.builtin.wait_for` with both port and host?

- Waits for a specific service on a specific host to be ready.

688. How do you use `ansible.builtin.shell` with `warn: no` to suppress warnings?

- Example:
- - shell: some_command
- args:
- warn: no

689. How do you use `ansible.builtin.command` with `creates` for idempotency?

- Prevents re-running commands unnecessarily.

690. How do you use `ansible.builtin.user` to lock an account?

- Example:
- - user:
- name: testuser
- state: present
- password_lock: yes

691. How do you use `ansible.builtin.group` to add users to a group?

- Example:
- - group:
- name: devops
- state: present
- - user:
- name: alice
- groups: devops
- append: yes

692. How do you use `ansible.builtin.file` with `recurse: yes` to remove directories?

- Example:
- - file:
- path: /tmp/old
- state: absent
- recurse: yes

693. How do you use `ansible.builtin.copy` with `backup: yes`?

- Example:
- - copy:
- src: myconfig.conf
- dest: /etc/myconfig.conf
- backup: yes

Creates a backup of the existing file before replacing it.

694. How do you use `ansible.builtin.replace` with multiple regex patterns?

- Example:
- - replace:
- path: /etc/myapp.conf
- regexp: 'DEBUG=(True|False)'
- replace: 'DEBUG=False'

695. How do you use `ansible.builtin.lineinfile` with `state: absent`?

- Example:
- - lineinfile:
- path: /etc/myapp.conf
- regexp: '^OLD_SETTING='
- state: absent

Removes a specific line matching the regex.

696. How do you use `ansible.builtin.service` with `enabled: no` to disable a service?

- Example:
- - service:
- name: httpd
- state: stopped

- enabled: no

697. How do you use `ansible.builtin.systemd` to mask a service?

- Example:
- - systemd:
- name: apache2
- masked: yes

Prevents the service from being started manually or automatically.

698. How do you use `ansible.builtin.pip` with `virtualenv` to install packages?

- Example:
- - pip:
- name: flask
- virtualenv: /opt/venv
- state: present

699. How do you use `ansible.builtin.get_url` with `force: no`?

- Example:
- - get_url:
- url: http://example.com/file.tar.gz
- dest: /tmp/file.tar.gz
- force: no

Downloads the file only if it does not already exist or has changed.

700. How do you use `ansible.builtin.uri` to perform authenticated API requests?

- Example:
 - - uri:
 - url: http://api.example.com/data
 - method: GET
 - user: myuser
 - password: mypass
 - return_content: yes
-

701. How do you use `ansible.builtin.include_role` with `tasks_from`?

- Example:
- - `include_role:`
- `name: webserver`
- `tasks_from: setup.yml`

702. How do you use `ansible.builtin.import_role` for static role inclusion?

- Example:
- - `import_role:`
- `name: database`

703. How do you use `ansible.builtin.meta: noop`?

- Used to mark a point in playbooks without performing any action.

704. How do you use `ansible.builtin.meta: clear_facts`?

- Removes previously set facts from hosts.

705. How do you use `ansible.builtin.assert` with multiple conditions?

- Example:
- - `assert:`
- `that:`
- `- ansible_os_family == "Debian"`
- `- ansible_processor_cores > 2`

706. How do you use `ansible.builtin.set_fact` with a default value?

- Example:
- - `set_fact:`
- `my_var: "{{ my_var | default('default_value') }}"`

707. How do you use `ansible.builtin.group_by` to categorize hosts by memory size?

- Example:
- - `group_by:`
- `key: "memory_{{ ansible_memtotal_mb }}"`

708. How do you use `ansible.builtin.slurp` with large files?

- Example:
- - `slurp:`
- `src: /var/log/large.log`

- register: content
- - set_fact:
- decoded_content: "{{ content.content | b64decode }}"

709. How do you use ansible.builtin.fetch with host-specific folders?

- Example:
- - fetch:
- src: /var/log/syslog
- dest: ./logs/{{ inventory_hostname }}/

710. How do you use ansible.builtin.wait_for to check a port with delay?

- Example:
- - wait_for:
- host: "{{ inventory_hostname }}"
- port: 443
- delay: 10
- timeout: 120

711. How do you use ansible.builtin.template for environment-specific configuration?

- Example:
- - template:
- src: app_{{ env }}.conf.j2
- dest: /etc/myapp/app.conf

712. How do you use ansible.builtin.copy with remote_src and backup?

- Example:
- - copy:
- src: /tmp/config.cfg
- dest: /etc/myapp/config.cfg
- remote_src: yes
- backup: yes

713. How do you use ansible.builtin.yum with multiple packages?

- Example:
- - yum:
- name:

- - httpd
- - git
- state: present

714. How do you use `ansible.builtin.apt` with `upgrade: dist`?

- Example:
- - apt:
- upgrade: dist

715. How do you use `ansible.builtin.pip` to install requirements from a file?

- Example:
- - pip:
- requirements: /opt/myapp/requirements.txt

716. How do you use `ansible.builtin.systemd` to enable and start multiple services?

- Example:
- - systemd:
- name: "{{ item }}"
- enabled: yes
- state: started
- loop:
- - nginx
- - mysql

717. How do you use `ansible.builtin.user` to create users with SSH keys?

- Example:
- - user:
- name: deploy
- state: present
- generate_ssh_key: yes

718. How do you use `ansible.builtin.group` with multiple users?

- Example:
- - group:
- name: devops
- state: present

- - user:
- name: alice
- groups: devops
- append: yes

719. How do you use `ansible.builtin.file` with `mode: 0644` for files?

- Example:
- - file:
- path: /etc/myapp/config.cfg
- state: file
- mode: '0644'

720. How do you use `ansible.builtin.lineinfile` to insert a line only if absent?

- Example:
- - lineinfile:
- path: /etc/myapp/config.cfg
- line: "MAX_CONNECTIONS=100"
- state: present

721. How do you use `ansible.builtin.replace` to update a configuration value?

- Example:
- - replace:
- path: /etc/myapp/config.cfg
- regexp: '^PORT=.*'
- replace: 'PORT=8080'

722. How do you use `ansible.builtin.shell` with `creates` to avoid re-running commands?

- Example:
- - shell: tar -xzf app.tar.gz -C /opt/app
- args:
- creates: /opt/app/app_binary

723. How do you use `ansible.builtin.command` with `removes` for idempotency?

- Example:
- - command: rm /tmp/tempfile
- args:

- removes: /tmp/tempfile

724. How do you use ansible.builtin.get_url with timeout and retries?

- Example:
- - get_url:
- url: http://example.com/file.tar.gz
- dest: /tmp/file.tar.gz
- timeout: 30
- retries: 3

725. How do you use ansible.builtin.unarchive with remote_src: yes and creates?

- Example:
- - unarchive:
- src: /tmp/app.tar.gz
- dest: /opt/app
- remote_src: yes
- creates: /opt/app/app_binary

726. How do you use ansible.builtin.meta: flush_handlers?

- Immediately executes handlers defined in the playbook.

727. How do you use ansible.builtin.meta: end_play?

- Stops the current play for certain conditions.

728. How do you use ansible.builtin.pause to wait between tasks?

- Example:
- - pause:
- seconds: 10

729. How do you use ansible.builtin.wait_for_connection?

- Waits until SSH connection is available for the host.

730. How do you use ansible.builtin.set_fact with hostvars?

- Example:
- - set_fact:
- combined_memory: "{{ hostvars['host1'].ansible_memtotal_mb + hostvars['host2'].ansible_memtotal_mb }}"

731. How do you use ansible.builtin.include_tasks with loop?

- Example:
- - include_tasks: tasks.yml
- loop:
- - item1
- - item2

732. How do you use `ansible.builtin.include_vars` for environment-specific files?

- Example:
- - include_vars: "{{ env }}.yml"

733. How do you use `ansible.builtin.raw` to run commands on hosts without Python?

- Example:
- - raw: uname -a

734. How do you use `ansible.builtin.debug` with verbosity?

- Example:
- - debug:
- msg: "Detailed info"
- verbosity: 2

735. How do you use `ansible.builtin.expect` to handle interactive commands?

- Example:
- - expect:
- command: passwd user
- responses:
- 'New password:': 'MyPassword'
- 'Retype new password:': 'MyPassword'

736. How do you use `ansible.builtin.uri` for REST API calls?

- Example:
- - uri:
- url: http://api.example.com/status
- method: GET
- return_content: yes

737. How do you use `ansible.builtin.systemd` to reload services without restarting?

- Example:

- - systemd:
- name: nginx
- state: reloaded

738. How do you use ansible.builtin.pip with virtualenv?

- Example:
- - pip:
- name: flask
- virtualenv: /opt/venv
- state: present

739. How do you use ansible.builtin.lineinfile with backrefs?

- Example:
- - lineinfile:
- path: /etc/myapp.conf
- regexp: '(DEBUG=).*'
- line: '\1False'
- backrefs: yes

740. How do you use ansible.builtin.set_fact with conditional assignment?

- Example:
- - set_fact:
- log_dir: "{{ '/var/log' if ansible_os_family == 'RedHat' else '/tmp/logs' }}"

741. How do you use ansible.builtin.file to create symlinks?

- Example:
- - file:
- src: /opt/app/current
- dest: /opt/app/latest
- state: link

742. How do you use ansible.builtin.user to lock accounts?

- Example:
- - user:
- name: testuser
- password_lock: yes

743. How do you use `ansible.builtin.group` to manage multiple users?

- Example:
- - group:
- name: devops
- state: present
- - user:
- name: alice
- groups: devops
- append: yes

744. How do you use `ansible.builtin.file` with `recurse: yes`?

- Removes directories recursively:
- - file:
- path: /tmp/old
- state: absent
- recurse: yes

745. How do you use `ansible.builtin.copy` with `force: no`?

- Prevents overwriting unchanged files.

746. How do you use `ansible.builtin.replace` for multi-line patterns?

- Example:
- - replace:
- path: /etc/myapp.conf
- regexp: 'start=.*\nend=.*'
- replace: 'start=1\nend=2'

747. How do you use `ansible.builtin.meta: noop`?

- Marks a point in the playbook without executing anything.

748. How do you use `ansible.builtin.meta: clear_facts`?

- Clears previously set facts for a host.

749. How do you use `ansible.builtin.assert` to validate a directory exists?

- Example:
- - assert:
- that:

- - ansible_facts.stat.exists

750. How do you use ansible.builtin.wait_for with search_regex in logs?

- Example:
 - - wait_for:
 - path: /var/log/app.log
 - search_regex: "Service started"
 - timeout: 120
-

751. How do you use ansible.builtin.slurp to read binary files?

- Example:
- - slurp:
- src: /path/to/binary
- register: bin_content
- - set_fact:
- decoded_bin: "{{ bin_content.content | b64decode }}"

752. How do you use ansible.builtin.fetch to organize files per host?

- Example:
- - fetch:
- src: /var/log/syslog
- dest: ./logs/{{ inventory_hostname }}/

753. How do you use ansible.builtin.copy with validate command?

- Example:
- - copy:
- src: nginx.conf
- dest: /etc/nginx/nginx.conf
- validate: 'nginx -t -c %s'

754. How do you use ansible.builtin.uri with status_code validation?

- Example:
- - uri:
- url: http://example.com/api
- method: GET

- status_code: 200

755. How do you use ansible.builtin.wait_for with both port and host?

- Example:
- - wait_for:
- host: "{{ inventory_hostname }}"
- port: 443
- timeout: 120

756. How do you use ansible.builtin.set_fact with hostvars for aggregated data?

- Example:
- - set_fact:
- total_memory: "{{ hostvars['host1'].ansible_memtotal_mb + hostvars['host2'].ansible_memtotal_mb }}"

757. How do you use ansible.builtin.include_tasks with loop_control?

- Example:
- - include_tasks: tasks.yml
- loop: "{{ items }}"
- loop_control:
- loop_var: item

758. How do you use ansible.builtin.include_vars with dir to include multiple files?

- Example:
- - include_vars:
- dir: vars/

759. How do you use ansible.builtin.raw to run commands without Python?

- Example:
- - raw: uname -r

760. How do you use ansible.builtin.expect for interactive commands?

- Example:
- - expect:
- command: passwd user
- responses:
- 'New password:': 'MyPassword'

- 'Retype new password:': 'MyPassword'

761. How do you use `ansible.builtin.template` with `force: no`?

- Avoids replacing files if unchanged.

762. How do you use `ansible.builtin.unarchive` with `extra_opts`?

- Example:
- - unarchive:
- src: app.tar.gz
- dest: /opt/app
- extra_opts: [--strip-components=1]

763. How do you use `ansible.builtin.pip` with `executable`?

- Example:
- - pip:
- name: flask
- executable: /opt/venv/bin/pip

764. How do you use `ansible.builtin.systemd` to enable, start, and mask services?

- Example:
- - systemd:
- name: nginx
- state: started
- enabled: yes
- masked: no

765. How do you use `ansible.builtin.lineinfile` with backrefs for regex groups?

- Example:
- - lineinfile:
- path: /etc/myapp.conf
- regexp: '(DEBUG=).*'
- line: '\1False'
- backrefs: yes

766. How do you use `ansible.builtin.replace` with multiline patterns?

- Example:
- - replace:

- path: /etc/myapp.conf
- regexp: 'start=.*\nend=.*'
- replace: 'start=1\nend=2'

767. How do you use `ansible.builtin.assert` to check service status?

- Example:
- - assert:
- that:
- - "'nginx.service' in ansible_facts.services"
- - ansible_facts.services['nginx.service'].state == 'running'

768. How do you use `ansible.builtin.meta: flush_handlers` for immediate handler execution?

- Forces all pending handlers to run at that point in playbook.

769. How do you use `ansible.builtin.meta: end_play` to stop execution?

- Stops the playbook for hosts meeting certain conditions.

770. How do you use `ansible.builtin.pause` with prompt?

- Example:
- - pause:
- prompt: "Press ENTER to continue"

771. How do you use `ansible.builtin.wait_for_connection` to ensure SSH is ready?

- Example:
- - wait_for_connection:
- delay: 10
- timeout: 60

772. How do you use `ansible.builtin.set_fact` with conditional expressions?

- Example:
- - set_fact:
- log_dir: "{{ '/var/log' if ansible_os_family == 'RedHat' else '/tmp/logs' }}"

773. How do you use `ansible.builtin.file` with state: link?

- Example:
- - file:
- src: /opt/app/current
- dest: /opt/app/latest

- state: link

774. How do you use `ansible.builtin.user` to create a user with SSH key?

- Example:
- - user:
- name: deploy
- generate_ssh_key: yes
- state: present

775. How do you use `ansible.builtin.group` to add multiple users?

- Example:
- - group:
- name: devops
- state: present
- - user:
- name: alice
- groups: devops
- append: yes

776. How do you use `ansible.builtin.file` with `recurse: yes` to remove directories?

- Example:
- - file:
- path: /tmp/old
- state: absent
- recurse: yes

777. How do you use `ansible.builtin.copy` with `force: no`?

- Avoid overwriting unchanged files.

778. How do you use `ansible.builtin.get_url` with `force: no`?

- Downloads only if file is missing or changed.

779. How do you use `ansible.builtin.shell` with `warn: no`?

- Suppresses warnings for shell commands.

780. How do you use `ansible.builtin.command` with `creates` for idempotency?

- Avoids re-running commands unnecessarily.

781. How do you use `ansible.builtin.yum` with `enablerepo` option?

- Example:
- - yum:
- name: httpd
- enablerepo: epel
- state: present

782. How do you use `ansible.builtin.apt` with `update_cache`?

- Example:
- - apt:
- name: nginx
- state: present
- update_cache: yes

783. How do you use `ansible.builtin.pip` to install from Git repository?

- Example:
- - pip:
- name: git+https://github.com/pallets/flask.git

784. How do you use `ansible.builtin.systemd` to restart a service safely?

- Example:
- - systemd:
- name: nginx
- state: restarted

785. How do you use `ansible.builtin.lineinfile` with `insertbefore`?

- Example:
- - lineinfile:
- path: /etc/myapp.conf
- line: "NEW_SETTING=yes"
- insertbefore: '^END\$'

786. How do you use `ansible.builtin.replace` with `backup: yes`?

- Example:
- - replace:
- path: /etc/myapp.conf
- regexp: 'DEBUG=.*'

- replace: 'DEBUG=False'
- backup: yes

787. How do you use `ansible.builtin.assert` with multiple variables?

- Ensures environment variables exist before deployment.

788. How do you use `ansible.builtin.include_tasks` with `when`?

- Conditionally include tasks based on host variables.

789. How do you use `ansible.builtin.import_tasks` with static inclusion?

- Example:
- - import_tasks: setup.yml

790. How do you use `ansible.builtin.meta: noop` in a playbook?

- Marks a point in the playbook without action.

791. How do you use `ansible.builtin.meta: clear_facts`?

- Clears all previously set facts.

792. How do you use `ansible.builtin.debug` with verbosity control?

- Example:
- - debug:
- msg: "Detailed info"
- verbosity: 2

793. How do you use `ansible.builtin.raw` to execute commands on hosts without Python?

- Example:
- - raw: uname -r

794. How do you use `ansible.builtin.pause` with seconds?

- Example:
- - pause:
- seconds: 15

795. How do you use `ansible.builtin.wait_for_connection` with delay?

- Ensures host is ready before tasks.

796. How do you use `ansible.builtin.set_fact` to calculate totals dynamically?

- Example:
- - set_fact:
- total_cpu: "{{ ansible_processor_cores * ansible_processor_count }}"

797. How do you use `ansible.builtin.get_url` with checksum?

- Example:
- - `get_url`:
- `url: http://example.com/file.tar.gz`
- `dest: /tmp/file.tar.gz`
- `checksum: sha256:abcd1234`

798. How do you use `ansible.builtin.unarchive` with `remote_src: yes`?

- Example:
- - `unarchive`:
- `src: /tmp/app.tar.gz`
- `dest: /opt/app`
- `remote_src: yes`

799. How do you use `ansible.builtin.systemd` to reload units?

- Example:
- - `systemd`:
- `name: nginx`
- `state: reloaded`

800. How do you use `ansible.builtin.pip` to upgrade a package inside a virtualenv?

- Example:
- - `pip`:
- `name: requests`
- `virtualenv: /opt/venv`
- `state: latest`

801. How do you use `ansible.builtin.lineinfile` with `insertafter`?

- Example:

- `lineinfile`:

path: /etc/myapp.conf

line: "LOG_LEVEL=DEBUG"

insertafter: '^# Settings\$'

- Inserts the line after a matching pattern.

802. How do you use `ansible.builtin.replace` with `multiple: yes`?

- Example:

- `replace:`

`path: /etc/myapp.conf`

`regexp: 'PORT=.*'`

`replace: 'PORT=8080'`

`multiple: yes`

- Replaces all occurrences of the pattern.

803. How do you use `ansible.builtin.assert` to check multiple conditions?

- Example:

- `assert:`

`that:`

`- ansible_os_family == "RedHat"`

`- ansible_processor_cores >= 2`

- Ensures all conditions are true before continuing.

804. How do you use `ansible.builtin.include_tasks` with tags?

- Example:

- `include_tasks: deploy.yml`

`tags: deploy`

- Only runs the included tasks if the specified tag is invoked.

805. How do you use `ansible.builtin.import_tasks` with tags?

- Example:

- `import_tasks: setup.yml`

`tags: setup`

- Static task inclusion, executed when the tag matches.

806. How do you use `ansible.builtin.meta: flush_handlers` in the middle of a play?

- Example:

- `meta: flush_handlers`

- Forces all pending handlers to execute immediately.

807. How do you use `ansible.builtin.meta: end_play` conditionally?

- Example:

- meta: end_play

when: ansible_os_family != "RedHat"

- Stops the playbook for hosts meeting the condition.

808. How do you use ansible.builtin.pause with prompt and echo: yes?

- Example:

- pause:

prompt: "Enter confirmation to continue:"

echo: yes

- Pauses playbook execution and optionally displays user input.

809. How do you use ansible.builtin.wait_for with path and search_regex?

- Example:

- wait_for:

path: /var/log/app.log

search_regex: "Service started"

timeout: 120

- Waits until a file contains a specific regex string.

810. How do you use ansible.builtin.wait_for_connection with retries?

- Example:

- wait_for_connection:

delay: 10

timeout: 60

sleep: 5

- Ensures SSH connection is ready with retries.

811. How do you use ansible.builtin.set_fact with conditional defaults?

- Example:

- set_fact:

env: "{{ env | default('prod') }}"

- Sets variable if it's not defined.

812. How do you use ansible.builtin.file to create directories recursively?

- Example:

- file:

path: /opt/myapp/logs

state: directory

mode: '0755'

recurse: yes

- Creates nested directories with permissions.

813. How do you use `ansible.builtin.file` to remove directories?

- Example:

- file:

path: /tmp/old_logs

state: absent

- Deletes the directory and its contents.

814. How do you use `ansible.builtin.user` to update a user shell?

- Example:

- user:

name: deploy

shell: /bin/bash

- Updates an existing user's shell.

815. How do you use `ansible.builtin.group` to ensure a group exists?

- Example:

- group:

name: devops

state: present

- Creates a group if not already present.

816. How do you use `ansible.builtin.copy` with `remote_src`: yes?

- Example:

- copy:

src: /tmp/config.cfg

dest: /etc/myapp/config.cfg

remote_src: yes

- Copies a file from the remote machine instead of local.

817. How do you use `ansible.builtin.get_url` with `validate_certs`: no?

- Example:

- get_url:

url: https://example.com/file.tar.gz

dest: /tmp/file.tar.gz

validate_certs: no

- Downloads a file ignoring SSL certificate verification.

818. How do you use ansible.builtin.unarchive with extra_opts?

- Example:

- unarchive:

src: /tmp/app.tar.gz

dest: /opt/app

remote_src: yes

extra_opts: [--strip-components=1]

- Extracts archive and strips top-level directory.

819. How do you use ansible.builtin.pip in a specific virtualenv?

- Example:

- pip:

name: requests

virtualenv: /opt/venv

- Installs Python packages inside a virtual environment.

820. How do you use ansible.builtin.systemd to reload a unit?

- Example:

- systemd:

name: nginx

state: reloaded

- Reloads a systemd service configuration without stopping it.

821. How do you use ansible.builtin.shell with creates for idempotency?

- Example:

- shell: tar -xzf app.tar.gz -C /opt/app

args:

creates: /opt/app/app_binary

- Ensures command runs only if the target file doesn't exist.

822. How do you use `ansible.builtin.command` with `removes`?

- Example:

- command: `rm /tmp/tempfile`

args:

`removes: /tmp/tempfile`

- Executes command only if the file exists.

823. How do you use `ansible.builtin.lineinfile` with `backrefs`?

- Example:

- lineinfile:

`path: /etc/myapp.conf`

`regexp: '(DEBUG=).*'`

`line: '\1False'`

`backrefs: yes`

- Updates lines using captured regex groups.

824. How do you use `ansible.builtin.replace` with `backup: yes`?

- Example:

- replace:

`path: /etc/myapp.conf`

`regexp: 'PORT=.*'`

`replace: 'PORT=8080'`

`backup: yes`

- Replaces lines with backup of original file.

825. How do you use `ansible.builtin.assert` to validate variables?

- Example:

- assert:

that:

- `my_var` is defined

- `my_var != ''`

- Checks that required variables are set.

826. How do you use `ansible.builtin.include_tasks` conditionally?

- Example:

- include_tasks: optional_tasks.yml

when: env == 'dev'

- Includes tasks only if condition is met.

827. How do you use ansible.builtin.import_tasks statically?

- Example:

- import_tasks: setup.yml

- Static import, always executed in play.

828. How do you use ansible.builtin.meta: noop?

- Marks a point in playbook with no operation.

829. How do you use ansible.builtin.meta: clear_facts?

- Clears all previously set facts for the host.

830. How do you use ansible.builtin.debug with var?

- Example:

- debug:

var: my_var

- Prints variable content for debugging.

831. How do you use ansible.builtin.raw for hosts without Python?

- Example:

- raw: uname -r

- Executes command without requiring Python module support.

832. How do you use ansible.builtin.pause with seconds?

- Example:

- pause:

seconds: 10

- Pauses playbook for a specified number of seconds.

833. How do you use ansible.builtin.wait_for_connection with delay?

- Example:

- wait_for_connection:

delay: 10

timeout: 60

- Waits until SSH connection is ready.

834. How do you use `ansible.builtin.set_fact` with arithmetic?

- Example:

- `set_fact`:

```
total_cpu: "{{ ansible_processor_cores * ansible_processor_count }}"
```

- Calculates dynamic facts using variables.

835. How do you use `ansible.builtin.get_url` with checksum?

- Example:

- `get_url`:

```
url: http://example.com/file.tar.gz
```

```
dest: /tmp/file.tar.gz
```

```
checksum: sha256:abcd1234
```

- Ensures downloaded file matches checksum.

836. How do you use `ansible.builtin.unarchive` with `remote_src`?

- Example:

- `unarchive`:

```
src: /tmp/app.tar.gz
```

```
dest: /opt/app
```

```
remote_src: yes
```

- Extracts an archive already present on remote host.

837. How do you use `ansible.builtin.systemd` to restart services safely?

- Example:

- `systemd`:

```
name: nginx
```

```
state: restarted
```

- Restarts a systemd service.

838. How do you use `ansible.builtin.pip` with `state: latest`?

- Example:

- `pip`:

```
name: requests
```

```
virtualenv: /opt/venv
```


state: latest

- Ensures package is upgraded to the latest version.

839. How do you use ansible.builtin.lineinfile with insertbefore?

- Example:

- lineinfile:

path: /etc/myapp.conf

line: "NEW_SETTING=yes"

insertbefore: '^END\$'

- Inserts line before a matching pattern.

840. How do you use ansible.builtin.replace with multi-line patterns?

- Example:

- replace:

path: /etc/myapp.conf

regexp: 'START=.*\nEND=.*'

replace: 'START=1\nEND=2'

841. How do you use ansible.builtin.assert to check service states?

- Example:

- assert:

that:

- "'nginx.service' in ansible_facts.services"

- ansible_facts.services['nginx.service'].state == 'running'

842. How do you use ansible.builtin.include_tasks with tags?

- Example:

- include_tasks: deploy.yml

tags: deploy

843. How do you use ansible.builtin.import_tasks with tags?

- Example:

- import_tasks: setup.yml

tags: setup

844. How do you use ansible.builtin.meta: flush_handlers?

- Example:

- meta: flush_handlers

845. How do you use ansible.builtin.meta: end_play?

- Example:

- meta: end_play

when: ansible_os_family != "RedHat"

846. How do you use ansible.builtin.pause with prompt?

- Example:

- pause:

prompt: "Press ENTER to continue"

847. How do you use ansible.builtin.wait_for with host and port?

- Example:

- wait_for:

host: "{{ inventory_hostname }}"

port: 443

timeout: 120

848. How do you use ansible.builtin.wait_for_connection with retries?

- Example:

- wait_for_connection:

delay: 10

timeout: 60

sleep: 5

849. How do you use ansible.builtin.set_fact with default values?

- Example:

- set_fact:

env: "{{ env | default('prod') }}"

850. How do you use ansible.builtin.file to create symlinks?

- Example:

- file:

src: /opt/app/current

dest: /opt/app/latest

state: link

851. How do you use `ansible.builtin.yum` to install multiple packages?

- Example:

- yum:

name:

- git

- nginx

state: present

852. How do you use `ansible.builtin.apt` to update cache before installing?

- Example:

- apt:

name: python3-pip

state: present

update_cache: yes

853. How do you use `ansible.builtin.yum` with `enablerepo`?

- Example:

- yum:

name: epel-release

state: present

enablerepo: epel

854. How do you use `ansible.builtin.apt` with `upgrade: dist`?

- Example:

- apt:

upgrade: dist

- Performs a distribution upgrade on Debian/Ubuntu.

855. How do you use `ansible.builtin.yum` with `disable_gpg_check`?

- Example:

- yum:

name: mypackage

state: present

disable_gpg_check: yes

856. How do you use `ansible.builtin.apt_key` to add a repository key?

- Example:

- `apt_key`:

`url: https://download.example.com/key.gpg`

`state: present`

857. How do you use `ansible.builtin.apt_repository` to add a PPA?

- Example:

- `apt_repository`:

`repo: ppa:deadsnakes/ppa`

`state: present`

858. How do you use `ansible.builtin.yum_repository` to add a repo?

- Example:

- `yum_repository`:

`name: epel`

`description: EPEL repo`

`baseurl: http://download.fedoraproject.org/pub/epel/$releasever/$basearch/`

`gpgcheck: yes`

`state: present`

859. How do you use `ansible.builtin.pip` to install a package globally?

- Example:

- `pip`:

`name: boto3`

`state: present`

860. How do you use `ansible.builtin.pip` to install in a virtual environment?

- Example:

- `pip`:

`name: awscli`

`virtualenv: /opt/venv`

861. How do you use `ansible.builtin.service` to enable and start a service?

- Example:

- `service`:

name: nginx

state: started

enabled: yes

862. How do you use ansible.builtin.systemd with daemon_reload?

- Example:

- systemd:

daemon_reload: yes

863. How do you use ansible.builtin.systemd to stop and disable a service?

- Example:

- systemd:

name: firewalld

state: stopped

enabled: no

864. How do you use ansible.builtin.command with chdir?

- Example:

- command: make install

args:

chdir: /opt/app

865. How do you use ansible.builtin.shell with environment variables?

- Example:

- shell: echo \$MYVAR

environment:

MYVAR: "value"

866. How do you use ansible.builtin.copy with owner and group?

- Example:

- copy:

src: config.cfg

dest: /etc/myapp/config.cfg

owner: root

group: root

mode: '0644'

867. How do you use `ansible.builtin.template` with Jinja2 variables?

- Example:

- template:

src: config.j2

dest: /etc/myapp/config.cfg

868. How do you use `ansible.builtin.unarchive` with `remote_src: yes`?

- Example:

- unarchive:

src: /tmp/app.tar.gz

dest: /opt/app

remote_src: yes

869. How do you use `ansible.builtin.get_url` with `force: yes`?

- Example:

- get_url:

url: https://example.com/app.tar.gz

dest: /tmp/app.tar.gz

force: yes

870. How do you use `ansible.builtin.uri` to check a URL status?

- Example:

- uri:

url: https://example.com/health

return_content: yes

status_code: 200

871. How do you use `ansible.builtin.debug` to show a message?

- Example:

- debug:

msg: "Deployment started on {{ inventory_hostname }}"

872. How do you use `ansible.builtin.debug` with variables?

- Example:

- debug:

var: ansible_facts['os_family']

873. How do you use `ansible.builtin.assert` to validate AWS variables?

- Example:

- `assert`:

that:

- `aws_region` is defined
- `aws_access_key` is defined

874. How do you use `ansible.builtin.wait_for` with TCP port?

- Example:

- `wait_for`:

host: "{{ inventory_hostname }}"

port: 22

timeout: 60

875. How do you use `ansible.builtin.wait_for_connection` with timeout?

- Example:

- `wait_for_connection`:

timeout: 120

876. How do you use `ansible.builtin.set_fact` to define dynamic variables?

- Example:

- `set_fact`:

app_path: "/opt/{{ app_name }}"

877. How do you use `ansible.builtin.stat` to check if a file exists?

- Example:

- `stat`:

path: /etc/myapp/config.cfg

register: config_file

878. How do you use `ansible.builtin.fail` to stop playbook execution?

- Example:

- `fail`:

msg: "Required file is missing!"

when: not config_file.stat.exists

879. How do you use `ansible.builtin.include_role` with `tasks_from`?

- Example:

- include_role:

name: webserver

tasks_from: install.yml

880. How do you use ansible.builtin.import_role statically?

- Example:

- import_role:

name: database

881. How do you use ansible.builtin.meta: flush_handlers to execute handlers immediately?

- Example:

- meta: flush_handlers

882. How do you use ansible.builtin.meta: end_play conditionally?

- Example:

- meta: end_play

when: env != "prod"

883. How do you use ansible.builtin.pause for human interaction?

- Example:

- pause:

prompt: "Press ENTER to continue deployment"

884. How do you use ansible.builtin.uri to download a file via HTTP?

- Example:

- uri:

url: https://example.com/app.tar.gz

dest: /tmp/app.tar.gz

method: GET

885. How do you use ansible.builtin.pip with extra arguments?

- Example:

- pip:

name: boto3

extra_args: "--upgrade"

886. How do you use ansible.builtin.user to create a system user?

- Example:

- user:

name: deploy

shell: /bin/bash

system: yes

887. How do you use `ansible.builtin.group` to remove a group?

- Example:

- group:

name: oldgroup

state: absent

888. How do you use `ansible.builtin.file` to set ownership and permissions?

- Example:

- file:

path: /opt/myapp

owner: deploy

group: devops

mode: '0755'

889. How do you use `ansible.builtin.copy` with mode and backup?

- Example:

- copy:

src: config.cfg

dest: /etc/myapp/config.cfg

mode: '0644'

backup: yes

890. How do you use `ansible.builtin.template` with Jinja2 loops?

- Example:

- template:

src: nginx.conf.j2

dest: /etc/nginx/nginx.conf

- Template can include Jinja2 loops for dynamic content.

891. How do you use `ansible.builtin.unarchive` to extract .zip files?

- Example:

- unarchive:

src: /tmp/app.zip

dest: /opt/app

remote_src: yes

892. How do you use ansible.builtin.get_url with AWS S3 signed URL?

- Example:

- get_url:

url: "{{ s3_signed_url }}"

dest: /tmp/app.tar.gz

893. How do you use ansible.builtin.systemd with enabled and masked?

- Example:

- systemd:

name: firewalld

state: stopped

masked: yes

894. How do you use ansible.builtin.shell with creates for idempotency?

- Example:

- shell: ./install.sh

args:

creates: /opt/app/app_binary

895. How do you use ansible.builtin.command safely for scripts?

- Example:

- command: /usr/local/bin/deploy.sh

896. How do you use ansible.builtin.debug with verbosity?

- Example:

- debug:

msg: "Debug info"

verbosity: 2

897. How do you use ansible.builtin.stat to get file checksum?

- Example:

- stat:

path: /etc/myapp/config.cfg

checksum_algorithm: sha256

register: file_stat

898. How do you use ansible.builtin.fail with dynamic message?

- Example:

- fail:

msg: "File {{ file_stat.stat.path }} is missing!"

when: not file_stat.stat.exists

899. How do you use ansible.builtin.include_role with defaults?

- Example:

- include_role:

name: webserver

defaults_from: default_vars.yml

900. How do you use ansible.builtin.import_role with tasks_from?

- Example:

- import_role:

name: database

tasks_from: install.yml

901. How do you use ansible.builtin.yum to remove multiple packages?

- Example:

- yum:

name:

- httpd

- nginx

state: absent

902. How do you use ansible.builtin.apt to remove a package?

- Example:

- apt:

name: apache2

state: absent

903. How do you use ansible.builtin.yum to install a package with a specific version?

- Example:

- yum:

name: nginx-1.20.1

state: present

904. How do you use ansible.builtin.apt to install a specific version?

- Example:

- apt:

name: python3=3.11.4-1

state: present

905. How do you use ansible.builtin.yum with exclude?

- Example:

- yum:

name: '*'

state: latest

exclude: kernel*

906. How do you use ansible.builtin.apt with autoremove?

- Example:

- apt:

autoremove: yes

907. How do you use ansible.builtin.pip to uninstall a package?

- Example:

- pip:

name: boto3

state: absent

908. How do you use ansible.builtin.service to restart multiple services?

- Example:

- service:

name:

- nginx

- redis

state: restarted

909. How do you use ansible.builtin.systemd with scope?

- Example:

- systemd:

name: myapp

scope: system

910. How do you use ansible.builtin.command with warn: no?

- Example:

- command: rpm -qa

args:

warn: no

911. How do you use ansible.builtin.shell with executable?

- Example:

- shell: echo \$HOME

args:

executable: /bin/bash

912. How do you use ansible.builtin.copy to set SELinux context?

- Example:

- copy:

src: myapp.conf

dest: /etc/myapp/myapp.conf

seuser: system_u

serole: object_r

setype: etc_t

913. How do you use ansible.builtin.template with conditionals in Jinja2?

- Example:

- template:

src: config.j2

dest: /etc/myapp/config.cfg

- Template can include {% if env == "prod" %} statements.

914. How do you use ansible.builtin.unarchive with creates?

- Example:

- unarchive:

```
src: /tmp/app.tar.gz
dest: /opt/app
remote_src: yes
creates: /opt/app/app_binary
```

915. How do you use ansible.builtin.get_url with timeout?

- Example:

- get_url:

```
url: https://example.com/app.tar.gz
dest: /tmp/app.tar.gz
timeout: 60
```

916. How do you use ansible.builtin.uri to POST JSON data?

- Example:

- uri:

```
url: https://api.example.com/deploy
method: POST
body: '{"app": "myapp"}'
body_format: json
```

917. How do you use ansible.builtin.debug with msg and condition?

- Example:

- debug:

```
msg: "Deploying app on production"
when: env == "prod"
```

918. How do you use ansible.builtin.assert to validate AWS facts?

- Example:

- assert:

```
that:
  - ansible_facts['ec2_ami_id'] is defined
```

919. How do you use ansible.builtin.wait_for with path and state?

- Example:

- wait_for:

path: /var/log/app.log

state: present

timeout: 60

920. How do you use ansible.builtin.wait_for_connection with delay and sleep?

- Example:

- wait_for_connection:

delay: 10

sleep: 5

timeout: 120

921. How do you use ansible.builtin.set_fact with hostvars?

- Example:

- set_fact:

db_host: "{{ hostvars['db-server'].ansible_host }}"

922. How do you use ansible.builtin.stat to get file permissions?

- Example:

- stat:

path: /etc/myapp/config.cfg

register: config_stat

- config_stat.stat.mode shows permissions.

923. How do you use ansible.builtin.fail with a condition?

- Example:

- fail:

msg: "Config file missing"

when: not config_stat.stat.exists

924. How do you use ansible.builtin.include_role with tasks_from?

- Example:

- include_role:

name: webserver

tasks_from: install.yml

925. How do you use `ansible.builtin.import_role` with `defaults_from`?

- Example:

- `import_role`:

name: database

defaults_from: default_vars.yml

926. How do you use `ansible.builtin.meta: flush_handlers` mid-play?

- Example:

- `meta: flush_handlers`

927. How do you use `ansible.builtin.meta: end_play` based on a variable?

- Example:

- `meta: end_play`

when: aws_region != "us-east-1"

928. How do you use `ansible.builtin.pause` with a custom message?

- Example:

- `pause`:

prompt: "Confirm deployment to production"

929. How do you use `ansible.builtin.uri` to check SSL certificate?

- Example:

- `uri`:

url: https://example.com

validate_certs: yes

930. How do you use `ansible.builtin.pip` with requirements file?

- Example:

- `pip`:

requirements: /opt/requirements.txt

931. How do you use `ansible.builtin.user` with groups?

- Example:

- `user`:

name: deploy

groups: devops

932. How do you use `ansible.builtin.group` with state present?

- Example:

- group:

name: devops

state: present

933. How do you use `ansible.builtin.file` to create directories with mode?

- Example:

- file:

path: /opt/myapp/logs

state: directory

mode: '0755'

934. How do you use `ansible.builtin.copy` with `remote_src` and `owner`?

- Example:

- copy:

src: /tmp/config.cfg

dest: /etc/myapp/config.cfg

remote_src: yes

owner: root

935. How do you use `ansible.builtin.template` with `hostvars`?

- Example:

- template:

src: nginx.conf.j2

dest: /etc/nginx/nginx.conf

936. How do you use `ansible.builtin.unarchive` with mode?

- Example:

- unarchive:

src: /tmp/app.tar.gz

dest: /opt/app

mode: '0755'

937. How do you use `ansible.builtin.get_url` with `force` and `checksum`?

- Example:

- get_url:

url: https://example.com/app.tar.gz

dest: /tmp/app.tar.gz

force: yes

checksum: sha256:abcd1234

938. How do you use ansible.builtin.systemd to reload a service?

- Example:

- systemd:

name: nginx

state: reloaded

939. How do you use ansible.builtin.shell with creates for idempotency?

- Example:

- shell: tar -xzf app.tar.gz -C /opt/app

args:

creates: /opt/app/app_binary

940. How do you use ansible.builtin.command with removes?

- Example:

- command: rm /tmp/tempfile

args:

removes: /tmp/tempfile

941. How do you use ansible.builtin.lineinfile with regex and backrefs?

- Example:

- lineinfile:

path: /etc/myapp.conf

regexp: '(DEBUG=).*'

line: '\1False'

backrefs: yes

942. How do you use ansible.builtin.replace with backup?

- Example:

- replace:

path: /etc/myapp.conf

regexp: 'PORT=.*'

replace: 'PORT=8080'

backup: yes

943. How do you use `ansible.builtin.assert` to check a list variable?

- Example:

- assert:

that:

- my_list | length > 0

944. How do you use `ansible.builtin.include_tasks` with a loop?

- Example:

- include_tasks: deploy.yml

loop:

- app1

- app2

loop_control:

loop_var: app_name

945. How do you use `ansible.builtin.import_tasks` with a loop?

- Example:

- import_tasks: setup.yml

loop:

- db

- cache

946. How do you use `ansible.builtin.meta: noop`?

- Example:

- meta: noop

947. How do you use `ansible.builtin.meta: clear_facts`?

- Example:

- meta: clear_facts

948. How do you use `ansible.builtin.debug` with verbosity?

- Example:

- debug:

msg: "Verbose output"

verbosity: 2

949. How do you use `ansible.builtin.raw` to run a command on hosts without Python?

- Example:

- raw: uname -r

950. How do you use `ansible.builtin.pause` to wait for a number of seconds?

- Example:

- pause:

seconds: 10

951. How do you use `ansible.builtin.yum` to check if a package is installed?

- Example:

- yum:

name: nginx

state: present

register: nginx_status

952. How do you use `ansible.builtin.apt` to check if a package is installed?

- Example:

- apt:

name: apache2

state: present

register: apache_status

953. How do you use `ansible.builtin.yum` to upgrade a specific package?

- Example:

- yum:

name: nginx

state: latest

954. How do you use `ansible.builtin.apt` to upgrade a specific package?

- Example:

- apt:

name: python3

state: latest

955. How do you use `ansible.builtin.pip` to upgrade a package?

- Example:

- pip:

name: boto3

state: latest

956. How do you use `ansible.builtin.service` with `enabled: no` and `state: stopped`?

- Example:

- service:

name: firewalld

state: stopped

enabled: no

957. How do you use `ansible.builtin.systemd` to mask a service?

- Example:

- systemd:

name: firewalld

masked: yes

958. How do you use `ansible.builtin.command` with `creates` to ensure idempotency?

- Example:

- command: ./install.sh

args:

creates: /opt/app/app_binary

959. How do you use `ansible.builtin.shell` with `warn: no`?

- Example:

- shell: echo \$HOME

args:

warn: no

960. How do you use `ansible.builtin.copy` with `backup: yes`?

- Example:

- copy:

src: config.cfg

dest: /etc/myapp/config.cfg

backup: yes

961. How do you use ansible.builtin.template with force: yes?

- Example:

- template:

src: config.j2

dest: /etc/myapp/config.cfg

force: yes

962. How do you use ansible.builtin.unarchive with remote_src: yes?

- Example:

- unarchive:

src: /tmp/app.tar.gz

dest: /opt/app

remote_src: yes

963. How do you use ansible.builtin.get_url with mode?

- Example:

- get_url:

url: https://example.com/app.tar.gz

dest: /tmp/app.tar.gz

mode: '0644'

964. How do you use ansible.builtin.uri to send headers?

- Example:

- uri:

url: https://api.example.com

method: GET

headers:

Authorization: "Bearer {{ api_token }}"

965. How do you use ansible.builtin.debug with var?

- Example:

- debug:

var: ansible_facts['distribution']

966. How do you use ansible.builtin.assert to check variable type?

- Example:

- assert:

that:

- my_var is string

967. How do you use `ansible.builtin.wait_for` with path and state: absent?

- Example:

- wait_for:

path: /tmp/tempfile

state: absent

968. How do you use `ansible.builtin.wait_for_connection` with retries?

- Example:

- wait_for_connection:

delay: 5

timeout: 60

sleep: 10

969. How do you use `ansible.builtin.set_fact` to calculate a value?

- Example:

- set_fact:

total_memory_gb: "{{ ansible_facts['memtotal_mb'] / 1024 }}"

970. How do you use `ansible.builtin.stat` to check if directory exists?

- Example:

- stat:

path: /opt/myapp

register: dir_stat

971. How do you use `ansible.builtin.fail` to stop play if a condition is met?

- Example:

- fail:

msg: "Deployment aborted!"

when: env != "prod"

972. How do you use `ansible.builtin.include_role` with `tasks_from` for modular tasks?

- Example:

- include_role:
 name: webserver
 tasks_from: setup.yml

973. How do you use ansible.builtin.import_role for static inclusion?

- Example:

- import_role:
 name: database

974. How do you use ansible.builtin.meta: flush_handlers to run handlers immediately?

- Example:

- meta: flush_handlers

975. How do you use ansible.builtin.meta: end_play with a variable?

- Example:

- meta: end_play
 when: skip_deploy | default(false)

976. How do you use ansible.builtin.pause with timeout?

- Example:

- pause:
 seconds: 15

977. How do you use ansible.builtin.uri to check URL with status code 200?

- Example:

- uri:
 url: https://example.com
 status_code: 200

978. How do you use ansible.builtin.pip with virtualenv and requirements?

- Example:

- pip:
 requirements: /opt/requirements.txt
 virtualenv: /opt/venv

979. How do you use ansible.builtin.user to set UID and shell?

- Example:

- user:

name: deploy
uid: 1500
shell: /bin/bash

980. How do you use `ansible.builtin.group` with `system: yes`?

- Example:

- group:

name: devops
system: yes

981. How do you use `ansible.builtin.file` to create a symlink?

- Example:

- file:

src: /opt/app/current
dest: /opt/app/latest
state: link

982. How do you use `ansible.builtin.copy` with `force: yes`?

- Example:

- copy:

src: config.cfg
dest: /etc/myapp/config.cfg
force: yes

983. How do you use `ansible.builtin.template` with Jinja2 filters?

- Example:

- template:

src: config.j2
dest: /etc/myapp/config.cfg

- Filters like `{{ var | default('value') }}` can be used.

984. How do you use `ansible.builtin.unarchive` with `remote_src` and `creates`?

- Example:

- unarchive:

src: /tmp/app.tar.gz
dest: /opt/app

remote_src: yes

creates: /opt/app/app_binary

985. How do you use ansible.builtin.get_url with validate_certs: no?

- Example:

- get_url:

url: https://example.com/app.tar.gz

dest: /tmp/app.tar.gz

validate_certs: no

986. How do you use ansible.builtin.systemd with reloaded and enabled?

- Example:

- systemd:

name: nginx

state: reloaded

enabled: yes

987. How do you use ansible.builtin.shell with stdin?

- Example:

- shell: mysql -u root -p

args:

stdin: "{{ mysql_password }}"

988. How do you use ansible.builtin.command with chdir?

- Example:

- command: make install

args:

chdir: /opt/app

989. How do you use ansible.builtin.lineinfile with insertbefore?

- Example:

- lineinfile:

path: /etc/myapp/config.cfg

line: "DEBUG=False"

insertbefore: '^#'

990. How do you use ansible.builtin.replace with multiple occurrences?

- Example:

- replace:

path: /etc/myapp/config.cfg

regexp: 'PORT=.*'

replace: 'PORT=8080'

replace_all: yes

991. How do you use ansible.builtin.assert with multiple conditions?

- Example:

- assert:

that:

- my_var is defined

- my_var != ""

992. How do you use ansible.builtin.include_tasks with tags?

- Example:

- include_tasks: deploy.yml

tags: deploy

993. How do you use ansible.builtin.import_tasks with tags?

- Example:

- import_tasks: setup.yml

tags: setup

994. How do you use ansible.builtin.meta: noop for conditional execution?

- Example:

- meta: noop

995. How do you use ansible.builtin.meta: clear_facts to remove variables?

- Example:

- meta: clear_facts

996. How do you use ansible.builtin.debug with conditional verbosity?

- Example:

- debug:

msg: "Debug info"

verbosity: 2

997. How do you use `ansible.builtin.raw` to run a command on hosts without Python?

- Example:

- raw: uname -a

998. How do you use `ansible.builtin.pause` to wait for user confirmation?

- Example:

- pause:

prompt: "Press ENTER to continue"

999. How do you use `ansible.builtin.yum` to update all packages except kernel?

- Example:

- yum:

name: '*'

state: latest

exclude: kernel*

1000. How do you use `ansible.builtin.apt` to upgrade all packages on Debian/Ubuntu?

- Example:

- apt:

upgrade: dist

update_cache: yes

Complex Ansible Q&A 1001–1500 for AWS DevOps:

1001. How do you use `ansible.builtin.block` with `rescue` and `always` for error handling?

- Example:

- block:

- command: /opt/deploy.sh

rescue:

- debug: msg="Deployment failed, rolling back"

always:

- debug: msg="Cleaning temporary files"

1002. How do you use `ansible.builtin.include_role` conditionally based on a variable?

- Example:

- include_role:

name: database

when: db_required

1003. How do you use ansible.builtin.import_role with vars?

- Example:

- import_role:

name: webserver

vars:

port: 8080

1004. How do you use ansible.builtin.template with nested dictionaries in Jinja2?

- Example:

- template:

src: nginx.conf.j2

dest: /etc/nginx/nginx.conf

- Jinja2 template can access {{ servers.web[0].ip }}.

1005. How do you use ansible.builtin.lineinfile with multiple: yes?

- Example:

- lineinfile:

path: /etc/myapp.conf

regexp: '^PORT='

line: 'PORT=8080'

multiple: yes

1006. How do you use ansible.builtin.replace with regex groups?

- Example:

- replace:

path: /etc/myapp.conf

regexp: '(PORT=).*'

replace: '\18080'

backup: yes

1007. How do you use ansible.builtin.shell with async and poll for background execution?

- Example:

- shell: long_running_task.sh

async: 600

poll: 0

1008. How do you use ansible.builtin.async_status to check background job status?

- Example:

- async_status:

jid: "{{ async_job.ansible_job_id }}"

register: job_result

until: job_result.finished

retries: 10

delay: 15

1009. How do you use ansible.builtin.stat to check symbolic links?

- Example:

- stat:

path: /opt/app/latest

follow: no

register: symlink_stat

1010. How do you use ansible.builtin.set_fact with complex calculations?

- Example:

- set_fact:

free_memory_gb: "{{ ansible_facts['memfree_mb'] / 1024 | round(2) }}"

1011. How do you use ansible.builtin.assert to validate a list of dictionaries?

- Example:

- assert:

that:

- apps | selectattr('enabled', 'equalto', true) | list | length > 0

1012. How do you use ansible.builtin.wait_for with nested conditions?

- Example:

- wait_for:

path: /var/run/app.sock

state: started

timeout: 120

1013. How do you use ansible.builtin.get_url with retries and delay?

- Example:

- get_url:

url: https://example.com/app.tar.gz

dest: /tmp/app.tar.gz

retries: 3

delay: 10

1014. How do you use ansible.builtin.pip with environment variables and virtualenv?

- Example:

- pip:

name: awscli

virtualenv: /opt/venv

extra_args: "--upgrade"

environment:

AWS_REGION: us-east-1

1015. How do you use ansible.builtin.user with append to add groups?

- Example:

- user:

name: deploy

groups: devops

append: yes

1016. How do you use ansible.builtin.systemd to start multiple services using a loop?

- Example:

- systemd:

name: "{{ item }}"

state: started

loop:

- nginx

- redis

1017. How do you use `ansible.builtin.copy` with content dynamically generated?

- Example:

- copy:

dest: /etc/myapp/config.cfg

content: |

HOST={{ host_ip }}

PORT={{ port }}

1018. How do you use `ansible.builtin.unarchive` with checksum verification?

- Example:

- unarchive:

src: /tmp/app.tar.gz

dest: /opt/app

checksum: sha256:abcd1234

1019. How do you use `ansible.builtin.uri` to send a POST request with JSON body and headers?

- Example:

- uri:

url: https://api.example.com/deploy

method: POST

body: '{"app": "myapp"}'

body_format: json

headers:

Authorization: "Bearer {{ api_token }}"

1020. How do you use `ansible.builtin.debug` with complex dictionaries?

- Example:

- debug:

var: apps

1021. How do you use `ansible.builtin.include_tasks` with loop and dynamic file names?

- Example:

- include_tasks: "{{ item }}.yaml"

loop:

- setup

- deploy

1022. How do you use `ansible.builtin.import_tasks` with loop?

- Example:

- import_tasks: tasks/{{ item }}.yaml

loop:

- db
- cache

1023. How do you use `ansible.builtin.meta: clear_facts` to remove variables across hosts?

- Example:

- meta: clear_facts

1024. How do you use `ansible.builtin.pause` with dynamic timeout?

- Example:

- pause:

seconds: "{{ deploy_timeout }}"

1025. How do you use `ansible.builtin.raw` to execute multiple commands sequentially?

- Example:

- raw: |

cd /opt/app

./install.sh

1026. How do you use `ansible.builtin.lineinfile` to manage multiple lines with regex and backrefs?

- Example:

- lineinfile:

path: /etc/myapp.conf

regexp: '(DEBUG=).*'

line: '\1False'

backrefs: yes

1027. How do you use `ansible.builtin.replace` for multiline patterns?

- Example:

- replace:

path: /etc/myapp.conf

regexp: 'START=.*\nEND=.*'

replace: 'START=1\nEND=2'

1028. How do you use ansible.builtin.yum with exclude and disablerepo?

- Example:

- yum:

name: '*'

state: latest

exclude: kernel*

disablerepo: epel

1029. How do you use ansible.builtin.apt with force_apt_get for legacy systems?

- Example:

- apt:

name: python3

state: latest

force_apt_get: yes

1030. How do you use ansible.builtin.service with pattern to manage multiple services?

- Example:

- service:

name: "nginx*"

state: restarted

1031. How do you use ansible.builtin.set_fact with conditional logic?

- Example:

- set_fact:

deploy_path: "/opt/app-prod"

when: env == "prod"

1032. How do you use ansible.builtin.stat with checksum_algorithm?

- Example:

- stat:

path: /etc/myapp/config.cfg

checksum_algorithm: sha256

1033. How do you use ansible.builtin.fail with dynamic message formatting?

- Example:

- fail:

msg: "File {{ file_path }} missing!"

when: not file_stat.stat.exists

1034. How do you use ansible.builtin.include_role with multiple variables?

- Example:

- include_role:

name: webserver

vars:

port: 8080

ssl: yes

1035. How do you use ansible.builtin.import_role to include tasks and vars from a role?

- Example:

- import_role:

name: database

vars:

db_name: mydb

1036. How do you use ansible.builtin.meta: flush_handlers in the middle of play?

- Example:

- meta: flush_handlers

1037. How do you use ansible.builtin.meta: end_play with complex condition?

- Example:

- meta: end_play

when: env not in ["prod", "staging"]

1038. How do you use ansible.builtin.pause to wait for AWS resources creation?

- Example:

- pause:

seconds: "{{ rds_creation_time }}"

1039. How do you use ansible.builtin.uri with retries and status codes?

- Example:

- uri:

url: https://api.example.com/status

status_code: 200

retries: 3

delay: 10

1040. How do you use ansible.builtin.pip with multiple packages and options?

- Example:

- pip:

name:

- awscli

- boto3

virtualenv: /opt/venv

extra_args: "--upgrade"

****1041. How do you use `ansible.builtin.user` to manage multiple users in a loop?****

- Example:

``yaml

- user:

name: "{{ item.name }}"

state: present

loop:

- { name: "deploy" }

- { name: "admin" }

1042. How do you use ansible.builtin.group with multiple group options?

- Example:

- group:

name: devops

gid: 500

system: yes

1043. How do you use ansible.builtin.file to manage directory, owner, group, and mode?

- Example:

- file:

path: /opt/app/logs

state: directory

owner: deploy

group: devops

mode: '0755'

1044. How do you use ansible.builtin.copy with content, mode, and owner dynamically?

- Example:

- copy:

dest: /etc/myapp/config.cfg

content: |

HOST={{ host_ip }}

PORT={{ port }}

owner: deploy

mode: '0644'

1045. How do you use ansible.builtin.template with default filter for undefined variables?

- Example:

- template:

src: config.j2

dest: /etc/myapp/config.cfg

- In template: {{ var | default('default_value') }}

1046. How do you use ansible.builtin.unarchive with remote_src, creates, and mode?

- Example:

- unarchive:

src: /tmp/app.tar.gz

dest: /opt/app

remote_src: yes

creates: /opt/app/app_binary

mode: '0755'

1047. How do you use ansible.builtin.get_url with checksum, retries, and headers?

- Example:

- get_url:

url: https://example.com/app.tar.gz

dest: /tmp/app.tar.gz
checksum: sha256:abcd1234

retries: 3

headers:

Authorization: "Bearer {{ api_token }}"

1048. How do you use `ansible.builtin.systemd` to reload multiple services with loop?

- Example:

- systemd:

name: "{{ item }}"

state: reloaded

loop:

- nginx
- redis

1049. How do you use `ansible.builtin.shell` with `async`, `poll`, `chdir`, and `environment`?

- Example:

- shell: ./deploy.sh

async: 600

poll: 0

chdir: /opt/app

environment:

ENV: prod

1050. How do you use `ansible.builtin.command` with `creates` and `removes` for idempotency?

- Example:

- command: ./setup.sh

args:

creates: /opt/app/setup_done

removes: /tmp/setup_tmp

1051. How do you use `ansible.builtin.block` with multiple tasks and conditional rescue?

- Example:

- block:

- command: /opt/deploy.sh
- command: /opt/configure.sh

rescue:

- debug: msg="Deployment failed, manual intervention required"

always:

- debug: msg="Cleanup temp files"

1052. How do you use `ansible.builtin.include_role` with multiple conditional variables?

- Example:

- include_role:

name: database

when: db_required and env == "prod"

1053. How do you use `ansible.builtin.import_role` with `tasks_from` dynamically?

- Example:

- import_role:

name: webserver

tasks_from: "{{ task_file }}"

1054. How do you use `ansible.builtin.template` with complex Jinja2 loops and conditions?

- Example:

- template:

src: nginx.conf.j2

dest: /etc/nginx/nginx.conf

- Template example:

{% for server in servers %}

server {{ server.ip }}:{{ server.port }};

{% endfor %}

1055. How do you use `ansible.builtin.lineinfile` to insert before and after patterns?

- Example:

- lineinfile:

path: /etc/myapp/config.cfg

line: 'PORT=8080'

insertbefore: '^# Start'

insertafter: '^# End'

1056. How do you use `ansible.builtin.replace` to update multiple regex patterns in one file?

- Example:

- replace:

path: /etc/myapp/config.cfg

regexp: '(PORT|DEBUG)=.*'

replace: '\1=8080'

1057. How do you use `ansible.builtin.shell` with `async`, `poll`, and `register` results?

- Example:

- shell: long_task.sh

async: 600

poll: 0

register: job

1058. How do you use `ansible.builtin.async_status` to monitor a background job?

- Example:

- async_status:

jid: "{{ job.ansible_job_id }}"

register: job_result

until: job_result.finished

retries: 10

delay: 15

1059. How do you use `ansible.builtin.stat` with multiple attributes like `isdir` and `exists`?

- Example:

- stat:

path: /opt/app

register: app_dir

1060. How do you use `ansible.builtin.set_fact` with nested dictionaries?

- Example:

- set_fact:

app_config:

host: "{{ ansible_facts['default_ipv4']['address'] }}"

port: 8080

1061. How do you use ansible.builtin.assert to validate multiple conditions?

- Example:

- assert:

that:

- var1 is defined
- var2 != ""
- var3 | length > 0

1062. How do you use ansible.builtin.wait_for with nested conditions and timeout?

- Example:

- wait_for:

path: /var/run/app.sock

state: started

timeout: 120

1063. How do you use ansible.builtin.get_url with retries, delay, and checksum verification?

- Example:

- get_url:

url: https://example.com/app.tar.gz

dest: /tmp/app.tar.gz

retries: 3

delay: 10

checksum: sha256:abcd1234

1064. How do you use ansible.builtin.pip to install packages with virtualenv and extra args?

- Example:

- pip:

name: boto3

virtualenv: /opt/venv

extra_args: "--upgrade"

1065. How do you use `ansible.builtin.user` with multiple group memberships and append?

- Example:

- user:

name: deploy

groups: devops,admin

append: yes

1066. How do you use `ansible.builtin.group` with GID and system flag?

- Example:

- group:

name: devops

gid: 500

system: yes

1067. How do you use `ansible.builtin.file` to manage directories with owner, group, and mode?

- Example:

- file:

path: /opt/app/logs

state: directory

owner: deploy

group: devops

mode: '0755'

1068. How do you use `ansible.builtin.copy` to create a file from dynamic content and set permissions?

- Example:

- copy:

dest: /etc/myapp/config.cfg

content: |

HOST={{ host_ip }}

PORT={{ port }}

owner: deploy

mode: '0644'

1069. How do you use `ansible.builtin.template` with default filter and conditions?

- Example:

- template:

src: config.j2

dest: /etc/myapp/config.cfg

- Template: {{ var | default('default_value') }}

1070. How do you use ansible.builtin.unarchive with remote_src, creates, and mode?

- Example:

- unarchive:

src: /tmp/app.tar.gz

dest: /opt/app

remote_src: yes

creates: /opt/app/app_binary

mode: '0755'

1071. How do you use ansible.builtin.get_url with headers, retries, and validation?

- Example:

- get_url:

url: https://api.example.com/app.tar.gz

dest: /tmp/app.tar.gz

retries: 3

headers:

Authorization: "Bearer {{ api_token }}"

validate_certs: yes

1072. How do you use ansible.builtin.systemd with loop to reload multiple services?

- Example:

- systemd:

name: "{{ item }}"

state: reloaded

loop:

- nginx

- redis

1073. How do you use ansible.builtin.shell with async, poll, chdir, and environment variables?

- Example:

- shell: ./deploy.sh

async: 600

poll: 0

chdir: /opt/app

environment:

ENV: prod

1074. How do you use `ansible.builtin.command` with `creates` and `removes` for idempotency?

- Example:

- command: ./setup.sh

args:

creates: /opt/app/setup_done

removes: /tmp/setup_tmp

1075. How do you use `ansible.builtin.meta: flush_handlers` mid-play for immediate handler execution?

- Example:

- meta: flush_handlers

1076. How do you use `ansible.builtin.meta: end_play` with complex conditions?

- Example:

- meta: end_play

when: env not in ['prod', 'staging']

1077. How do you use `ansible.builtin.pause` to wait for user input dynamically?

- Example:

- pause:

prompt: "Press ENTER to continue deployment"

1078. How do you use `ansible.builtin.uri` with retries and status code validation?

- Example:

- uri:

url: https://api.example.com/status

status_code: 200

retries: 5

delay: 10

1079. How do you use ansible.builtin.pip with multiple packages in virtualenv?

- Example:

- pip:

name:

- awscli

- boto3

virtualenv: /opt/venv

1080. How do you use ansible.builtin.user in a loop with dynamic usernames and groups?

- Example:

- user:

name: "{{ item.name }}"

groups: "{{ item.groups }}"

append: yes

loop:

- { name: "deploy", groups: "devops" }

- { name: "admin", groups: "wheel" }

1081. How do you use ansible.builtin.group to manage multiple system groups?

- Example:

- group:

name: "{{ item }}"

system: yes

loop:

- devops

- admin

1082. How do you use ansible.builtin.file to manage symlinks with force?

- Example:

- file:

src: /opt/app/current

dest: /opt/app/latest

state: link

force: yes

1083. How do you use ansible.builtin.copy with backup and owner permissions?

- Example:

- copy:

src: config.cfg

dest: /etc/myapp/config.cfg

backup: yes

owner: deploy

1084. How do you use ansible.builtin.template with loops and conditional blocks?

- Example:

- template:

src: config.j2

dest: /etc/myapp/config.cfg

- Template example:

{% for server in servers %}

{% if server.active %}

server {{ server.ip }};

{% endif %}

{% endfor %}

1085. How do you use ansible.builtin.unarchive to extract files with mode and owner?

- Example:

- unarchive:

src: /tmp/app.tar.gz

dest: /opt/app

remote_src: yes

owner: deploy

mode: '0755'

1086. How do you use ansible.builtin.get_url to download multiple files with loop?

- Example:

- get_url:

url: "{{ item.url }}"

dest: "{{ item.dest }}"

loop:

- { url: "https://example.com/file1", dest: "/tmp/file1" }

- { url: "https://example.com/file2", dest: "/tmp/file2" }

1087. How do you use ansible.builtin.systemd to restart services with conditionals?

- Example:

- systemd:

name: nginx

state: restarted

when: nginx_installed

1088. How do you use ansible.builtin.shell to execute multiple commands with environment and chdir?

- Example:

- shell: |

./deploy.sh

./migrate.sh

environment:

ENV: prod

chdir: /opt/app

1089. How do you use ansible.builtin.command to run scripts only if a file does not exist?

- Example:

- command: ./install.sh

args:

creates: /opt/app/installed

1090. How do you use ansible.builtin.meta: noop to skip tasks dynamically?

- Example:

- meta: noop

when: skip_tasks | default(false)

1091. How do you use ansible.builtin.assert to enforce environment variable rules?

- Example:

- assert:

that:

- env in ['dev', 'staging', 'prod']

1092. How do you use `ansible.builtin.debug` to show nested dictionary values?

- Example:

- debug:

```
var: apps[0].config.port
```

1093. How do you use `ansible.builtin.include_tasks` with conditional looping?

- Example:

- include_tasks: "{{ item }}.yaml"

```
loop: "{{ task_list }}"
```

1094. How do you use `ansible.builtin.import_tasks` with dynamic file names?

- Example:

- import_tasks: tasks/{{ item }}.yaml

```
loop:
```

```
- db
```

```
- cache
```

1095. How do you use `ansible.builtin.raw` to execute multiple commands in a single SSH session?

- Example:

- raw: |

```
cd /opt/app
```

```
./install.sh
```

1096. How do you use `ansible.builtin.lineinfile` with backrefs to modify configuration dynamically?

- Example:

- lineinfile:

```
path: /etc/myapp.conf
```

```
regexp: '(DEBUG=).*
```

```
line: '\1False'
```

```
backrefs: yes
```

1097. How do you use `ansible.builtin.replace` with multiline regex and backup?

- Example:

- replace:


```
path: /etc/myapp.conf
regexp: 'START=.*\nEND=.*'
replace: 'START=1\nEND=2'
backup: yes
```

1098. How do you use `ansible.builtin.yum` to update all packages except excluded ones and disable repo?

- Example:

- yum:

```
name: '*'
state: latest
exclude: kernel*
disablerepo: epel
```

1099. How do you use `ansible.builtin.apt` with `force_apt_get` for legacy systems?

- Example:

- apt:

```
name: python3
state: latest
force_apt_get: yes
```

1100. How do you use `ansible.builtin.service` with `pattern` to manage multiple services at once?

- Example:

- service:

```
name: "nginx*"
state: restarted
```

1101. How do you use `ansible.builtin.set_fact` with conditional expressions?

- Example:

- set_fact:

```
deploy_path: "/opt/app-prod"
when: env == "prod"
```

1102. How do you use `ansible.builtin.stat` to get checksum and size of a file?

- Example:

- stat:

```
path: /etc/myapp/config.cfg  
checksum_algorithm: sha256  
register: file_info
```

1103. How do you use ansible.builtin.fail with dynamic messages?

- Example:

- fail:

```
msg: "File {{ file_path }} missing!"  
when: not file_info.stat.exists
```

1104. How do you use ansible.builtin.include_role with multiple variables for conditional execution?

- Example:

- include_role:

```
name: webserver  
vars:  
  port: 8080  
  ssl: yes  
when: env == "prod"
```

1105. How do you use ansible.builtin.import_role to include tasks with variable overrides?

- Example:

- import_role:

```
name: database  
vars:  
  db_name: mydb  
  db_user: admin
```

1106. How do you use ansible.builtin.meta: flush_handlers in the middle of a play?

- Example:

- meta: flush_handlers

1107. How do you use ansible.builtin.meta: end_play to skip the remaining tasks dynamically?

- Example:

- meta: end_play

when: env not in ["prod", "staging"]

1108. How do you use ansible.builtin.pause to wait for AWS resource creation dynamically?

- Example:

- pause:

seconds: "{{ rds_creation_time }}"

1109. How do you use ansible.builtin.uri with retries, status codes, and delay?

- Example:

- uri:

url: https://api.example.com/status

status_code: 200

retries: 3

delay: 10

1110. How do you use ansible.builtin.pip to install multiple packages in a virtual environment?

- Example:

- pip:

name:

- awscli

- boto3

virtualenv: /opt/venv

extra_args: "--upgrade"

1111. How do you use ansible.builtin.user with multiple groups in a loop?

- Example:

- user:

name: "{{ item.name }}"

groups: "{{ item.groups }}"

append: yes

loop:

- { name: "deploy", groups: "devops" }

- { name: "admin", groups: "wheel" }

1112. How do you use ansible.builtin.group to manage multiple system groups?

- Example:

- group:
 name: "{{ item }}"
 system: yes

loop:
 - devops
 - admin

1113. How do you use `ansible.builtin.file` to create directories with owner, group, and mode?

- Example:

- file:
 path: /opt/app/logs
 state: directory
 owner: deploy
 group: devops
 mode: '0755'

1114. How do you use `ansible.builtin.copy` to write dynamic content with permissions?

- Example:

- copy:
 dest: /etc/myapp/config.cfg
 content: |
 HOST={{ host_ip }}
 PORT={{ port }}
 owner: deploy
 mode: '0644'

1115. How do you use `ansible.builtin.template` with loops and conditional logic?

- Example:

- template:
 src: config.j2
 dest: /etc/myapp/config.cfg

- Template snippet:

```
{% for server in servers %}  
{% if server.active %}
```

```
server {{ server.ip }};
```

```
{% endif %}
```

```
{% endfor %}
```

1116. How do you use `ansible.builtin.unarchive` to extract remote files with ownership and mode?

- Example:

- unarchive:

```
src: /tmp/app.tar.gz
```

```
dest: /opt/app
```

```
remote_src: yes
```

```
owner: deploy
```

```
mode: '0755'
```

1117. How do you use `ansible.builtin.get_url` to download multiple files using a loop?

- Example:

- get_url:

```
url: "{{ item.url }}"
```

```
dest: "{{ item.dest }}"
```

loop:

```
- { url: "https://example.com/file1", dest: "/tmp/file1" }
```

```
- { url: "https://example.com/file2", dest: "/tmp/file2" }
```

1118. How do you use `ansible.builtin.systemd` with loop and conditional service management?

- Example:

- systemd:

```
name: "{{ item }}"
```

```
state: restarted
```

loop:

```
- nginx
```

```
- redis
```

```
when: services_installed
```

1119. How do you use `ansible.builtin.shell` to run multiple scripts in a specific directory with environment variables?

- Example:

- shell: |
./deploy.sh
./migrate.sh
chdir: /opt/app
environment:
ENV: prod

1120. How do you use `ansible.builtin.command` to run scripts only if specific files do not exist?

- Example:

- command: ./install.sh
args:
creates: /opt/app/installed

1121. How do you use `ansible.builtin.meta: noop` to skip tasks dynamically?

- Example:

- meta: noop
when: skip_tasks | default(false)

1122. How do you use `ansible.builtin.assert` to enforce variable rules and environment checks?

- Example:

- assert:
that:
- env in ['dev', 'staging', 'prod']

1123. How do you use `ansible.builtin.debug` to display nested dictionary values?

- Example:

- debug:
var: apps[0].config.port

1124. How do you use `ansible.builtin.include_tasks` with loop and dynamic file inclusion?

- Example:

- include_tasks: "{{ item }}.yaml"
loop: "{{ task_list }}"

1125. How do you use `ansible.builtin.import_tasks` with dynamic file names in a loop?

- Example:

- import_tasks: tasks/{{ item }}.yaml

loop:

- db
- cache

1126. How do you use `ansible.builtin.raw` to execute multiple commands in one SSH session?

- Example:

```
- raw: |
    cd /opt/app
    ./install.sh
```

1127. How do you use `ansible.builtin.lineinfile` with backrefs and regex to modify configs?

- Example:

```
- lineinfile:
    path: /etc/myapp.conf
    regexp: '(DEBUG=).*'
    line: '\1False'
    backrefs: yes
```

1128. How do you use `ansible.builtin.replace` to modify multiline patterns with backup?

- Example:

```
- replace:
    path: /etc/myapp.conf
    regexp: 'START=.*\nEND=.*'
    replace: 'START=1\nEND=2'
    backup: yes
```

1129. How do you use `ansible.builtin.yum` to update packages excluding certain ones and disabling a repo?

- Example:

```
- yum:
    name: '*'
    state: latest
    exclude: kernel*
    disablerepo: epel
```

1130. How do you use `ansible.builtin.apt` with `force_apt_get` on legacy systems?

- Example:

- apt:

name: python3

state: latest

force_apt_get: yes

1131. How do you use `ansible.builtin.service` with `pattern` to manage multiple services?

- Example:

- service:

name: "nginx*"

state: restarted

1132. How do you use `ansible.builtin.set_fact` with calculations for memory or CPU?

- Example:

- set_fact:

free_memory_gb: "{{ ansible_facts.memfree_mb / 1024 | round(2) }}"

1133. How do you use `ansible.builtin.stat` to check if a symlink exists without following it?

- Example:

- stat:

path: /opt/app/current

follow: no

register: symlink_info

1134. How do you use `ansible.builtin.assert` with filters to validate list of dictionaries?

- Example:

- assert:

that:

- apps | selectattr('enabled', 'equalto', true) | list | length > 0

1135. How do you use `ansible.builtin.wait_for` to wait for TCP port availability?

- Example:

- wait_for:

host: "{{ inventory_hostname }}"

port: 8080

timeout: 120

1136. How do you use `ansible.builtin.get_url` with validation and headers for REST APIs?

- Example:

- `get_url`:

`url: https://example.com/app.tar.gz`

`dest: /tmp/app.tar.gz`

`headers:`

`Authorization: "Bearer {{ api_token }}"`

`validate_certs: yes`

1137. How do you use `ansible.builtin.pip` with `extra_args` for upgrading packages?

- Example:

- `pip`:

`name: boto3`

`virtualenv: /opt/venv`

`extra_args: "--upgrade"`

1138. How do you use `ansible.builtin.user` to ensure idempotent creation of multiple users?

- Example:

- `user`:

`name: "{{ item }}"`

`state: present`

`loop:`

- `deploy`

- `admin`

1139. How do you use `ansible.builtin.group` with `loop` to create multiple groups?

- Example:

- `group`:

`name: "{{ item }}"`

`state: present`

`loop:`

- `devops`

- `admin`

1140. How do you use `ansible.builtin.file` to create symlinks safely with force?

- Example:

- file:

src: /opt/app/current

dest: /opt/app/latest

state: link

force: yes

1141. How do you use ansible.builtin.copy with backup and dynamic content?

- Example:

- copy:

src: config.cfg

dest: /etc/myapp/config.cfg

backup: yes

content: "{{ config_content }}"

1142. How do you use ansible.builtin.template to apply conditionals in Jinja2?

- Example:

- template:

src: nginx.conf.j2

dest: /etc/nginx/nginx.conf

1143. How do you use ansible.builtin.unarchive with creates to ensure idempotency?

- Example:

- unarchive:

src: /tmp/app.tar.gz

dest: /opt/app

remote_src: yes

creates: /opt/app/app_binary

1144. How do you use ansible.builtin.get_url with retries and delay?

- Example:

- get_url:

url: https://example.com/file

dest: /tmp/file

retries: 3

delay: 10

1145. How do you use `ansible.builtin.systemd` with `enabled` and `state` to ensure service startup?

- Example:

- systemd:

name: nginx

state: started

enabled: yes

1146. How do you use `ansible.builtin.shell` to run scripts asynchronously and check results?

- Example:

- shell: ./deploy.sh

async: 600

poll: 0

register: job

1147. How do you use `ansible.builtin.async_status` to monitor async jobs?

- Example:

- async_status:

jid: "{{ job.ansible_job_id }}"

register: result

until: result.finished

retries: 10

delay: 15

1148. How do you use `ansible.builtin.command` with `chdir` and `creates` for idempotency?

- Example:

- command: ./setup.sh

args:

chdir: /opt/app

creates: /opt/app/setup_done

1149. How do you use `ansible.builtin.meta: end_host` to skip remaining tasks for a host?

- Example:

- meta: end_host

when: host_offline

1150. How do you use `ansible.builtin.assert` to ensure AWS environment variables are set?

- Example:

- `assert`:

that:

- `aws_access_key` is defined
 - `aws_secret_key` is defined
-

1151. How do you use `ansible.builtin.set_fact` with multiple conditions in one task?

- Example:

- `set_fact`:

`deploy_path: "{{ '/opt/app-prod' if env == 'prod' else '/opt/app-dev' }}"`

1152. How do you use `ansible.builtin.stat` to check file existence and type simultaneously?

- Example:

- `stat`:

`path: /opt/app/config.cfg`

`register: config_file`

1153. How do you use `ansible.builtin.fail` with multiple conditional checks?

- Example:

- `fail`:

`msg: "Required file missing!"`

`when: not config_file.stat.exists or config_file.stat.isdir`

1154. How do you use `ansible.builtin.include_role` with tags and variable overrides?

- Example:

- `include_role`:

`name: webserver`

`vars:`

`port: 8080`

`ssl: yes`

`tags: deploy`

1155. How do you use `ansible.builtin.import_role` with dynamic `tasks_from` file?

- Example:

- import_role:

name: database

tasks_from: "{{ db_task_file }}"

1156. How do you use ansible.builtin.meta: flush_handlers to immediately trigger handlers?

- Example:

- meta: flush_handlers

1157. How do you use ansible.builtin.meta: end_play conditionally based on AWS tags?

- Example:

- meta: end_play

when: aws_env_tag not in ['prod', 'staging']

1158. How do you use ansible.builtin.pause with prompts for manual approval?

- Example:

- pause:

prompt: "Approve deployment to production? Press ENTER to continue"

1159. How do you use ansible.builtin.uri to check HTTP endpoints with retries and delay?

- Example:

- uri:

url: https://api.example.com/status

status_code: 200

retries: 5

delay: 10

1160. How do you use ansible.builtin.pip to install packages with specific versions in virtualenv?

- Example:

- pip:

name:

- boto3==1.26.0

- awscli==2.11.0

virtualenv: /opt/venv

1161. How do you use ansible.builtin.user to manage multiple users with append option?

- Example:

- user:

```
name: "{{ item.name }}"
groups: "{{ item.groups }}"
append: yes
```

loop:

```
- { name: deploy, groups: devops }
- { name: admin, groups: wheel }
```

1162. How do you use `ansible.builtin.group` to create system groups with loop?

- Example:

- group:

```
name: "{{ item }}"
system: yes
```

loop:

```
- devops
- admin
```

1163. How do you use `ansible.builtin.file` to create directories recursively with ownership?

- Example:

- file:

```
path: /opt/app/logs
state: directory
owner: deploy
group: devops
mode: '0755'
```

1164. How do you use `ansible.builtin.copy` with dynamic content and backup enabled?

- Example:

- copy:

```
dest: /etc/myapp/config.cfg
content: "{{ config_content }}"
owner: deploy
mode: '0644'
backup: yes
```

1165. How do you use `ansible.builtin.template` with Jinja2 conditions and loops?

- Example:

- template:

src: nginx.conf.j2

dest: /etc/nginx/nginx.conf

- Template example:

{% for server in servers %}

{% if server.active %}

server {{ server.ip }};

{% endif %}

{% endfor %}

1166. How do you use ansible.builtin.unarchive with remote_src, creates, and mode?

- Example:

- unarchive:

src: /tmp/app.tar.gz

dest: /opt/app

remote_src: yes

creates: /opt/app/app_binary

mode: '0755'

1167. How do you use ansible.builtin.get_url with headers, retries, and validate_certs?

- Example:

- get_url:

url: https://example.com/app.tar.gz

dest: /tmp/app.tar.gz

headers:

Authorization: "Bearer {{ api_token }}"

retries: 3

validate_certs: yes

1168. How do you use ansible.builtin.systemd to enable, start, or restart multiple services?

- Example:

- systemd:

name: "{{ item }}"

state: restarted

enabled: yes

loop:

- nginx

- redis

1169. How do you use `ansible.builtin.shell` with `async` and environment variables?

- Example:

- shell: ./deploy.sh

async: 600

poll: 0

environment:

ENV: prod

1170. How do you use `ansible.builtin.command` with `creates` to ensure idempotency?

- Example:

- command: ./setup.sh

args:

creates: /opt/app/setup_done

1171. How do you use `ansible.builtin.meta: noop` to skip a task dynamically?

- Example:

- meta: noop

when: skip_tasks | default(false)

1172. How do you use `ansible.builtin.assert` to validate AWS credentials before deployment?

- Example:

- assert:

that:

- aws_access_key is defined

- aws_secret_key is defined

1173. How do you use `ansible.builtin.debug` to print complex nested variables?

- Example:

- debug:

var: apps[0].config.port

1174. How do you use `ansible.builtin.include_tasks` with a loop for dynamic inclusion?

- Example:

```
- include_tasks: "{{ item }}.yaml"
  loop: "{{ task_list }}"
```

1175. How do you use `ansible.builtin.import_tasks` with dynamic file names in AWS environments?

- Example:

```
- import_tasks: tasks/{{ item }}.yaml
  loop:
    - db
    - cache
```

1176. How do you use `ansible.builtin.raw` to execute commands on minimal hosts?

- Example:

```
- raw: |
  cd /opt/app
  ./install.sh
```

1177. How do you use `ansible.builtin.lineinfile` with regex and backrefs for configuration?

- Example:

```
- lineinfile:
  path: /etc/myapp.conf
  regexp: '(DEBUG=).*'
  line: '\1False'
  backrefs: yes
```

1178. How do you use `ansible.builtin.replace` with multiline regex patterns?

- Example:

```
- replace:
  path: /etc/myapp.conf
  regexp: 'START=.*\nEND=.*'
  replace: 'START=1\nEND=2'
  backup: yes
```

1179. How do you use `ansible.builtin.yum` to update all packages except excluded ones?

- Example:

- yum:

name: '*'

state: latest

exclude: kernel*

1180. How do you use ansible.builtin.apt with force_apt_get and multiple packages?

- Example:

- apt:

name:

- python3

- python3-pip

state: latest

force_apt_get: yes

1181. How do you use ansible.builtin.service with wildcard pattern for multiple services?

- Example:

- service:

name: "nginx*"

state: restarted

1182. How do you use ansible.builtin.set_fact with calculations for memory and CPU in AWS?

- Example:

- set_fact:

free_memory_gb: "{{ ansible_facts.memfree_mb / 1024 | round(2) }}"

1183. How do you use ansible.builtin.stat to check symlinks without following them?

- Example:

- stat:

path: /opt/app/current

follow: no

register: symlink_info

1184. How do you use ansible.builtin.assert with filters to validate lists of dictionaries?

- Example:

- assert:

that:

- apps | selectattr('enabled','equalto',true) | list | length > 0

1185. How do you use ansible.builtin.wait_for to wait for TCP port availability?

- Example:

- wait_for:

host: "{{ inventory_hostname }}"

port: 8080

timeout: 120

1186. How do you use ansible.builtin.get_url with validation and API headers?

- Example:

- get_url:

url: https://example.com/app.tar.gz

dest: /tmp/app.tar.gz

headers:

Authorization: "Bearer {{ api_token }}"

validate_certs: yes

1187. How do you use ansible.builtin.pip with extra_args for upgrading packages in virtualenv?

- Example:

- pip:

name: boto3

virtualenv: /opt/venv

extra_args: "--upgrade"

1188. How do you use ansible.builtin.user in a loop for idempotent user creation?

- Example:

- user:

name: "{{ item }}"

state: present

loop:

- deploy

- admin

1189. How do you use ansible.builtin.group in a loop to create multiple groups?

- Example:

```
- group:
  name: "{{ item }}"
  state: present

loop:
  - devops
  - admin
```

1190. How do you use ansible.builtin.file to create symlinks safely?

- Example:

```
- file:
  src: /opt/app/current
  dest: /opt/app/latest
  state: link
  force: yes
```

1191. How do you use ansible.builtin.copy with backup and dynamic content?

- Example:

```
- copy:
  src: config.cfg
  dest: /etc/myapp/config.cfg
  backup: yes
  content: "{{ config_content }}"
```

1192. How do you use ansible.builtin.template with conditions in Jinja2?

- Example:

```
- template:
  src: nginx.conf.j2
  dest: /etc/nginx/nginx.conf
```

1193. How do you use ansible.builtin.unarchive with creates to ensure idempotency?

- Example:

```
- unarchive:
  src: /tmp/app.tar.gz
  dest: /opt/app
  remote_src: yes
```

creates: /opt/app/app_binary

1194. How do you use ansible.builtin.get_url with retries and delay?

- Example:

- get_url:

url: https://example.com/file

dest: /tmp/file

retries: 3

delay: 10

1195. How do you use ansible.builtin.systemd with enabled and state?

- Example:

- systemd:

name: nginx

state: started

enabled: yes

1196. How do you use ansible.builtin.shell asynchronously and register results?

- Example:

- shell: ./deploy.sh

async: 600

poll: 0

register: job

1197. How do you use ansible.builtin.async_status to monitor async jobs?

- Example:

- async_status:

jid: "{{ job.ansible_job_id }}"

register: result

until: result.finished

retries: 10

delay: 15

1198. How do you use ansible.builtin.command with chdir and creates?

- Example:

- command: ./setup.sh

args:

chdir: /

opt/app

creates: /opt/app/setup_done

****1199. How do you use `ansible.builtin.meta: end\_host` to skip remaining tasks for a host?****

- Example:

``yaml

- meta: end_host

when: host_offline

1200. How do you use ansible.builtin.assert to ensure AWS environment variables are set?

- Example:

- assert:

that:

- aws_access_key is defined

- aws_secret_key is defined

1201. How do you use ansible.builtin.set_fact to create nested dictionaries dynamically?

- Example:

- set_fact:

app_config:

database:

host: "{{ db_host }}"

port: "{{ db_port }}"

1202. How do you use ansible.builtin.stat to check both file and directory existence?

- Example:

- stat:

path: /opt/app/config.cfg

register: config_stat

1203. How do you use ansible.builtin.fail with multiple conditions and a custom message?

- Example:

- fail:

msg: "Deployment failed: file missing or directory incorrect!"

when: not config_stat.stat.exists or config_stat.stat.isdir

1204. How do you include a role conditionally based on environment variables?

- Example:

- include_role:

name: webserver

when: env == "prod"

1205. How do you import a role with tasks_from for a specific AWS setup?

- Example:

- import_role:

name: database

tasks_from: rds_setup.yml

1206. How do you use meta: flush_handlers after critical configuration changes?

- Example:

- meta: flush_handlers

1207. How do you end a play dynamically for non-production hosts?

- Example:

- meta: end_play

when: env not in ["prod", "staging"]

1208. How do you use pause to wait for user confirmation in production deployment?

- Example:

- pause:

prompt: "Press ENTER to deploy to production"

1209. How do you use uri to check a REST API endpoint repeatedly until it succeeds?

- Example:

- uri:

url: https://api.example.com/health

status_code: 200

retries: 5

delay: 10

1210. How do you install Python packages in a virtualenv with specific versions using pip?

- Example:

- pip:

name:

- boto3==1.26.0

- awscli==2.11.0

virtualenv: /opt/venv

1211. How do you create multiple users with append to avoid overwriting groups?

- Example:

- user:

name: "{{ item.name }}"

groups: "{{ item.groups }}"

append: yes

loop:

- { name: deploy, groups: devops }

- { name: admin, groups: wheel }

1212. How do you create multiple system groups using a loop?

- Example:

- group:

name: "{{ item }}"

system: yes

loop:

- devops

- admin

1213. How do you create directories with specific permissions and ownership recursively?

- Example:

- file:

path: /opt/app/logs

state: directory

owner: deploy

group: devops

mode: '0755'

1214. How do you copy files with dynamic content and backup enabled?

- Example:

- copy:

dest: /etc/myapp/config.cfg

content: "{{ config_content }}"

owner: deploy

mode: '0644'

backup: yes

1215. How do you use a template with Jinja2 loops and conditions?

- Example:

- template:

src: nginx.conf.j2

dest: /etc/nginx/nginx.conf

- Template snippet:

{% for server in servers %}

{% if server.active %}

server {{ server.ip }};

{% endif %}

{% endfor %}

1216. How do you unarchive a tarball ensuring idempotency and setting permissions?

- Example:

- unarchive:

src: /tmp/app.tar.gz

dest: /opt/app

remote_src: yes

creates: /opt/app/app_binary

mode: '0755'

1217. How do you download a file with headers and retries using get_url?

- Example:

- get_url:

url: https://example.com/app.tar.gz

dest: /tmp/app.tar.gz

headers:

Authorization: "Bearer {{ api_token }}"

retries: 3

validate_certs: yes

1218. How do you manage multiple services using systemd in a loop?

- Example:

- systemd:

name: "{{ item }}"

state: restarted

enabled: yes

loop:

- nginx

- redis

1219. How do you run shell scripts asynchronously with environment variables?

- Example:

- shell: ./deploy.sh

async: 600

poll: 0

environment:

ENV: prod

1220. How do you ensure idempotent command execution using creates?

- Example:

- command: ./setup.sh

args:

creates: /opt/app/setup_done

1221. How do you skip tasks dynamically using meta: noop?

- Example:

- meta: noop

when: skip_tasks | default(false)

1222. How do you assert that AWS credentials exist before running tasks?

- Example:

- assert:

that:

- aws_access_key is defined
- aws_secret_key is defined

1223. How do you debug nested variables in Ansible?

- Example:

- debug:

var: apps[0].config.port

1224. How do you include tasks dynamically using a loop?

- Example:

- include_tasks: "{{ item }}.yaml"

loop: "{{ task_list }}"

1225. How do you import tasks dynamically for AWS deployments?

- Example:

- import_tasks: tasks/{{ item }}.yaml

loop:

- db
- cache

1226. How do you execute raw commands on minimal hosts using raw?

- Example:

- raw: |

cd /opt/app

./install.sh

1227. How do you use lineinfile with regex and backreferences to modify configs?

- Example:

- lineinfile:

path: /etc/myapp.conf

regexp: '(DEBUG=).*'

line: '\1False'

backrefs: yes

1228. How do you replace multiline patterns safely with replace?

- Example:

- replace:

path: /etc/myapp.conf

regexp: 'START=.*\nEND=.*'

replace: 'START=1\nEND=2'

backup: yes

1229. How do you update packages excluding some using yum?

- Example:

- yum:

name: '*'

state: latest

exclude: kernel*

1230. How do you install multiple packages with apt forcing apt-get?

- Example:

- apt:

name:

- python3

- python3-pip

state: latest

force_apt_get: yes

1231. How do you manage services with wildcard names using service?

- Example:

- service:

name: "nginx*"

state: restarted

1232. How do you calculate system memory facts with set_fact?

- Example:

- set_fact:

free_memory_gb: "{{ ansible_facts.memfree_mb / 1024 | round(2) }}"

1233. How do you check if a symlink exists without following it?

- Example:

- stat:

path: /opt/app/current

follow: no

register: symlink_info

1234. How do you assert that at least one enabled app exists in a list?

- Example:

- assert:

that:

- apps | selectattr('enabled','equalto',true) | list | length > 0

1235. How do you wait for a TCP port to be available using wait_for?

- Example:

- wait_for:

host: "{{ inventory_hostname }}"

port: 8080

timeout: 120

1236. How do you download files from API endpoints with headers using get_url?

- Example:

- get_url:

url: https://example.com/app.tar.gz

dest: /tmp/app.tar.gz

headers:

Authorization: "Bearer {{ api_token }}"

validate_certs: yes

1237. How do you upgrade pip packages inside virtualenv with extra args?

- Example:

- pip:

name: boto3

virtualenv: /opt/venv

extra_args: "--upgrade"

1238. How do you create users idempotently using a loop?

- Example:

- user:

name: "{{ item }}"

state: present

loop:

- deploy

- admin

1239. How do you create multiple groups in a loop?

- Example:

- group:

name: "{{ item }}"

state: present

loop:

- devops

- admin

1240. How do you create a symlink safely using file?

- Example:

- file:

src: /opt/app/current

dest: /opt/app/latest

state: link

force: yes

1241. How do you copy files with backup and dynamic content?

- Example:

- copy:

src: config.cfg

dest: /etc/myapp/config.cfg

backup: yes

content: "{{ config_content }}"

1242. How do you use templates with Jinja2 conditions?

- Example:

- template:

src: nginx.conf.j2

dest: /etc/nginx/nginx.conf

1243. How do you unarchive files idempotently using creates?

- Example:

- unarchive:

src: /tmp/app.tar.gz

dest: /opt/app

remote_src: yes

creates: /opt/app/app_binary

1244. How do you retry downloads with get_url?

- Example:

- get_url:

url: https://example.com/file

dest: /tmp/file

retries: 3

delay: 10

1245. How do you start and enable services with systemd?

- Example:

- systemd:

name: nginx

state: started

enabled: yes

1246. How do you run asynchronous shell commands and register results?

- Example:

- shell: ./deploy.sh

async: 600

poll: 0

register: job

1247. How do you monitor async jobs using async_status?

- Example:

- async_status:

jid: "{{ job.ansible_job_id }}"

register: result

until: result.finished

retries: 10

delay: 15

1248. How do you ensure idempotent execution using command with creates?

- Example:

- command: ./setup.sh

args:

chdir: /opt/app

creates: /opt/app/setup_done

1249. How do you skip remaining tasks for a host dynamically?

- Example:

- meta: end_host

when: host_offline

1250. How do you assert that AWS environment variables are defined before deployment?

- Example:

- assert:

that:

- aws_access_key is defined

- aws_secret_key is defined

1251. How do you use ansible.builtin.set_fact with conditional expressions?

- Example:

- set_fact:

deploy_path: "{{ '/opt/app-prod' if env == 'prod' else '/opt/app-dev' }}"

1252. How do you use ansible.builtin.stat to check file ownership and permissions?

- Example:

- stat:

path: /opt/app/config.cfg

register: config_file

1253. How do you fail a task if a required file is missing or empty?

- Example:

- fail:

msg: "Configuration file missing or empty!"

when: not config_file.stat.exists or config_file.stat.size == 0

1254. How do you include a role with variable overrides dynamically?

- Example:

- include_role:

name: webserver

vars:

port: 8080

ssl: yes

1255. How do you import a role from a specific task file dynamically?

- Example:

- import_role:

name: database

tasks_from: "{{ db_task_file }}"

1256. How do you immediately trigger handlers after a task?

- Example:

- meta: flush_handlers

1257. How do you end a play dynamically based on environment conditions?

- Example:

- meta: end_play

when: aws_env_tag not in ['prod', 'staging']

1258. How do you pause for manual approval before critical deployment?

- Example:

- pause:

prompt: "Approve deployment to production? Press ENTER to continue"

1259. How do you check HTTP endpoints repeatedly until they succeed using uri?

- Example:

- uri:

url: https://api.example.com/status

status_code: 200

retries: 5

delay: 10

1260. How do you install Python packages in a virtual environment with specific versions?

- Example:

- pip:

name:

- boto3==1.26.0

- awscli==2.11.0

virtualenv: /opt/venv

1261. How do you create multiple users idempotently using a loop?

- Example:

- user:

name: "{{ item.name }}"

groups: "{{ item.groups }}"

append: yes

loop:

- { name: deploy, groups: devops }

- { name: admin, groups: wheel }

1262. How do you create multiple system groups with a loop?

- Example:

- group:

name: "{{ item }}"

system: yes

loop:

- devops

- admin

1263. How do you create directories recursively with specific ownership and permissions?

- Example:

- file:

path: /opt/app/logs

state: directory

owner: deploy

group: devops

mode: '0755'

1264. How do you copy files with dynamic content and backup enabled?

- Example:

- copy:

dest: /etc/myapp/config.cfg

content: "{{ config_content }}"

owner: deploy

mode: '0644'

backup: yes

1265. How do you use a template with Jinja2 loops and conditional statements?

- Example:

- template:

src: nginx.conf.j2

dest: /etc/nginx/nginx.conf

- Template snippet:

{% for server in servers %}

{% if server.active %}

server {{ server.ip }};

{% endif %}

{% endfor %}

1266. How do you unarchive files idempotently using creates?

- Example:

- unarchive:

src: /tmp/app.tar.gz

dest: /opt/app

remote_src: yes

creates: /opt/app/app_binary

1267. How do you download files with headers and retries using get_url?

- Example:

- get_url:

url: https://example.com/app.tar.gz

dest: /tmp/app.tar.gz

headers:

Authorization: "Bearer {{ api_token }}"

retries: 3

validate_certs: yes

1268. How do you manage multiple services using systemd in a loop?

- Example:

- systemd:

name: "{{ item }}"

state: restarted

enabled: yes

loop:

- nginx

- redis

1269. How do you run shell scripts asynchronously with environment variables?

- Example:

- shell: ./deploy.sh

async: 600

poll: 0

environment:

ENV: prod

1270. How do you ensure idempotent command execution using creates?

- Example:

- command: ./setup.sh

args:

creates: /opt/app/setup_done

1271. How do you skip tasks dynamically using meta: noop?

- Example:

- meta: noop

when: skip_tasks | default(false)

1272. How do you assert that AWS credentials exist before executing tasks?

- Example:

- assert:

that:

- aws_access_key is defined

- aws_secret_key is defined

1273. How do you debug nested variables in Ansible?

- Example:

- debug:

var: apps[0].config.port

1274. How do you include tasks dynamically using a loop?

- Example:

- include_tasks: "{{ item }}.yaml"

loop: "{{ task_list }}"

1275. How do you import tasks dynamically for AWS deployments?

- Example:

- import_tasks: tasks/{{ item }}.yaml

loop:

- db

- cache

1276. How do you execute raw commands on minimal hosts using raw?

- Example:

- raw: |

cd /opt/app

./install.sh

1277. How do you use lineinfile with regex and backreferences to modify configuration?

- Example:

- lineinfile:

path: /etc/myapp.conf

regexp: '(DEBUG=).*'

line: '\1False'

backrefs: yes

1278. How do you replace multiline patterns safely with replace?

- Example:

- replace:

path: /etc/myapp.conf

regexp: 'START=.*\nEND=.*'

replace: 'START=1\nEND=2'

backup: yes

1279. How do you update packages excluding certain packages using yum?

- Example:

- yum:

name: '*'

state: latest

exclude: kernel*

1280. How do you install multiple packages with apt forcing apt-get?

- Example:

- apt:

name:

- python3

- python3-pip

state: latest

force_apt_get: yes

1281. How do you manage services with wildcard names using service?

- Example:

- service:

name: "nginx*"

state: restarted

1282. How do you calculate system memory facts dynamically?

- Example:

- set_fact:

free_memory_gb: "{{ ansible_facts.memfree_mb / 1024 | round(2) }}"

1283. How do you check if a symlink exists without following it?

- Example:

- stat:

path: /opt/app/current

follow: no

register: symlink_info

1284. How do you assert that at least one enabled app exists in a list?

- Example:

- assert:

that:

- apps | selectattr('enabled','equalto',true) | list | length > 0

1285. How do you wait for a TCP port to be available using wait_for?

- Example:

- wait_for:

host: "{{ inventory_hostname }}"

port: 8080

timeout: 120

1286. How do you download files from API endpoints with headers using get_url?

- Example:

- get_url:

url: https://example.com/app.tar.gz

dest: /tmp/app.tar.gz

headers:

Authorization: "Bearer {{ api_token }}"

validate_certs: yes

1287. How do you upgrade pip packages inside a virtualenv with extra args?

- Example:

- pip:

name: boto3

virtualenv: /opt/venv

extra_args: "--upgrade"

1288. How do you create users idempotently using a loop?

- Example:

- user:

name: "{{ item }}"

state: present

loop:

- deploy

- admin

1289. How do you create multiple groups in a loop?

- Example:

- group:

name: "{{ item }}"

state: present

loop:

- devops

- admin

1290. How do you create a symlink safely using file?

- Example:

- file:

src: /opt/app/current

dest: /opt/app/latest

state: link

force: yes

1291. How do you copy files with backup and dynamic content?

- Example:

- copy:


```
src: config.cfg
dest: /etc/myapp/config.cfg
backup: yes
content: "{{ config_content }}"
```

1292. How do you use templates with Jinja2 conditions?

- Example:

- template:

```
src: nginx.conf.j2
dest: /etc/nginx/nginx.conf
```

1293. How do you unarchive files idempotently using creates?

- Example:

- unarchive:

```
src: /tmp/app.tar.gz
dest: /opt/app
remote_src: yes
creates: /opt/app/app_binary
```

1294. How do you retry downloads with get_url?

- Example:

- get_url:

```
url: https://example.com/file
dest: /tmp/file
retries: 3
delay: 10
```

1295. How do you start and enable services with systemd?

- Example:

- systemd:

```
name: nginx
state: started
enabled: yes
```

1296. How do you run asynchronous shell commands and register results?

- Example:

- shell: ./deploy.sh

async: 600

poll: 0

register: job

1297. How do you monitor async jobs using async_status?

- Example:

- async_status:

jid: "{{ job.ansible_job_id }}"

register: result

until: result.finished

retries: 10

delay: 15

1298. How do you ensure idempotent execution using command with creates?

- Example:

- command: ./setup.sh

args:

chdir: /opt/app

creates: /opt/app/setup_done

1299. How do you skip remaining tasks for a host dynamically?

- Example:

- meta: end_host

when: host_offline

1300. How do you assert that AWS environment variables are defined before deployment?

- Example:

- assert:

that:

- aws_access_key is defined

- aws_secret_key is defined

1301. How do you create dynamic inventories using AWS EC2 plugin?

- Example:

plugin: aws_ec2

regions:

- us-east-1

filters:

tag:Env: prod

keyed_groups:

- key: tags.Role

prefix: role

1302. How do you use ec2_instance module to launch instances dynamically?

- Example:

- name: Launch EC2 instance

amazon.aws.ec2_instance:

name: my-instance

key_name: "{{ key_name }}"

instance_type: t2.micro

image_id: ami-123456

wait: yes

1303. How do you terminate multiple EC2 instances using a loop?

- Example:

- amazon.aws.ec2_instance:

state: absent

instance_ids: "{{ item }}"

loop: "{{ ec2_ids }}"

1304. How do you create security groups with multiple rules?

- Example:

- amazon.aws.ec2_group:

name: my-sg

description: "Security group"

region: us-east-1

rules:

- proto: tcp
- from_port: 22
- to_port: 22
- cidr_ip: 0.0.0.0/0
- proto: tcp
- from_port: 80
- to_port: 80
- cidr_ip: 0.0.0.0/0

1305. How do you create an S3 bucket with specific ACL and versioning?

- Example:

- amazon.aws.s3_bucket:

name: my-bucket
state: present
versioning: yes
acl: private

1306. How do you upload multiple files to S3 with a loop?

- Example:

- amazon.aws.s3:

bucket: my-bucket
object: "{{ item.name }}"
src: "{{ item.src }}"
mode: put
loop: "{{ files }}"

1307. How do you check EC2 instance facts dynamically?

- Example:

- amazon.aws.ec2_instance_info:

filters:
instance-state-name: running
register: ec2_info

1308. How do you create RDS instances with specific parameters?

- Example:

- amazon.aws.rds_instance:

```
db_instance_identifier: mydb
engine: mysql
size: db.t2.micro
username: admin
password: "{{ db_password }}"
state: present
```

1309. How do you create IAM users with policies attached?

- Example:

- amazon.aws.iam_user:

```
name: deploy
state: present
```

- amazon.aws.iam_policy_attachment:

```
name: deploy-policy
users: deploy
policy_arn: arn:aws:iam::aws:policy/AdministratorAccess
```

1310. How do you use wait_for to wait for RDS endpoint availability?

- Example:

- wait_for:

```
host: "{{ rds_endpoint }}"
port: 3306
timeout: 300
```

1311. How do you create AWS ELB with multiple listeners?

- Example:

- amazon.aws.elb_application_lb:

```
name: my-alb
subnets: "{{ subnets }}"
security_groups: "{{ sg_ids }}"
state: present
```

1312. How do you create target groups and attach them to ALB?

- Example:

- amazon.aws.elb_target_group:

name: my-target

port: 80

protocol: HTTP

vpc_id: "{{ vpc_id }}"

state: present

1313. How do you attach EC2 instances to a target group?

- Example:

- amazon.aws.elb_target:

target_group_arn: "{{ target_group_arn }}"

targets: "{{ ec2_ids }}"

1314. How do you configure Route53 records dynamically?

- Example:

- amazon.aws.route53:

state: present

zone: example.com

record: www

type: A

ttl: 300

value: "{{ ec2_public_ip }}"

1315. How do you create CloudWatch alarms for EC2 CPU usage?

- Example:

- amazon.aws.cloudwatch_alarm:

name: HighCPU

metric: CPUUtilization

namespace: AWS/EC2

statistic: Average

comparison: ">="

threshold: 80

evaluation_periods: 2

```
period: 300

alarm_actions:
  - "{{ sns_arn }}"
```

1316. How do you send notifications to an SNS topic?

- Example:
- ```
- amazon.aws.sns_topic:
 name: deploy-notifications
 state: present
```

**1317. How do you publish messages to an SNS topic?**

- Example:
- ```
- amazon.aws.sns:
  topic: deploy-notifications
  msg: "Deployment completed successfully"
```

1318. How do you launch a CloudFormation stack with parameters?

- Example:
- ```
- amazon.aws.cloudformation:
 stack_name: my-stack
 state: present
 template: /tmp/template.yaml
 parameters:
 InstanceType: t2.micro
 KeyName: "{{ key_name }}"
```

**1319. How do you delete a CloudFormation stack safely?**

- Example:
- ```
- amazon.aws.cloudformation:
  stack_name: my-stack
  state: absent
```

1320. How do you create DynamoDB tables with specific throughput?

- Example:
- ```
- amazon.aws.dynamodb_table:
 name: my-table
```

hash\_key\_name: id

hash\_key\_type: S

read\_capacity: 5

write\_capacity: 5

**1321. How do you insert items into DynamoDB table using aws\_dynamodb module?**

- Example:

- amazon.aws.dynamodb:

table\_name: my-table

hash\_key: id

item:

id: "1"

name: "Test"

**1322. How do you create SQS queues dynamically?**

- Example:

- amazon.aws.sqs\_queue:

name: my-queue

state: present

**1323. How do you send messages to SQS queue?**

- Example:

- amazon.aws.sqs\_message:

queue\_url: "{{ queue\_url }}"

message\_body: "Hello from Ansible"

**1324. How do you receive messages from SQS queue?**

- Example:

- amazon.aws.sqs\_message:

queue\_url: "{{ queue\_url }}"

wait\_time\_seconds: 5

register: sqs\_messages

**1325. How do you create AWS Lambda functions using Ansible?**

- Example:

- amazon.aws.lambda:



```
name: my-function
state: present
runtime: python3.9
role: "{{ lambda_role_arn }}"
handler: lambda_function.lambda_handler
zip_file: /tmp/lambda.zip
```

#### **1326. How do you invoke Lambda functions dynamically?**

- Example:

```
- amazon.aws.lambda_invoke:
 name: my-function
 payload: '{"key":"value"}'
 register: lambda_result
```

#### **1327. How do you create API Gateway endpoints using Ansible?**

- Example:

```
- amazon.aws.apigateway_rest_api:
 name: my-api
 state: present
```

#### **1328. How do you deploy API Gateway stages dynamically?**

- Example:

```
- amazon.aws.apigateway_deployment:
 rest_api_id: "{{ api_id }}"
 stage_name: prod
```

#### **1329. How do you attach CloudWatch logs to Lambda functions?**

- Example:

```
- amazon.aws.lambda:
 name: my-function
 log_group_name: "/aws/lambda/my-function"
```

#### **1330. How do you schedule Lambda functions using CloudWatch events?**

- Example:

```
- amazon.aws.cloudwatch_event_rule:
 name: daily-rule
```

```
schedule_expression: "rate(1 day)"
```

```
- amazon.aws.cloudwatch_event_target:
```

```
rule: daily-rule
```

```
target_arn: "{{ lambda_arn }}"
```

### **1331. How do you manage AWS Secrets Manager secrets using Ansible?**

- Example:

```
- amazon.aws.secretsmanager_secret:
```

```
name: db_password
```

```
state: present
```

```
- amazon.aws.secretsmanager_secret_value:
```

```
name: db_password
```

```
secret_string: "{{ db_password }}"
```

### **1332. How do you rotate secrets in AWS Secrets Manager dynamically?**

- Example:

```
- amazon.aws.secretsmanager_secret:
```

```
name: db_password
```

```
rotation_enabled: yes
```

```
rotation_lambda_arn: "{{ rotation_lambda_arn }}"
```

### **1333. How do you enable versioning for an S3 bucket dynamically?**

- Example:

```
- amazon.aws.s3_bucket:
```

```
name: my-bucket
```

```
versioning: yes
```

### **1334. How do you enable server-side encryption on S3 buckets?**

- Example:

```
- amazon.aws.s3_bucket:
```

```
name: my-bucket
```

```
encryption:
```

```
rules:
```

- apply\_server\_side\_encryption\_by\_default:

```
sse_algorithm: AES256
```

**1335. How do you delete multiple S3 buckets dynamically using a loop?**

- Example:

- amazon.aws.s3\_bucket:

name: "{{ item }}"

state: absent

loop: "{{ bucket\_list }}"

**1336. How do you attach IAM roles to EC2 instances?**

- Example:

- amazon.aws.ec2\_instance:

instance\_ids: "{{ ec2\_ids }}"

iam\_instance\_profile\_name: my-role

**1337. How do you attach policies to IAM roles dynamically?**

- Example:

- amazon.aws.iam\_role:

name: my-role

managed\_policies:

- arn:aws:iam::aws:policy/AmazonS3FullAccess

**1338. How do you detach policies from IAM roles using Ansible?**

- Example:

- amazon.aws.iam\_role:

name: my-role

managed\_policies: []

**1339. How do you delete IAM users with associated policies?**

- Example:

- amazon.aws.iam\_user:

name: old-user

state: absent

**1340. How do you automate EC2 tagging dynamically during creation?**

- Example:

- amazon.aws.ec2\_instance:

name: my-instance

tags:

Environment: prod

Role: web

#### **1341. How do you deregister old AMIs automatically?**

- Example:

- amazon.aws.ec2\_ami:

image\_id: "{{ ami\_id }}"

state: absent

#### **1342. How do you create EBS volumes and attach them dynamically?**

- Example:

- amazon.aws.ec2\_vol:

instance: "{{ ec2\_id }}"

volume\_size: 20

device\_name: /dev/sdf

state: present

#### **1343. How do you create snapshots of EBS volumes?**

- Example:

- amazon.aws.ec2\_snapshot:

volume\_id: "{{ volume\_id }}"

description: "Backup snapshot"

#### **1344. How do you share EBS snapshots across AWS accounts?**

- Example:

- amazon.aws.ec2\_snapshot:

snapshot\_id: "{{ snapshot\_id }}"

state: present

share\_accounts:

- 123456789012

#### **1345. How do you create CloudFront distributions dynamically?**

- Example:

- amazon.aws.cloudfront\_distribution:

state: present

origin:

domain\_name: my-bucket.s3.amazonaws.com

enabled: yes

**1346. How do you invalidate CloudFront cache programmatically?**

- Example:

- amazon.aws.cloudfront\_invalidation:

distribution\_id: "{{ cf\_id }}"

paths:

- "/"

**1347. How do you enable logging for CloudFront distributions?**

- Example:

- amazon.aws.cloudfront\_distribution:

distribution\_id: "{{ cf\_id }}"

logging\_config:

bucket: my-log-bucket.s3.amazonaws.com

include\_cookies: yes

**1348. How do you configure Elastic Beanstalk environments using Ansible?**

- Example:

- amazon.aws.elasticbeanstalk:

name: my-app

state: present

application\_name: my-app

environment\_name: my-app-env

**1349. How do you deploy application versions to Elastic Beanstalk?**

- Example:

- amazon.aws.elasticbeanstalk\_app\_version:

application\_name: my-app

version\_label: v1

source\_bundle:

s3\_bucket: my-bucket

s3\_key: app\_v1.zip

### 1350. How do you terminate Elastic Beanstalk environments safely?

- Example:

- amazon.aws.elasticbeanstalk:

environment\_name: my-app-env

state: absent

---

### 1351. How do you configure AWS VPCs dynamically with Ansible?

- Example:

- amazon.aws.ec2\_vpc:

cidr\_block: 10.0.0.0/16

state: present

tags:

Name: my-vpc

### 1352. How do you create subnets in a VPC dynamically?

- Example:

- amazon.aws.ec2\_vpc\_subnet:

vpc\_id: "{{ vpc\_id }}"

cidr: 10.0.1.0/24

availability\_zone: us-east-1a

state: present

### 1353. How do you configure Internet Gateways for a VPC?

- Example:

- amazon.aws.ec2\_vpc\_igw:

vpc\_id: "{{ vpc\_id }}"

state: present

### 1354. How do you configure route tables and associate them with subnets?

- Example:

- amazon.aws.ec2\_vpc\_route\_table:

vpc\_id: "{{ vpc\_id }}"

routes:

- dest: 0.0.0.0/0

```
gateway_id: "{{ igw_id }}"
```

```
subnets:
```

```
- "{{ subnet_id }}"
```

### **1355. How do you create NAT gateways for private subnets?**

- Example:

```
- amazon.aws.ec2_vpc_nat_gateway:
```

```
subnet_id: "{{ public_subnet }}"
```

```
state: present
```

### **1356. How do you attach Elastic IPs to NAT gateways?**

- Example:

```
- amazon.aws.ec2_eip:
```

```
in_vpc: yes
```

```
state: present
```

```
instance_id: "{{ nat_gateway_id }}"
```

### **1357. How do you automate security group rules dynamically?**

- Example:

```
- amazon.aws.ec2_group:
```

```
name: web-sg
```

```
description: "Web servers"
```

```
vpc_id: "{{ vpc_id }}"
```

```
rules:
```

```
- proto: tcp
```

```
from_port: 80
```

```
to_port: 80
```

```
cidr_ip: 0.0.0.0/0
```

```
- proto: tcp
```

```
from_port: 443
```

```
to_port: 443
```

```
cidr_ip: 0.0.0.0/0
```

### **1358. How do you manage security group egress rules?**

- Example:

- amazon.aws.ec2\_group:

name: web-sg

rules\_egress:

- proto: tcp

from\_port: 0

to\_port: 65535

cidr\_ip: 0.0.0.0/0

**1359. How do you dynamically tag EC2 instances based on environment?**

- Example:

- amazon.aws.ec2\_tag:

resource: "{{ ec2\_id }}"

state: present

tags:

Environment: "{{ env }}"

Role: web

**1360. How do you launch EC2 instances using a launch template?**

- Example:

- amazon.aws.ec2\_instance:

launch\_template:

name: web-template

count: 3

wait: yes

**1361. How do you create auto-scaling groups with dynamic launch configurations?**

- Example:

- amazon.aws.autoscaling\_group:

name: web-asg

launch\_template: web-template

min\_size: 2

max\_size: 5

desired\_capacity: 3

**1362. How do you attach instances to an auto-scaling group dynamically?**



- Example:

- amazon.aws.autoscaling\_group:

name: web-asg

instance\_ids: "{{ ec2\_ids }}"

### 1363. How do you configure CloudWatch metrics for auto-scaling triggers?

- Example:

- amazon.aws.cloudwatch\_metric\_alarm:

name: cpu-high

metric: CPUUtilization

namespace: AWS/EC2

statistic: Average

threshold: 80

comparison: ">="

evaluation\_periods: 2

period: 300

alarm\_actions:

- "{{ asg\_policy\_arn }}"

### 1364. How do you create Elastic IPs and associate them with EC2 instances?

- Example:

- amazon.aws.ec2\_eip:

state: present

in\_vpc: yes

instance\_id: "{{ ec2\_id }}"

### 1365. How do you detach and release Elastic IPs?

- Example:

- amazon.aws.ec2\_eip:

state: absent

public\_ip: "{{ eip }}"

### 1366. How do you create AWS CloudTrail trails with logging enabled?

- Example:

- amazon.aws.cloudtrail:

name: my-trail

s3\_bucket\_name: my-trail-logs

state: present

**1367. How do you configure CloudTrail to log all global events?**

- Example:

- amazon.aws.cloudtrail:

name: my-trail

include\_global\_service\_events: yes

**1368. How do you enable S3 bucket logging for audit purposes?**

- Example:

- amazon.aws.s3\_bucket:

name: my-bucket

logging:

target\_bucket: log-bucket

target\_prefix: logs/

**1369. How do you create AWS Config rules dynamically?**

- Example:

- amazon.aws.aws\_config:

state: present

config\_rule\_name: s3-bucket-public-read-prohibited

**1370. How do you manage multiple AWS Config rules using a loop?**

- Example:

- amazon.aws.aws\_config:

state: present

config\_rule\_name: "{{ item }}"

loop:

- s3-bucket-public-read-prohibited

- cloudtrail-enabled

**1371. How do you automate EBS volume snapshots for backup?**

- Example:

- amazon.aws.ec2\_snapshot:

volume\_id: "{{ volume\_id }}"

description: "Daily backup"

### 1372. How do you delete old EBS snapshots older than X days?

- Example:

- amazon.aws.ec2\_snapshot\_facts:

filters:

volume-id: "{{ volume\_id }}"

register: snapshots

- amazon.aws.ec2\_snapshot:

snapshot\_id: "{{ item.id }}"

state: absent

loop: "{{ snapshots.snapshots | selectattr('start\_time', 'lt', cutoff\_date) | list }}"

### 1373. How do you enable S3 lifecycle policies for automatic deletion?

- Example:

- amazon.aws.s3\_bucket:

name: my-bucket

lifecycle\_rules:

- id: delete-old

prefix: logs/

status: Enabled

expiration:

days: 30

### 1374. How do you configure AWS WAF rules dynamically?

- Example:

- amazon.aws.waf\_web\_acl:

name: my-waf

default\_action: ALLOW

### 1375. How do you attach WAF ACLs to CloudFront distributions?

- Example:

- amazon.aws.waf\_web\_acl\_association:

resource\_arn: "{{ cf\_distribution\_arn }}"

```
web_acl_arn: "{{ waf_acl_arn }}"
```

### **1376. How do you create Amazon MQ brokers dynamically?**

- Example:

```
- amazon.aws.mq_broker:
```

```
broker_name: my-broker
```

```
engine_type: ActiveMQ
```

```
host_instance_type: mq.t3.micro
```

```
users:
```

```
- username: admin
```

```
password: "{{ mq_password }}"
```

### **1377. How do you create Amazon ElastiCache clusters dynamically?**

- Example:

```
- amazon.aws.elasticache:
```

```
name: my-redis
```

```
state: present
```

```
engine: redis
```

```
node_type: cache.t3.micro
```

```
num_nodes: 1
```

### **1378. How do you update ElastiCache cluster parameters dynamically?**

- Example:

```
- amazon.aws.elasticache:
```

```
name: my-redis
```

```
parameter_group_name: my-redis-params
```

### **1379. How do you configure AWS KMS keys for encryption dynamically?**

- Example:

```
- amazon.aws.kms_key:
```

```
alias: my-key
```

```
state: present
```

### **1380. How do you enable key rotation for AWS KMS keys?**

- Example:

```
- amazon.aws.kms_key:
```

alias: my-key

enable\_key\_rotation: yes

### **1381. How do you automate Amazon Athena queries using Ansible?**

- Example:

- amazon.aws.athena\_query:

query: "SELECT \* FROM my\_table LIMIT 10;"

database: my\_db

output\_location: s3://my-athena-results/

### **1382. How do you create Glue crawlers dynamically?**

- Example:

- amazon.aws.glue\_crawler:

name: my-crawler

role: "{{ glue\_role }}"

database\_name: my\_db

targets:

s3\_targets:

- path: s3://my-bucket/data/

### **1383. How do you start Glue crawlers dynamically?**

- Example:

- amazon.aws.glue\_crawler:

name: my-crawler

state: started

### **1384. How do you stop Glue crawlers dynamically?**

- Example:

- amazon.aws.glue\_crawler:

name: my-crawler

state: stopped

### **1385. How do you manage AWS Step Functions workflows with Ansible?**

- Example:

- amazon.aws.stepfunctions\_state\_machine:

name: my-workflow

definition: "{{ lookup('file', 'workflow.json') }}"

state: present

#### **1386. How do you start Step Functions executions dynamically?**

- Example:

- amazon.aws.stepfunctions\_execution:

state\_machine\_arn: "{{ workflow\_arn }}"

input: '{"key": "value"}'

#### **1387. How do you delete Step Functions state machines?**

- Example:

- amazon.aws.stepfunctions\_state\_machine:

state\_machine\_arn: "{{ workflow\_arn }}"

state: absent

#### **1388. How do you create Amazon Cognito user pools dynamically?**

- Example:

- amazon.aws.cognito\_user\_pool:

name: my-user-pool

state: present

#### **1389. How do you create Cognito app clients dynamically?**

- Example:

- amazon.aws.cognito\_user\_pool\_client:

name: my-app-client

user\_pool\_id: "{{ user\_pool\_id }}"

#### **1390. How do you enable MFA for Cognito users?**

- Example:

- amazon.aws.cognito\_user\_pool:

user\_pool\_id: "{{ user\_pool\_id }}"

mfa\_configuration: ON

#### **1391. How do you configure AWS ElasticSearch domains dynamically?**

- Example:

- amazon.aws.es\_domain:

domain\_name: my-es

elasticsearch\_version: 7.10

node\_count: 2

**1392. How do you configure access policies for ElasticSearch domains?**

- Example:

- amazon.aws.es\_domain:

domain\_name: my-es

access\_policies: "{{ lookup('file', 'policy.json') }}"

**1393. How do you create AWS EventBridge rules dynamically?**

- Example:

- amazon.aws.eventbridge\_rule:

name: my-rule

state: present

schedule\_expression: "rate(1 hour)"

**1394. How do you attach targets to EventBridge rules?**

- Example:

- amazon.aws.eventbridge\_target:

rule: my-rule

arn: "{{ lambda\_arn }}"

**1395. How do you create Amazon Kinesis streams dynamically?**

- Example:

- amazon.aws.kinesis\_stream:

name: my-stream

shard\_count: 2

**1396. How do you put records into Kinesis streams?**

- Example:

- amazon.aws.kinesis:

stream\_name: my-stream

partition\_key: "key1"

data: '{"message": "Hello"}'

**1397. How do you create Amazon Redshift clusters dynamically?**

- Example:

```
- amazon.aws.redshift_cluster:

 cluster_identifier: my-redshift

 node_type: dc2.large

 master_username: admin

 master_password: "{{ redshift_pass }}"

 number_of_nodes: 2
```

#### **1398. How do you delete Redshift clusters safely?**

- Example:

```
- amazon.aws.redshift_cluster:

 cluster_identifier: my-redshift

 state: absent
```

#### **1399. How do you configure AWS Data Pipeline dynamically?**

- Example:

```
- amazon.aws.data_pipeline:

 name: my-pipeline

 state: present
```

#### **1400. How do you activate AWS Data Pipeline tasks dynamically?**

- Example:

```
- amazon.aws.data_pipeline:

 pipeline_id: "{{ pipeline_id }}"

 state: active
```

---

#### **1401. How do you create multiple EC2 instances across different regions dynamically?**

- Example:

```
- amazon.aws.ec2_instance:

 region: "{{ item.region }}"

 instance_type: t2.micro

 image_id: "{{ item.ami }}"

 count: "{{ item.count }}"

loop:

- { region: us-east-1, ami: ami-123456, count: 2 }
```



- { region: us-west-2, ami: ami-789012, count: 3 }

**1402. How do you dynamically attach multiple IAM roles to EC2 instances?**

- Example:

- amazon.aws.ec2\_instance:

instance\_ids: "{{ ec2\_ids }}"

iam\_instance\_profile\_name: "{{ item }}"

loop: "{{ iam\_roles }}"

**1403. How do you create multiple S3 buckets dynamically with different ACLs?**

- Example:

- amazon.aws.s3\_bucket:

name: "{{ item.name }}"

acl: "{{ item.acl }}"

state: present

loop:

- { name: 'bucket1', acl: 'private' }

- { name: 'bucket2', acl: 'public-read' }

**1404. How do you configure cross-region replication for S3 buckets?**

- Example:

- amazon.aws.s3\_bucket:

name: source-bucket

replication:

role: "{{ replication\_role\_arn }}"

rules:

- destination\_bucket: dest-bucket

status: Enabled

**1405. How do you automate creation of multiple RDS instances in different AZs?**

- Example:

- amazon.aws.rds\_instance:

db\_instance\_identifier: "{{ item.id }}"

engine: mysql

db\_subnet\_group\_name: "{{ subnet\_group }}"

instance\_type: db.t3.micro

multi\_az: yes

state: present

loop:

- { id: 'db1' }

- { id: 'db2' }

#### **1406. How do you modify RDS instance parameters dynamically?**

- Example:

- amazon.aws.rds\_instance:

db\_instance\_identifier: mydb

allocated\_storage: 20

backup\_retention\_period: 7

#### **1407. How do you create multiple CloudWatch alarms using a loop?**

- Example:

- amazon.aws.cloudwatch\_alarm:

name: "{{ item.name }}"

metric: CPUUtilization

namespace: AWS/EC2

threshold: 70

comparison: ">="

evaluation\_periods: 2

period: 300

loop:

- { name: 'EC2-1-CPU' }

- { name: 'EC2-2-CPU' }

#### **1408. How do you enable CloudTrail logging for multiple regions?**

- Example:

- amazon.aws.cloudtrail:

name: multi-region-trail

s3\_bucket\_name: trail-logs

is\_multi\_region\_trail: yes

**1409. How do you manage multiple Elastic IPs and associate with EC2 instances?**

- Example:

- amazon.aws.ec2\_eip:

state: present

in\_vpc: yes

instance\_id: "{{ item }}"

loop: "{{ ec2\_ids }}"

**1410. How do you automate creation of multiple security groups dynamically?**

- Example:

- amazon.aws.ec2\_group:

name: "{{ item.name }}"

description: "{{ item.desc }}"

vpc\_id: "{{ vpc\_id }}"

rules: "{{ item.rules }}"

loop:

- { name: 'web-sg', desc: 'Web servers', rules: [{ proto: tcp, from\_port: 80, to\_port: 80, cidr\_ip: '0.0.0.0/0' }] }

- { name: 'db-sg', desc: 'DB servers', rules: [{ proto: tcp, from\_port: 3306, to\_port: 3306, cidr\_ip: '10.0.0.0/16' }] }

**1411. How do you dynamically attach multiple security groups to EC2 instances?**

- Example:

- amazon.aws.ec2\_instance:

instance\_ids: "{{ ec2\_ids }}"

group: "{{ item }}"

loop: "{{ sg\_list }}"

**1412. How do you configure multiple ELBs and attach EC2 instances?**

- Example:

- amazon.aws.elb\_application\_lb:

name: "{{ item.name }}"

subnets: "{{ subnets }}"

security\_groups: "{{ sg\_ids }}"

loop:

- { name: 'web-alb' }
- { name: 'app-alb' }

**1413. How do you dynamically create target groups for multiple ALBs?**

- Example:

- amazon.aws.elb\_target\_group:

name: "{{ item.name }}"

port: 80

protocol: HTTP

vpc\_id: "{{ vpc\_id }}"

loop:

- { name: 'web-tg' }
- { name: 'app-tg' }

**1414. How do you attach EC2 instances to multiple target groups dynamically?**

- Example:

- amazon.aws.elb\_target:

target\_group\_arn: "{{ item.tg\_arn }}"

targets: "{{ item.ec2\_ids }}"

loop: "{{ target\_groups }}"

**1415. How do you configure Route53 health checks dynamically?**

- Example:

- amazon.aws.route53\_health\_check:

state: present

fqdn: www.example.com

type: HTTP

resource\_path: /

**1416. How do you attach Route53 records to multiple ALBs?**

- Example:

- amazon.aws.route53:

zone: example.com

record: "{{ item.name }}"

type: A

alias:

name: "{{ item.alb\_dns }}"

zone\_id: "{{ item.zone\_id }}"

loop:

- { name: 'web', alb\_dns: web-alb.amazonaws.com, zone\_id: 'Z1' }

- { name: 'app', alb\_dns: app-alb.amazonaws.com, zone\_id: 'Z2' }

#### **1417. How do you automate S3 bucket policy attachments dynamically?**

- Example:

- amazon.aws.s3\_bucket\_policy:

bucket: "{{ item.name }}"

policy: "{{ item.policy }}"

loop: "{{ buckets }}"

#### **1418. How do you automate deployment of multiple Lambda functions?**

- Example:

- amazon.aws.lambda:

name: "{{ item.name }}"

state: present

runtime: python3.9

role: "{{ lambda\_role }}"

handler: lambda\_function.lambda\_handler

zip\_file: "{{ item.zip }}"

loop:

- { name: 'func1', zip: '/tmp/f1.zip' }

- { name: 'func2', zip: '/tmp/f2.zip' }

#### **1419. How do you invoke multiple Lambda functions dynamically?**

- Example:

- amazon.aws.lambda\_invoke:

name: "{{ item }}"

payload: '{"key": "value"}'

loop:

- func1

- func2

**1420. How do you manage multiple SNS topics dynamically?**

- Example:

- amazon.aws.sns\_topic:

name: "{{ item }}"

state: present

loop:

- deploy-notifications

- alert-notifications

**1421. How do you publish messages to multiple SNS topics dynamically?**

- Example:

- amazon.aws.sns:

topic: "{{ item.topic }}"

msg: "{{ item.message }}"

loop:

- { topic: 'deploy-notifications', message: 'Deployment Done' }

- { topic: 'alert-notifications', message: 'CPU High' }

**1422. How do you automate creation of multiple CloudFormation stacks dynamically?**

- Example:

- amazon.aws.cloudformation:

stack\_name: "{{ item.name }}"

state: present

template: "{{ item.template }}"

parameters: "{{ item.params }}"

loop:

- { name: 'stack1', template: '/tmp/t1.yaml', params: { InstanceType: t2.micro } }

- { name: 'stack2', template: '/tmp/t2.yaml', params: { InstanceType: t2.small } }

**1423. How do you delete multiple CloudFormation stacks dynamically?**

- Example:

- amazon.aws.cloudformation:

stack\_name: "{{ item }}"

state: absent

loop:

- stack1

- stack2

#### **1424. How do you automate DynamoDB table creation with different throughput settings?**

- Example:

- amazon.aws.dynamodb\_table:

name: "{{ item.name }}"

hash\_key\_name: id

hash\_key\_type: S

read\_capacity: "{{ item.read }}"

write\_capacity: "{{ item.write }}"

loop:

- { name: 'table1', read: 5, write: 5 }

- { name: 'table2', read: 10, write: 10 }

#### **1425. How do you insert multiple items into DynamoDB tables dynamically?**

- Example:

- amazon.aws.dynamodb:

table\_name: "{{ item.table }}"

hash\_key: id

item: "{{ item.data }}"

loop:

- { table: 'table1', data: { id: '1', name: 'A' } }

- { table: 'table2', data: { id: '2', name: 'B' } }

#### **1426. How do you create multiple SQS queues dynamically?**

- Example:

- amazon.aws.sqs\_queue:

name: "{{ item }}"

state: present

loop:

- queue1

- queue2

**1427. How do you send messages to multiple SQS queues dynamically?**

- Example:

- amazon.aws.sqs\_message:

queue\_url: "{{ item.queue }}"

message\_body: "{{ item.msg }}"

loop:

- { queue: 'queue1', msg: 'Hello Q1' }

- { queue: 'queue2', msg: 'Hello Q2' }

**1428. How do you receive messages from multiple SQS queues dynamically?**

- Example:

- amazon.aws.sqs\_message:

queue\_url: "{{ item }}"

wait\_time\_seconds: 5

loop:

- queue1

- queue2

**1429. How do you manage multiple Lambda environment variables dynamically?**

- Example:

- amazon.aws.lambda:

name: my-function

environment\_variables:

ENV: prod

DEBUG: 'false'

**1430. How do you manage multiple Cognito user pools dynamically?**

- Example:

- amazon.aws.cognito\_user\_pool:

name: "{{ item }}"

state: present

loop:

- pool1



- pool2

**1431. How do you attach multiple app clients to Cognito pools dynamically?**

- Example:

- amazon.aws.cognito\_user\_pool\_client:

name: "{{ item.client }}"

user\_pool\_id: "{{ item.pool\_id }}"

loop:

- { client: 'app1', pool\_id: pool1\_id }

- { client: 'app2', pool\_id: pool2\_id }

**1432. How do you enable MFA for multiple Cognito user pools?**

- Example:

- amazon.aws.cognito\_user\_pool:

user\_pool\_id: "{{ item }}"

mfa\_configuration: ON

loop:

- pool1\_id

- pool2\_id

**1433. How do you automate creation of multiple Elasticsearch domains?**

- Example:

- amazon.aws.es\_domain:

domain\_name: "{{ item }}"

elasticsearch\_version: 7.10

node\_count: 2

loop:

- es1

- es2

**1434. How do you configure access policies for multiple Elasticsearch domains?**

- Example:

- amazon.aws.es\_domain:

domain\_name: "{{ item.name }}"

access\_policies: "{{ item.policy }}"

loop:

- { name: 'es1', policy: '{{ policy1 }}' }
- { name: 'es2', policy: '{{ policy2 }}' }

#### **1435. How do you automate EventBridge rule creation for multiple events?**

- Example:

- amazon.aws.eventbridge\_rule:

```
name: "{{ item.name }}"
state: present
schedule_expression: "{{ item.schedule }}"
```

loop:

- { name: 'rule1', schedule: 'rate(1 hour)' }
- { name: 'rule2', schedule: 'rate(2 hours)' }

#### **1436. How do you attach multiple Lambda targets to EventBridge rules dynamically?**

- Example:

- amazon.aws.eventbridge\_target:

```
rule: "{{ item.rule }}"
arn: "{{ item.arn }}"
```

loop:

- { rule: 'rule1', arn: lambda1\_arn }
- { rule: 'rule2', arn: lambda2\_arn }

#### **1437. How do you automate Kinesis stream creation for multiple applications?**

- Example:

- amazon.aws.kinesis\_stream:

```
name: "{{ item.name }}"
shard_count: "{{ item.shards }}"
```

loop:

- { name: 'stream1', shards: 2 }
- { name: 'stream2', shards: 3 }

#### **1438. How do you send multiple records to different Kinesis streams dynamically?**

- Example:

- amazon.aws.kinesis:

```
stream_name: "{{ item.stream }}"
```

```
partition_key: "key1"
```

```
data: "{{ item.data }}"
```

```
loop:
```

```
- { stream: 'stream1', data: '{"msg":"Hello1"}' }
```

```
- { stream: 'stream2', data: '{"msg":"Hello2"}' }
```

#### **1439. How do you manage multiple Redshift clusters dynamically?**

- Example:

```
- amazon.aws.redshift_cluster:
```

```
cluster_identifier: "{{ item.name }}"
```

```
node_type: dc2.large
```

```
master_username: admin
```

```
master_password: "{{ redshift_pass }}"
```

```
number_of_nodes: 2
```

```
loop:
```

```
- { name: 'rs1' }
```

```
- { name: 'rs2' }
```

#### **1440. How do you delete multiple Redshift clusters safely?**

- Example:

```
- amazon.aws.redshift_cluster:
```

```
cluster_identifier: "{{ item }}"
```

```
state: absent
```

```
loop:
```

```
- rs1
```

```
- rs2
```

#### **1441. How do you automate Glue crawler creation for multiple data sources?**

- Example:

```
- amazon.aws.glue_crawler:
```

```
name: "{{ item.name }}"
```

```
role: "{{ glue_role }}"
```

```
database_name: "{{ item.db }}"
```

```

targets:
 s3_targets:
 - path: "{{ item.s3_path }}"
loop:
 - { name: 'crawler1', db: 'db1', s3_path: 's3://bucket1/' }
 - { name: 'crawler2', db: 'db2', s3_path: 's3://bucket2/' }

```

#### 1442. How do you start multiple Glue crawlers dynamically?

- Example:

```

- amazon.aws.glue_crawler:
 name: "{{ item }}"
 state: started
loop:
 - crawler1
 - crawler2

```

#### 1443. How do you stop multiple Glue crawlers dynamically?

- Example:

```

- amazon.aws.glue_crawler:
 name: "{{ item }}"
 state: stopped
loop:
 - crawler1
 - crawler2

```

#### 1444. How do you automate creation of multiple Step Functions workflows?

- Example:

```

- amazon.aws.stepfunctions_state_machine:
 name: "{{ item.name }}"
 definition: "{{ item.definition }}"
 state: present
loop:
 - { name: 'workflow1', definition: '{{ lookup("file","wf1.json") }}' }
 - { name: 'workflow2', definition: '{{ lookup("file","wf2.json") }}' }

```

#### 1445. How do you start multiple Step Functions executions dynamically?

- Example:

- amazon.aws.stepfunctions\_execution:

state\_machine\_arn: "{{ item.arn }}"

input: "{{ item.input }}"

loop:

- { arn: wf1\_arn, input: '{"key": "val1"}' }

- { arn: wf2\_arn, input: '{"key": "val2"}' }

#### 1446. How do you terminate multiple Step Functions state machines?

- Example:

- amazon.aws.stepfunctions\_state\_machine:

state\_machine\_arn: "{{ item }}"

state: absent

loop:

- wf1\_arn

- wf2\_arn

#### 1447. How do you automate management of multiple AWS MQ brokers?

- Example:

- amazon.aws.mq\_broker:

broker\_name: "{{ item.name }}"

engine\_type: ActiveMQ

host\_instance\_type: mq.t3.micro

users:

- username: admin

password: "{{ mq\_password }}"

loop:

- { name: 'broker1' }

- { name: 'broker2' }

#### 1448. How do you start multiple MQ brokers dynamically?

- Example:

- amazon.aws.mq\_broker:

```
broker_name: "{{ item }}"
```

```
state: started
```

```
loop:
```

```
- broker1
```

```
- broker2
```

#### **1449. How do you stop multiple MQ brokers dynamically?**

- Example:

```
- amazon.aws.mq_broker:
```

```
broker_name: "{{ item }}"
```

```
state: stopped
```

```
loop:
```

```
- broker1
```

```
- broker2
```

#### **1450. How do you delete multiple MQ brokers dynamically?**

- Example:

```
- amazon.aws.mq_broker:
```

```
broker_name: "{{
```

Here's **Complex Ansible Q&A 1451–1500** for AWS DevOps:

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#### **1451. How do you create multiple AWS VPCs dynamically with different CIDRs?**

- Example:

```
- amazon.aws.ec2_vpc:
```

```
cidr_block: "{{ item.cidr }}"
```

```
state: present
```

```
tags:
```

```
 Name: "{{ item.name }}"
```

```
loop:
```

```
- { cidr: '10.0.0.0/16', name: 'VPC1' }
```

```
- { cidr: '10.1.0.0/16', name: 'VPC2' }
```

#### **1452. How do you create multiple subnets across VPCs dynamically?**

- Example:

```
- amazon.aws.ec2_vpc_subnet:
 vpc_id: "{{ item.vpc_id }}"
 cidr: "{{ item.cidr }}"
 availability_zone: "{{ item.az }}"
 state: present
loop:
 - { vpc_id: vpc1, cidr: '10.0.1.0/24', az: 'us-east-1a' }
 - { vpc_id: vpc2, cidr: '10.1.1.0/24', az: 'us-east-1b' }
```

#### 1453. How do you attach Internet Gateways to multiple VPCs dynamically?

- Example:

```
- amazon.aws.ec2_vpc_igw:
 vpc_id: "{{ item.vpc_id }}"
 state: present
loop:
 - { vpc_id: vpc1 }
 - { vpc_id: vpc2 }
```

#### 1454. How do you configure multiple route tables and associate with subnets dynamically?

- Example:

```
- amazon.aws.ec2_vpc_route_table:
 vpc_id: "{{ item.vpc_id }}"
 routes:
 - dest: 0.0.0.0/0
 gateway_id: "{{ item.igw_id }}"
 subnets:
 - "{{ item.subnet_id }}"
loop:
 - { vpc_id: vpc1, igw_id: igw1, subnet_id: subnet1 }
 - { vpc_id: vpc2, igw_id: igw2, subnet_id: subnet2 }
```

#### 1455. How do you configure multiple NAT gateways dynamically for private subnets?

- Example:

```
- amazon.aws.ec2_vpc_nat_gateway:
```

```
subnet_id: "{{ item.subnet_id }}"
```

```
state: present
```

```
loop:
```

```
- { subnet_id: public_subnet1 }
```

```
- { subnet_id: public_subnet2 }
```

#### **1456. How do you dynamically assign Elastic IPs to multiple NAT gateways?**

- Example:

```
- amazon.aws.ec2_eip:
```

```
in_vpc: yes
```

```
state: present
```

```
instance_id: "{{ item.nat_id }}"
```

```
loop:
```

```
- { nat_id: nat1 }
```

```
- { nat_id: nat2 }
```

#### **1457. How do you create multiple security groups dynamically with rules?**

- Example:

```
- amazon.aws.ec2_group:
```

```
name: "{{ item.name }}"
```

```
description: "{{ item.desc }}"
```

```
vpc_id: "{{ item.vpc_id }}"
```

```
rules: "{{ item.rules }}"
```

```
loop:
```

```
- { name: 'web-sg', desc: 'Web servers', vpc_id: vpc1, rules: [{ proto: tcp, from_port: 80, to_port: 80, cidr_ip: '0.0.0.0/0' }] }
```

```
- { name: 'db-sg', desc: 'DB servers', vpc_id: vpc2, rules: [{ proto: tcp, from_port: 3306, to_port: 3306, cidr_ip: '10.0.0.0/16' }] }
```

#### **1458. How do you attach multiple security groups to EC2 instances dynamically?**

- Example:

```
- amazon.aws.ec2_instance:
```

```
instance_ids: "{{ item.ec2_id }}"
```

```
group: "{{ item.sg_ids }}"
```

```
loop:
```



- { ec2\_id: ec2\_1, sg\_ids: [web\_sg, db\_sg] }

- { ec2\_id: ec2\_2, sg\_ids: [web\_sg] }

#### **1459. How do you create multiple Auto Scaling Groups dynamically?**

- Example:

- amazon.aws.autoscaling\_group:

name: "{{ item.name }}"

launch\_template: "{{ item.launch\_template }}"

min\_size: "{{ item.min }}"

max\_size: "{{ item.max }}"

desired\_capacity: "{{ item.desired }}"

loop:

- { name: 'asg1', launch\_template: lt1, min: 2, max: 5, desired: 3 }

- { name: 'asg2', launch\_template: lt2, min: 1, max: 4, desired: 2 }

#### **1460. How do you attach multiple EC2 instances to ASGs dynamically?**

- Example:

- amazon.aws.autoscaling\_group:

name: "{{ item.name }}"

instance\_ids: "{{ item.ec2\_ids }}"

loop:

- { name: 'asg1', ec2\_ids: [ec2\_1, ec2\_2] }

- { name: 'asg2', ec2\_ids: [ec2\_3] }

#### **1461. How do you configure multiple CloudWatch alarms for scaling policies?**

- Example:

- amazon.aws.cloudwatch\_metric\_alarm:

name: "{{ item.name }}"

metric: CPUUtilization

namespace: AWS/EC2

threshold: "{{ item.threshold }}"

comparison: ">="

evaluation\_periods: 2

period: 300

```
alarm_actions: "{{ item.actions }}"
```

```
loop:
```

```
- { name: 'cpu-high-1', threshold: 80, actions: [scale_up_policy_1] }
```

```
- { name: 'cpu-high-2', threshold: 75, actions: [scale_up_policy_2] }
```

#### **1462. How do you dynamically launch multiple EC2 instances using launch templates?**

- Example:

```
- amazon.aws.ec2_instance:
```

```
 launch_template:
```

```
 name: "{{ item.template }}"
```

```
 count: "{{ item.count }}"
```

```
 wait: yes
```

```
loop:
```

```
- { template: lt1, count: 2 }
```

```
- { template: lt2, count: 3 }
```

#### **1463. How do you dynamically create multiple RDS instances across AZs?**

- Example:

```
- amazon.aws.rds_instance:
```

```
 db_instance_identifier: "{{ item.name }}"
```

```
 engine: mysql
```

```
 db_subnet_group_name: "{{ subnet_group }}"
```

```
 instance_type: db.t3.micro
```

```
 multi_az: yes
```

```
 state: present
```

```
loop:
```

```
- { name: 'db1' }
```

```
- { name: 'db2' }
```

#### **1464. How do you modify multiple RDS instance configurations dynamically?**

- Example:

```
- amazon.aws.rds_instance:
```

```
 db_instance_identifier: "{{ item.name }}"
```

```
 allocated_storage: "{{ item.storage }}"
```

```
backup_retention_period: "{{ item.backup }}"
```

```
loop:
```

```
- { name: 'db1', storage: 20, backup: 7 }
```

```
- { name: 'db2', storage: 25, backup: 10 }
```

**1465. How do you create multiple S3 buckets with versioning enabled?**

- Example:

```
- amazon.aws.s3_bucket:
```

```
name: "{{ item.name }}"
```

```
versioning: yes
```

```
state: present
```

```
loop:
```

```
- { name: 'bucket1' }
```

```
- { name: 'bucket2' }
```

**1466. How do you apply bucket policies to multiple S3 buckets dynamically?**

- Example:

```
- amazon.aws.s3_bucket_policy:
```

```
bucket: "{{ item.name }}"
```

```
policy: "{{ item.policy }}"
```

```
loop:
```

```
- { name: 'bucket1', policy: policy1 }
```

```
- { name: 'bucket2', policy: policy2 }
```

**1467. How do you create multiple CloudTrail trails dynamically across regions?**

- Example:

```
- amazon.aws.cloudtrail:
```

```
name: "{{ item.name }}"
```

```
s3_bucket_name: "{{ item.bucket }}"
```

```
is_multi_region_trail: yes
```

```
loop:
```

```
- { name: 'trail1', bucket: 'logs1' }
```

```
- { name: 'trail2', bucket: 'logs2' }
```

**1468. How do you automate multiple Lambda function deployments?**

- Example:

- amazon.aws.lambda:

name: "{{ item.name }}"

runtime: python3.9

role: "{{ lambda\_role }}"

handler: lambda\_function.lambda\_handler

zip\_file: "{{ item.zip }}"

loop:

- { name: 'func1', zip: '/tmp/f1.zip' }

- { name: 'func2', zip: '/tmp/f2.zip' }

#### 1469. How do you update multiple Lambda environment variables dynamically?

- Example:

- amazon.aws.lambda:

name: "{{ item.name }}"

environment\_variables: "{{ item.env }}"

loop:

- { name: 'func1', env: { ENV: 'prod', DEBUG: 'false' } }

- { name: 'func2', env: { ENV: 'dev', DEBUG: 'true' } }

#### 1470. How do you invoke multiple Lambda functions with different payloads?

- Example:

- amazon.aws.lambda\_invoke:

name: "{{ item.name }}"

payload: "{{ item.payload }}"

loop:

- { name: 'func1', payload: '{"key": "val1"}' }

- { name: 'func2', payload: '{"key": "val2"}' }

#### 1471. How do you create multiple SNS topics dynamically?

- Example:

- amazon.aws.sns\_topic:

name: "{{ item }}"

state: present

loop:

- topic1
- topic2

**1472. How do you publish messages to multiple SNS topics?**

- Example:

- amazon.aws.sns:

topic: "{{ item.topic }}"

msg: "{{ item.message }}"

loop:

- { topic: 'topic1', message: 'Deploy done' }
- { topic: 'topic2', message: 'Alert triggered' }

**1473. How do you create multiple SQS queues dynamically?**

- Example:

- amazon.aws.sqs\_queue:

name: "{{ item }}"

state: present

loop:

- queue1
- queue2

**1474. How do you send messages to multiple SQS queues dynamically?**

- Example:

- amazon.aws.sqs\_message:

queue\_url: "{{ item.queue }}"

message\_body: "{{ item.msg }}"

loop:

- { queue: 'queue1', msg: 'Hello Q1' }
- { queue: 'queue2', msg: 'Hello Q2' }

**1475. How do you receive messages from multiple SQS queues dynamically?**

- Example:

- amazon.aws.sqs\_message:

queue\_url: "{{ item }}"

wait\_time\_seconds: 5

loop:

- queue1
- queue2

#### **1476. How do you create multiple CloudFormation stacks dynamically?**

- Example:

- amazon.aws.cloudformation:

stack\_name: "{{ item.name }}"

template: "{{ item.template }}"

parameters: "{{ item.params }}"

state: present

loop:

- { name: 'stack1', template: '/tmp/t1.yaml', params: { InstanceType: t2.micro } }
- { name: 'stack2', template: '/tmp/t2.yaml', params: { InstanceType: t2.small } }

#### **1477. How do you delete multiple CloudFormation stacks dynamically?**

- Example:

- amazon.aws.cloudformation:

stack\_name: "{{ item }}"

state: absent

loop:

- stack1
- stack2

#### **1478. How do you create multiple DynamoDB tables dynamically?**

- Example:

- amazon.aws.dynamodb\_table:

name: "{{ item.name }}"

hash\_key\_name: id

hash\_key\_type: S

read\_capacity: "{{ item.read }}"

write\_capacity: "{{ item.write }}"

loop:

- { name: 'table1', read: 5, write: 5 }
- { name: 'table2', read: 10, write: 10 }

#### **1479. How do you insert multiple items into DynamoDB tables dynamically?**

- Example:

- amazon.aws.dynamodb:

table\_name: "{{ item.table }}"

hash\_key: id

item: "{{ item.data }}"

loop:

- { table: 'table1', data: { id: '1', name: 'A' } }
- { table: 'table2', data: { id: '2', name: 'B' } }

#### **1480. How do you create multiple Kinesis streams dynamically?**

- Example:

- amazon.aws.kinesis\_stream:

name: "{{ item.name }}"

shard\_count: "{{ item.shards }}"

loop:

- { name: 'stream1', shards: 2 }
- { name: 'stream2', shards: 3 }

#### **1481. How do you send records to multiple Kinesis streams dynamically?**

- Example:

- amazon.aws.kinesis:

stream\_name: "{{ item.stream }}"

partition\_key: "key1"

data: "{{ item.data }}"

loop:

- { stream: 'stream1', data: '{"msg": "Hello1"}' }

- { stream: 'stream2', data: '{"msg": "Hello2"}' }

**\*\*1482. How do you create multiple Redshift clusters dynamically?\*\***

- Example:

```
``yaml
```

- amazon.aws.redshift\_cluster:

```
cluster_identifier: "{{ item.name }}"
```

```
node_type: dc2.large
```

```
master_username: admin
```

```
master_password: "{{ redshift_pass }}"
```

```
number_of_nodes: 2
```

loop:

```
- { name: 'rs1' }
```

```
- { name: 'rs2' }
```

#### **1483. How do you delete multiple Redshift clusters dynamically?**

- Example:

- amazon.aws.redshift\_cluster:

```
cluster_identifier: "{{ item }}"
```

```
state: absent
```

loop:

```
- rs1
```

```
- rs2
```

#### **1484. How do you create multiple Glue crawlers dynamically?**

- Example:

- amazon.aws.glue\_crawler:

```
name: "{{ item.name }}"
```

```
role: "{{ glue_role }}"
```

```
database_name: "{{ item.db }}"
```

```
targets:
```

```
 s3_targets:
```

```
 - path: "{{ item.s3_path }}"
```

loop:

```
- { name: 'crawler1', db: 'db1', s3_path: 's3://bucket1/' }
```

```
- { name: 'crawler2', db: 'db2', s3_path: 's3://bucket2/' }
```



#### 1485. How do you start multiple Glue crawlers dynamically?

- Example:

- amazon.aws.glue\_crawler:

name: "{{ item }}"

state: started

loop:

- crawler1

- crawler2

#### 1486. How do you stop multiple Glue crawlers dynamically?

- Example:

- amazon.aws.glue\_crawler:

name: "{{ item }}"

state: stopped

loop:

- crawler1

- crawler2

#### 1487. How do you create multiple Step Functions workflows dynamically?

- Example:

- amazon.aws.stepfunctions\_state\_machine:

name: "{{ item.name }}"

definition: "{{ item.definition }}"

state: present

loop:

- { name: 'workflow1', definition: '{{ lookup("file","wf1.json") }}' }

- { name: 'workflow2', definition: '{{ lookup("file","wf2.json") }}' }

#### 1488. How do you start multiple Step Functions executions dynamically?

- Example:

- amazon.aws.stepfunctions\_execution:

state\_machine\_arn: "{{ item.arn }}"

input: "{{ item.input }}"

loop:

- { arn: wf1\_arn, input: '{"key": "val1"}' }

- { arn: wf2\_arn, input: '{"key": "val2"}' }

#### 1489. How do you terminate multiple Step Functions state machines?

- Example:

- amazon.aws.stepfunctions\_state\_machine:

state\_machine\_arn: "{{ item }}"

state: absent

loop:

- wf1\_arn

- wf2\_arn

#### 1490. How do you create multiple AWS MQ brokers dynamically?

- Example:

- amazon.aws.mq\_broker:

broker\_name: "{{ item.name }}"

engine\_type: ActiveMQ

host\_instance\_type: mq.t3.micro

users:

- username: admin

password: "{{ mq\_password }}"

loop:

- { name: 'broker1' }

- { name: 'broker2' }

#### 1491. How do you start multiple MQ brokers dynamically?

- Example:

- amazon.aws.mq\_broker:

broker\_name: "{{ item }}"

state: started

loop:

- broker1

- broker2

#### 1492. How do you stop multiple MQ brokers dynamically?

- Example:

- amazon.aws.mq\_broker:

broker\_name: "{{ item }}"

state: stopped

loop:

- broker1

- broker2

#### **1493. How do you delete multiple MQ brokers dynamically?**

- Example:

- amazon.aws.mq\_broker:

broker\_name: "{{ item }}"

state: absent

loop:

- broker1

- broker2

#### **1494. How do you automate creation of multiple Elasticsearch domains?**

- Example:

- amazon.aws.es\_domain:

domain\_name: "{{ item.name }}"

elasticsearch\_version: 7.10

node\_count: 2

loop:

- es1

- es2

#### **1495. How do you configure access policies for multiple Elasticsearch domains?**

- Example:

- amazon.aws.es\_domain:

domain\_name: "{{ item.name }}"

access\_policies: "{{ item.policy }}"

loop:

- { name: 'es1', policy: '{{ policy1 }}' }

- { name: 'es2', policy: '{{ policy2 }}' }

**1496. How do you create multiple EventBridge rules dynamically?**

- Example:

- amazon.aws.eventbridge\_rule:

name: "{{ item.name }}"

state: present

schedule\_expression: "{{ item.schedule }}"

loop:

- { name: 'rule1', schedule: 'rate(1 hour)' }

- { name: 'rule2', schedule: 'rate(2 hours)' }

**1497. How do you attach multiple Lambda targets to EventBridge rules dynamically?**

- Example:

- amazon.aws.eventbridge\_target:

rule: "{{ item.rule }}"

arn: "{{ item.arn }}"

loop:

- { rule: 'rule1', arn: lambda1\_arn }

- { rule: 'rule2', arn: lambda2\_arn }

**1498. How do you automate creation of multiple Kinesis Firehose delivery streams?**

- Example:

- amazon.aws.kinesis\_firehose:

name: "{{ item.name }}"

s3\_destination: "{{ item.bucket }}"

state: present

loop:

- { name: 'fh1', bucket: 'bucket1' }

- { name: 'fh2', bucket: 'bucket2' }

**1499. How do you start multiple Firehose delivery streams?**

- Example:

- amazon.aws.kinesis\_firehose:

name: "{{ item }}"

```
state: started
```

```
loop:
```

```
- fh1
```

```
- fh2
```

#### **1500. How do you stop multiple Firehose delivery streams?**

- Example:

```
- amazon.aws.kinesis_firehose:
```

```
name: "{{ item }}"
```

```
state: stopped
```

```
loop:
```

```
- fh1
```

```
- fh2
```

---

### **Max Complex Ansible Q&A 1501–2000 for AWS DevOps:**

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#### **1501. How do you design a fully automated multi-region VPC setup with Ansible?**

- Example: Use loops with `amazon.aws.ec2_vpc` for each region dynamically:

```
- name: Create VPCs in multiple regions
```

```
amazon.aws.ec2_vpc:
```

```
cidr_block: "{{ item.cidr }}"
```

```
region: "{{ item.region }}"
```

```
state: present
```

```
loop:
```

```
- { cidr: '10.0.0.0/16', region: 'us-east-1' }
```

```
- { cidr: '10.1.0.0/16', region: 'us-west-2' }
```

#### **1502. How do you automate multi-region subnets, route tables, and IGWs with Ansible?**

- Combine `ec2_vpc_subnet`, `ec2_vpc_route_table`, and `ec2_vpc_igw` modules with loops for each region.

#### **1503. How do you create multi-region NAT gateways and assign Elastic IPs dynamically?**

- Example: Use `ec2_vpc_nat_gateway` with `ec2_eip` and region-specific variables.

#### **1504. How do you deploy a multi-region Auto Scaling architecture with Ansible?**

- Example: Use `autoscaling_group` module with different `launch_templates` for each region and configure CloudWatch alarms dynamically.

**1505. How do you orchestrate multi-region RDS deployment with read replicas?**

- Use `rds_instance` module with `multi_az: yes` and create cross-region read replicas using `replicate_source_db`.

**1506. How do you dynamically configure VPC peering between regions?**

- Use `ec2_vpc_peering` module, loop over VPC IDs from different regions, and handle route table updates dynamically.

**1507. How do you automate cross-region S3 replication with Ansible?**

- Use `s3_bucket` module with `replication_configuration` and loop over source and destination buckets in different regions.

**1508. How do you configure multiple CloudFront distributions with dynamic origins?**

- Use `cloudfront_distribution` module with loops to create distributions per S3 bucket or ALB dynamically.

**1509. How do you deploy multi-region Lambda functions with dynamic environment variables?**

- Use `lambda` module, loop over regions, and dynamically set environment variables per region.

**1510. How do you orchestrate multi-region Step Functions workflows dynamically?**

- Use `stepfunctions_state_machine` module for each region and dynamically deploy JSON definitions per workflow.

**1511. How do you create multiple EventBridge rules with cross-region Lambda targets?**

- Use `eventbridge_rule` and `eventbridge_target` modules with loops over regions and Lambda ARNs.

**1512. How do you deploy multi-region Kinesis streams with dynamic shard allocation?**

- Use `kinesis_stream` module and dynamically assign `shard_count` based on region traffic requirements.

**1513. How do you deploy multi-region Firehose delivery streams with dynamic S3 destinations?**

- Use `kinesis_firehose` module, loop over buckets in each region, and start streams dynamically.

**1514. How do you automate CloudTrail logging for multi-region audit compliance?**

- Use `cloudtrail` module with `is_multi_region_trail: yes` and enable S3 logging dynamically per region.

**1515. How do you orchestrate multiple Elasticsearch domains in multi-region architecture?**

- Use `es_domain` module with loops, region-specific node counts, and access policies.

**1516. How do you dynamically deploy multiple Redshift clusters with multi-AZ configuration?**

- Use `redshift_cluster` module with region-specific parameters and automated deletion policies.

**1517. How do you create multi-region MQ brokers dynamically?**

- Use `mq_broker` module and loop over regions with dynamic usernames and passwords.

**1518. How do you dynamically update multiple Lambda functions across regions with zero downtime?**

- Use `lambda` module, versioning, and aliases for blue-green deployment automation.

**1519. How do you implement cross-region DynamoDB table replication with Ansible?**

- Use `dynamodb_table` module with `replica_regions` and automate item insertions per region.

**1520. How do you dynamically orchestrate multi-region Glue crawlers and jobs?**

- Use `glue_crawler` and `glue_job` modules, loop over regions, and schedule jobs dynamically.

**1521. How do you manage multi-region IAM roles and policies with Ansible?**

- Use `iam_role` and `iam_policy` modules with loops over regions, attach policies dynamically, and enforce least privilege.

**1522. How do you dynamically configure AWS Config rules across multiple accounts and regions?**

- Use `aws_config_rule` module, loop over accounts and regions, and enforce compliance.

**1523. How do you automate multi-region CodePipeline deployment with Ansible?**

- Use `codepipeline` module, loop over regions, and integrate CodeBuild/CodeDeploy dynamically.

**1524. How do you orchestrate multi-region CloudFormation stack sets dynamically?**

- Use `cloudformation` module with `stack_set` and loop over target regions for deployment.

**1525. How do you create multi-region SSM parameter store entries dynamically?**

- Use `ssm_parameter_store` module with loops over regions and dynamic values.

**1526. How do you deploy multi-region Elastic Beanstalk environments dynamically?**

- Use `elasticbeanstalk` module, loop over regions, and dynamically configure environment variables and scaling.

**1527. How do you dynamically configure Route53 hosted zones and cross-region failover?**

- Use `route53_zone` and `route53_record` modules with health checks and dynamic failover policies.

**1528. How do you automate multi-region ACM certificate provisioning and deployment?**

- Use `acm_certificate` module with DNS validation and dynamic region assignment.

**1529. How do you orchestrate multi-region SES email sending domains with DKIM?**

- Use `ses_domain_identity` and `ses_domain_dkim` modules, loop over regions and domains.

**1530. How do you deploy multi-region CloudWatch dashboards dynamically?**

- Use `cloudwatch_dashboard` module with region-specific metrics dynamically.

**1531. How do you automate multi-region AWS Batch job queues and compute environments?**

- Use `batch_compute_environment` and `batch_job_queue` modules with dynamic region configuration.

**1532. How do you deploy multi-region VPC endpoints dynamically?**

- Use `ec2_vpc_endpoint` module with loops over service names and regions.

**1533. How do you manage multi-region Secrets Manager secrets dynamically?**

- Use `secretsmanager_secret` and `secretsmanager_secret_version` modules with region loops.

**1534. How do you deploy multi-region API Gateway endpoints dynamically?**

- Use `apigateway_rest_api` and `apigateway_deployment` modules with loops over regions.

**1535. How do you automate multi-region CodeBuild projects dynamically?**

- Use `codebuild_project` module with loops for different build environments per region.

**1536. How do you orchestrate multi-region CodeDeploy applications and deployment groups?**

- Use `codedeploy_app` and `codedeploy_deployment_group` modules with dynamic region deployment.

**1537. How do you create multi-region ElastiCache clusters dynamically?**

- Use `elasticache` module with loops over regions, node types, and cluster configurations.

**1538. How do you orchestrate multi-region VPC flow logs dynamically?**

- Use `ec2_vpc_flow_log` module and loop over VPC IDs and regions.

**1539. How do you deploy multi-region FSx file systems dynamically?**

- Use `fsx` module with loops over regions, storage types, and throughput.

**1540. How do you dynamically configure AWS Backup plans across multiple regions?**

- Use `backup_plan` and `backup_selection` modules with loops for each region.

**1541. How do you orchestrate multi-region ECR repositories dynamically?**

- Use `ecr_repository` module with loops for different regions and image tagging policies.

**1542. How do you automate multi-region ECS clusters and services?**

- Use `ecs_cluster` and `ecs_service` modules with dynamic region and scaling configuration.

**1543. How do you deploy multi-region EKS clusters dynamically?**

- Use `eks_cluster` module with loops for different regions, node groups, and scaling policies.

**1544. How do you dynamically configure CloudWatch Logs groups and subscriptions across regions?**



- Use `cloudwatch_log_group` and `cloudwatch_log_subscription_filter` modules with region loops.

**1545. How do you orchestrate multi-region VPC endpoints for S3 and DynamoDB?**

- Use `ec2_vpc_endpoint` module with service names `s3` and `dynamodb` per region.

**1546. How do you automate multi-region Elastic Load Balancers dynamically?**

- Use `elb_application_lb` or `elb_network_lb` modules with dynamic subnets per region.

**1547. How do you deploy multi-region NAT gateways with high availability dynamically?**

- Use `ec2_vpc_nat_gateway` and `ec2_eip` modules, loop over public subnets per region.

**1548. How do you orchestrate multi-region CloudFront invalidations dynamically?**

- Use `cloudfront_distribution` module with `invalidation_batch` loops for multiple distributions.

**1549. How do you deploy multi-region AWS WAF rules dynamically?**

- Use `waf_rule` and `waf_web_acl` modules with loops for multiple regions.

**1550. How do you implement a full multi-region disaster recovery architecture with Ansible?**

- Combine modules: `ec2_vpc`, `ec2_vpc_subnet`, `ec2_instance`, `rds_instance`, `s3_bucket`, `cloudfront_distribution`, `route53_record` dynamically with region loops and failover policies.
- 

**1551. How do you automate multi-region CloudFront distributions with Lambda@Edge functions?**

- Example: Use `cloudfront_distribution` module with `lambda_function_association` and loop over multiple distributions and regions.

**1552. How do you orchestrate multi-region S3 replication with KMS encryption?**

- Use `s3_bucket` with `replication_configuration` and `kms_master_key_id` dynamically for each source and destination bucket.

**1553. How do you deploy multi-region RDS Aurora clusters with failover using Ansible?**

- Use `rds_instance` module with `engine: aurora`, `multi_az: yes`, and configure cross-region read replicas.

**1554. How do you dynamically create multi-region VPC peering connections with route updates?**

- Use `ec2_vpc_peering` and `ec2_vpc_route_table` modules with region-specific variables and loops.

**1555. How do you configure multi-region CloudWatch Logs to central logging account dynamically?**

- Use `cloudwatch_log_group` and `cloudwatch_log_subscription_filter` with `destination_arn` pointing to central logging account per region.

**1556. How do you orchestrate multi-region Lambda function deployment with different memory and timeout settings?**

- Use `lambda` module with `memory_size` and `timeout` set per region dynamically using loop.

**1557. How do you automate multi-region API Gateway deployments with stage variables?**

- Use `apigateway_rest_api` and `apigateway_stage` modules, loop over regions, configure stage variables per region.

**1558. How do you deploy multi-region Elastic Beanstalk environments with blue/green deployment?**

- Use `elasticbeanstalk` module with `version_label` and `environment_name` loops per region for zero-downtime deployment.

**1559. How do you automate multi-region CodePipeline with cross-region CodeBuild projects?**

- Use `codepipeline` module, integrate `codebuild_project` in multiple regions dynamically using loops.

**1560. How do you orchestrate multi-region DynamoDB global tables with Ansible?**

- Use `dynamodb_table` module with `replica_regions` dynamically configured for each region.

**1561. How do you create multi-region Kinesis Firehose streams delivering to S3 and Redshift simultaneously?**

- Use `kinesis_firehose` module, loop over destinations per region, configure dynamic buffering and compression.

**1562. How do you automate multi-region CloudTrail with CloudWatch logs integration?**

- Use `cloudtrail` module with `cloud_watch_logs_group_arn` per region and enable multi-region logging.

**1563. How do you deploy multi-region Redshift clusters with automated snapshots and cross-region snapshot copy?**

- Use `redshift_cluster` module with `snapshot_schedule_identifier` and `snapshot_copy_grant_name` per region.

**1564. How do you orchestrate multi-region EKS clusters with node groups using different AMIs?**

- Use `eks_cluster` and `eks_nodegroup` modules, assign AMIs dynamically per region and configure scaling policies.

**1565. How do you dynamically deploy multi-region ECS services with service discovery enabled?**

- Use `ecs_service` module with `service_registries` per region and environment-specific configuration.

**1566. How do you automate multi-region Secrets Manager secret replication and rotation?**

- Use `secretsmanager_secret` with `add_replica_regions` and `rotation_lambda_arn` dynamically.

**1567. How do you configure multi-region EventBridge rules with Lambda and SQS targets dynamically?**

- Use `eventbridge_rule` and `eventbridge_target` modules, loop over regions, configure multiple target types.

**1568. How do you deploy multi-region AWS Backup vaults with cross-region backup plans?**

- Use backup\_vault and backup\_plan modules, loop over regions, attach selection of resources per region.

**1569. How do you automate multi-region Route53 failover routing with health checks?**

- Use route53\_record with set\_identifier, region, and health\_check\_id loops per region.

**1570. How do you orchestrate multi-region ACM certificate issuance and automatic renewal?**

- Use acm\_certificate with DNS validation and loop over multiple domains across regions.

**1571. How do you deploy multi-region MQ brokers with active/standby configuration?**

- Use mq\_broker module, configure high availability per region with different host instance types.

**1572. How do you dynamically configure multi-region FSx Lustre file systems with auto-import of S3 data?**

- Use fsx module, loop over regions, enable import\_path and throughput configuration.

**1573. How do you automate multi-region CloudFormation StackSets with different parameters?**

- Use cloudformation module with stack\_set\_name, loop over target regions, and pass parameter overrides per region.

**1574. How do you orchestrate multi-region CloudWatch dashboards aggregating metrics from all regions?**

- Use cloudwatch\_dashboard module, fetch region-specific metrics dynamically, and configure widgets per region.

**1575. How do you automate multi-region S3 bucket policies with cross-account access?**

- Use s3\_bucket\_policy module, loop over buckets and accounts, and assign different policies per region.

**1576. How do you orchestrate multi-region SNS topics with subscription to multiple endpoints dynamically?**

- Use sns\_topic and sns\_subscription modules, loop over regions, subscribe endpoints like Lambda, SQS, email.

**1577. How do you deploy multi-region Lambda destinations for success/failure asynchronously?**

- Use lambda module, configure destination\_config for each region dynamically.

**1578. How do you automate multi-region S3 bucket notifications triggering Lambda and SQS dynamically?**

- Use s3\_bucket\_notification module, loop over regions, assign multiple targets per bucket.

**1579. How do you orchestrate multi-region CloudFront distributions with different origins per region?**

- Use cloudfront\_distribution module, loop over regions, configure origin groups dynamically.

**1580. How do you deploy multi-region AWS WAF with shared rules dynamically?**

- Use waf\_rule and waf\_web\_acl modules, loop over regions, attach shared rule sets dynamically.

**1581. How do you automate multi-region Elastic Load Balancers with cross-zone load balancing?**

- Use elb\_application\_lb module, configure subnets and zones dynamically for each region.

**1582. How do you orchestrate multi-region Auto Scaling policies with CloudWatch alarms dynamically?**

- Use autoscaling\_policy module, loop over regions and groups, attach dynamic alarm metrics per region.

**1583. How do you deploy multi-region CloudWatch log metric filters dynamically?**

- Use cloudwatch\_log\_metric\_filter module, loop over regions, and create metrics per application.

**1584. How do you automate multi-region CodeCommit repositories dynamically?**

- Use codecommit\_repository module, loop over regions, create branches, and configure triggers.

**1585. How do you orchestrate multi-region AWS Glue workflows dynamically with dependencies?**

- Use glue\_workflow and glue\_trigger modules, loop over regions, configure job dependencies dynamically.

**1586. How do you deploy multi-region AWS Batch jobs with dynamic compute environments?**

- Use batch\_job\_queue and batch\_compute\_environment modules, loop over regions with dynamic vCPUs and memory.

**1587. How do you automate multi-region SES verified identities and DKIM configuration?**

- Use ses\_domain\_identity and ses\_domain\_dkim modules, loop over regions and domains dynamically.

**1588. How do you orchestrate multi-region ECR repositories with image replication?**

- Use ecr\_repository module with replication\_configuration loops per region.

**1589. How do you deploy multi-region ECS task definitions dynamically with environment overrides?**

- Use ecs\_taskdefinition module, loop over regions, configure environment variables per region.

**1590. How do you automate multi-region ElastiCache replication groups dynamically?**

- Use elasticache\_replication\_group module, loop over regions, assign node types and cluster configuration.

**1591. How do you orchestrate multi-region AWS Config aggregators dynamically?**

- Use aws\_config\_configuration\_aggregator module, loop over accounts and regions dynamically.

**1592. How do you deploy multi-region AWS Step Functions with cross-region Lambda invocation?**

- Use `stepfunctions_state_machine` module, loop over regions, reference Lambda ARNs in other regions.

**1593. How do you automate multi-region EC2 image (AMI) creation and sharing?**

- Use `ec2_ami` module, loop over regions, create images, and share with target accounts dynamically.

**1594. How do you orchestrate multi-region CloudWatch events rules triggering Step Functions and Lambda?**

- Use `eventbridge_rule` and `eventbridge_target` modules with cross-region ARNs dynamically.

**1595. How do you deploy multi-region Route53 weighted routing with automated failover?**

- Use `route53_record` module with `set_identifier`, `weights`, and `health checks` per region dynamically.

**1596. How do you automate multi-region ACM certificate renewals and CloudFront deployment updates?**

- Use `acm_certificate` module with `renewal_status` monitoring and trigger `cloudfront_distribution` updates dynamically.

**1597. How do you orchestrate multi-region CloudFormation drift detection dynamically?**

- Use `cloudformation` module with `detect_stack_drift` looped per region.

**1598. How do you deploy multi-region AWS Backup jobs with cross-account resource selection?**

- Use `backup_selection` and `backup_plan` modules, loop over accounts and regions dynamically.

**1599. How do you automate multi-region EKS cluster add-ons deployment (like VPC CNI, CoreDNS)?**

- Use `eks_addon` module, loop over clusters and regions, configure versions dynamically.

**1600. How do you implement a fully automated multi-region DR solution for all AWS services with Ansible?**

- Combine modules: `ec2_vpc`, `rds_instance`, `s3_bucket`, `cloudfront_distribution`, `route53_record`, `lambda`, `stepfunctions_state_machine`, `ecs_service`, `eks_cluster` dynamically with region loops, failover policies, and automation playbooks.

---

**1601. How do you automate multi-region S3 lifecycle policies dynamically?**

- Use `s3_bucket_lifecycle` module, loop over buckets in different regions, configure transitions and expiration rules.

**1602. How do you deploy multi-region Lambda functions with different VPC configurations?**

- Use `lambda` module, loop over regions, and dynamically assign VPC subnets and security groups.

**1603. How do you orchestrate multi-region API Gateway with stage-level throttling dynamically?**

- Use `apigateway_stage` module with `throttling_rate_limit` and `throttling_burst_limit` per region in loops.

**1604. How do you automate multi-region CloudFront distributions with origin failover dynamically?**

- Use `cloudfront_distribution` module, configure `origin_group` with primary and secondary origins per region.

**1605. How do you deploy multi-region RDS with automated backup and cross-region snapshot copy?**

- Use `rds_instance` module, enable `backup_retention_period`, `multi_az`, and configure `cross_region_snapshot_copy`.

**1606. How do you orchestrate multi-region DynamoDB streams to trigger Lambda functions dynamically?**

- Use `dynamodb_table` with `stream_enabled: yes`, `stream_view_type: NEW_AND_OLD_IMAGES` and attach Lambda per region.

**1607. How do you automate multi-region CloudWatch alarms with SNS notifications?**

- Use `cloudwatch_alarm` module, loop over regions, configure `alarm_actions` dynamically with SNS ARNs.

**1608. How do you deploy multi-region Elastic Beanstalk applications with environment tiers dynamically?**

- Use `elasticbeanstalk` module, loop over regions, configure `EnvironmentTier` as `WebServer` or `Worker`.

**1609. How do you orchestrate multi-region Kinesis Data Analytics applications dynamically?**

- Use `kinesis_analytics_application` module, loop over regions, configure input and output streams per region.

**1610. How do you automate multi-region SSM automation documents and execution?**

- Use `ssm_document` and `ssm_automation_execution` modules, loop over regions, assign targets dynamically.

**1611. How do you deploy multi-region ECS clusters with service auto-scaling policies dynamically?**

- Use `ecs_service` and `ecs_taskdefinition` modules, loop over regions, configure `scaling_policies` per service.

**1612. How do you orchestrate multi-region EKS clusters with Fargate profiles dynamically?**

- Use `eks_fargate_profile` module, loop over clusters and regions, configure selectors dynamically.

**1613. How do you automate multi-region CloudTrail with S3 and CloudWatch Logs integration?**

- Use cloudtrail module, enable is\_multi\_region\_trail: yes, configure S3 buckets and log groups per region.

**1614. How do you deploy multi-region AWS Batch jobs with different compute environments dynamically?**

- Use batch\_compute\_environment and batch\_job\_queue modules, loop over regions, assign vCPUs, memory, and instance types.

**1615. How do you orchestrate multi-region CodeBuild projects with dynamic environment variables?**

- Use codebuild\_project module, loop over regions, assign environment\_variables per region dynamically.

**1616. How do you automate multi-region CloudFormation StackSets with parameter overrides?**

- Use cloudformation module with stack\_set\_name and loop over target regions, provide dynamic parameter\_overrides.

**1617. How do you deploy multi-region ACM certificates with DNS validation automatically?**

- Use acm\_certificate module, loop over regions, configure validation\_method: DNS, and automate route53 records.

**1618. How do you orchestrate multi-region SNS topics with cross-region subscriptions dynamically?**

- Use sns\_topic and sns\_subscription modules, loop over regions, subscribe endpoints like Lambda, SQS, and email.

**1619. How do you automate multi-region AWS Config rules for compliance dynamically?**

- Use aws\_config\_config\_rule module, loop over regions, assign rule parameters and remediation actions.

**1620. How do you deploy multi-region CloudWatch dashboards with dynamic widgets?**

- Use cloudwatch\_dashboard module, loop over regions, configure widgets dynamically per application metrics.

**1621. How do you orchestrate multi-region S3 bucket notifications for multiple Lambda functions dynamically?**

- Use s3\_bucket\_notification module, loop over buckets and Lambda targets in different regions.

**1622. How do you deploy multi-region AWS Glue jobs with dynamic dependencies?**

- Use glue\_job module, loop over regions, configure depends\_on for job execution order.

**1623. How do you automate multi-region Redshift cluster snapshot copy schedules dynamically?**

- Use redshift\_cluster module, loop over clusters, assign snapshot\_copy\_grant\_name and snapshot\_schedule\_identifier.

**1624. How do you orchestrate multi-region EC2 instances with dynamic user-data scripts?**

- Use `ec2_instance` module, loop over regions, assign `user_data` scripts dynamically per environment.

**1625. How do you deploy multi-region AWS Lambda layers dynamically?**

- Use `lambda_layer` module, loop over regions, assign compatible runtimes and versions.

**1626. How do you automate multi-region ECR repository replication configuration dynamically?**

- Use `ecr_repository` module with `replication_configuration`, loop over regions, configure replication destinations.

**1627. How do you orchestrate multi-region ElastiCache clusters with automatic failover dynamically?**

- Use `elasticache_replication_group` module, loop over regions, enable `automatic_failover: yes`.

**1628. How do you deploy multi-region Step Functions workflows with dynamic input parameters?**

- Use `stepfunctions_state_machine` module, loop over regions, pass input dynamically.

**1629. How do you automate multi-region AWS Batch compute environments with spot and on-demand instances?**

- Use `batch_compute_environment` module, loop over regions, assign `instance_types` and `allocation_strategy`.

**1630. How do you orchestrate multi-region Route53 geolocation routing with failover dynamically?**

- Use `route53_record` module, loop over regions, configure `set_identifier`, `geo_location`, and `health_check_id`.

**1631. How do you deploy multi-region Elastic Load Balancers with listener rules dynamically?**

- Use `elb_application_lb` module, loop over regions, configure listeners and rules dynamically.

**1632. How do you automate multi-region CloudWatch log metric filters for application monitoring?**

- Use `cloudwatch_log_metric_filter` module, loop over regions, configure filters and metrics dynamically.

**1633. How do you orchestrate multi-region CloudFront distributions with multiple origins and caching policies?**

- Use `cloudfront_distribution` module, loop over regions, configure origins, behaviors, and caching policies dynamically.

**1634. How do you deploy multi-region AWS WAFv2 web ACLs with shared rules dynamically?**

- Use `wafv2_web_acl` and `wafv2_rule_group` modules, loop over regions, attach shared rules dynamically.

**1635. How do you automate multi-region DynamoDB global table creation with streams enabled?**

- Use `dynamodb_table` module, loop over regions, enable `stream_enabled` and `replica_regions`.



**1636. How do you orchestrate multi-region AWS Backup vaults with cross-account resource selection?**

- Use backup\_vault and backup\_selection modules, loop over regions and accounts dynamically.

**1637. How do you deploy multi-region SSM parameters with hierarchical names dynamically?**

- Use ssm\_parameter\_store module, loop over regions, configure parameter\_name hierarchy and types.

**1638. How do you automate multi-region Kinesis Data Streams with dynamic shard count?**

- Use kinesis\_stream module, loop over regions, configure shard\_count dynamically based on load.

**1639. How do you orchestrate multi-region CloudFormation stack updates with change sets?**

- Use cloudformation module with change\_set\_name, loop over regions, apply updates dynamically.

**1640. How do you deploy multi-region Elastic Beanstalk worker environments with dynamic queues?**

- Use elasticbeanstalk module, loop over regions, configure EnvironmentTier: Worker and SQS queues.

**1641. How do you automate multi-region S3 bucket encryption with KMS keys?**

- Use s3\_bucket module, loop over regions, configure server\_side\_encryption\_configuration dynamically.

**1642. How do you orchestrate multi-region AWS Glue crawlers with dynamic database targets?**

- Use glue\_crawler module, loop over regions, assign database\_name and S3 targets dynamically.

**1643. How do you deploy multi-region ECS clusters with multiple task definitions and environment overrides?**

- Use ecs\_taskdefinition and ecs\_service modules, loop over regions, configure environment variables and volumes.

**1644. How do you automate multi-region API Gateway stages with throttling and logging?**

- Use apigateway\_stage module, loop over regions, configure throttling\_rate\_limit, logging\_level, and metrics.

**1645. How do you orchestrate multi-region VPC peering with dynamic route table updates?**

- Use ec2\_vpc\_peering and ec2\_vpc\_route\_table modules, loop over regions, update routes dynamically.

**1646. How do you deploy multi-region Lambda function versions with alias routing dynamically?**

- Use lambda module with publish: yes, assign aliases, and route traffic per region dynamically.

**1647. How do you automate multi-region CloudTrail S3 bucket policy updates dynamically?**

- Use `s3_bucket_policy` module, loop over regions, configure bucket policies for CloudTrail logs dynamically.

**1648. How do you orchestrate multi-region ECS service discovery with Route53 private hosted zones?**

- Use `ecs_service` and `route53_record` modules, loop over regions, configure service discovery dynamically.

**1649. How do you deploy multi-region AWS Batch jobs with dynamic job definitions and environment variables?**

- Use `batch_job_definition` and `batch_job_queue` modules, loop over regions, configure parameters dynamically.

**1650. How do you implement multi-region disaster recovery for AWS Lambda, S3, RDS, and CloudFront using Ansible?**

- Combine modules: `lambda`, `s3_bucket`, `rds_instance`, `cloudfront_distribution` dynamically, loop over regions, configure replication, failover, and routing automatically.
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**1651. How do you automate multi-region S3 bucket cross-account replication dynamically?**

- Use `s3_bucket` with `replication_configuration`, loop over regions, configure destination buckets and IAM roles per account.

**1652. How do you deploy multi-region Lambda functions with VPC peering connectivity?**

- Use `lambda` module, loop over regions, configure `vpc_config` with subnets and security groups in peered VPCs.

**1653. How do you orchestrate multi-region API Gateway custom domain names with ACM certificates dynamically?**

- Use `apigateway_domain_name` module, loop over regions, assign `regional_certificate_arn` dynamically.

**1654. How do you automate multi-region CloudFront distributions with multiple origin access identities (OAI)?**

- Use `cloudfront_distribution` module, loop over regions, create OAI and assign S3 origins dynamically.

**1655. How do you deploy multi-region RDS Aurora clusters with automated failover and read/write splitting?**

- Use `rds_instance` module, enable `multi_az`: yes, configure endpoints for writer/reader dynamically per region.

**1656. How do you orchestrate multi-region DynamoDB global tables with TTL and streams enabled?**

- Use `dynamodb_table` module, loop over regions, configure `replica_regions`, `ttl_specification`, and `stream_specification`.

**1657. How do you automate multi-region CloudWatch alarms with composite alarms across regions?**

- Use `cloudwatch_composite_alarm` module, loop over regions, combine individual alarms for cross-region monitoring.

**1658. How do you deploy multi-region Elastic Beanstalk environments with rolling updates and zero downtime?**

- Use `elasticbeanstalk` module, loop over regions, configure `DeploymentPolicy: RollingWithAdditionalBatch` dynamically.

**1659. How do you orchestrate multi-region Kinesis Data Streams with dynamic shard scaling based on metrics?**

- Use `kinesis_stream` module, loop over regions, configure `shard_count` dynamically based on CloudWatch metrics.

**1660. How do you automate multi-region SSM documents for patching and maintenance tasks?**

- Use `ssm_document` module, loop over regions, configure `document_type: Automation`, assign targets dynamically.

**1661. How do you deploy multi-region ECS services with dynamic load balancer target groups?**

- Use `ecs_service` module, loop over regions, configure `load_balancers` dynamically per cluster.

**1662. How do you orchestrate multi-region EKS clusters with managed node groups using different instance types?**

- Use `eks_nodegroup` module, loop over regions, configure node group scaling and instance types dynamically.

**1663. How do you automate multi-region CloudTrail with CloudWatch Logs integration for compliance auditing?**

- Use `cloudtrail` module, loop over regions, configure `cloud_watch_logs_group_arn` and multi-region logging.

**1664. How do you deploy multi-region AWS Batch jobs with dynamic compute environments and instance fleets?**

- Use `batch_compute_environment` module, loop over regions, assign spot and on-demand instances dynamically.

**1665. How do you orchestrate multi-region CodeBuild projects with cross-region CodePipeline integration?**

- Use `codebuild_project` module, loop over regions, configure source, environment, and artifacts dynamically.

**1666. How do you automate multi-region CloudFormation StackSets with dynamic parameter overrides and deployment targets?**

- Use `cloudformation` module, loop over regions, apply `parameter_overrides` and target accounts dynamically.

**1667. How do you deploy multi-region ACM certificates with automatic renewal and validation for multiple domains?**

- Use `acm_certificate` module, loop over regions, configure DNS validation and renewals dynamically.

**1668. How do you orchestrate multi-region SNS topics with cross-region Lambda and SQS subscriptions?**

- Use `sns_topic` and `sns_subscription` modules, loop over regions, attach endpoints dynamically.

**1669. How do you automate multi-region AWS Config rules and remediation actions dynamically?**

- Use `aws_config_config_rule` module, loop over regions, assign rule parameters and remediation actions dynamically.

**1670. How do you deploy multi-region CloudWatch dashboards aggregating metrics from multiple regions?**

- Use `cloudwatch_dashboard` module, loop over regions, create widgets and dynamic metric queries.

**1671. How do you orchestrate multi-region S3 bucket notifications triggering multiple Lambda functions and SQS queues?**

- Use `s3_bucket_notification` module, loop over buckets and targets in different regions dynamically.

**1672. How do you deploy multi-region AWS Glue jobs with dynamic connections and job dependencies?**

- Use `glue_job` module, loop over regions, assign connections, role, and `depends_on` dynamically.

**1673. How do you automate multi-region Redshift clusters with snapshot copy and cross-region security groups?**

- Use `redshift_cluster` module, loop over regions, configure `snapshot_copy_grant_name` and `vpc_security_group_ids`.

**1674. How do you orchestrate multi-region EC2 instances with dynamic IAM roles, tags, and security groups?**

- Use `ec2_instance` module, loop over regions, assign IAM roles, tags, and security groups dynamically.

**1675. How do you deploy multi-region Lambda layers with versioning and compatible runtimes dynamically?**

- Use `lambda_layer` module, loop over regions, assign compatible runtimes and publish versions dynamically.

**1676. How do you automate multi-region ECR repositories with replication rules and lifecycle policies?**

- Use `ecr_repository` module, loop over regions, configure `replication_configuration` and `lifecycle_policy`.

**1677. How do you orchestrate multi-region ElastiCache replication groups with automatic failover and backups?**

- Use `elasticache_replication_group` module, loop over regions, enable `automatic_failover` and `backup_retention`.

**1678. How do you deploy multi-region Step Functions state machines with dynamic input parameters and Lambda integration?**

- Use `stepfunctions_state_machine` module, loop over regions, pass input dynamically and configure Lambda tasks.

**1679. How do you automate multi-region AWS Batch jobs with dynamic job definitions and environment variables?**

- Use `batch_job_definition` module, loop over regions, configure container properties dynamically.

**1680. How do you orchestrate multi-region Route53 geolocation routing with health checks and failover policies?**

- Use `route53_record` module, loop over regions, configure `geo_location`, `health_check_id`, and `set_identifier`.

**1681. How do you deploy multi-region Application Load Balancers with dynamic listener rules and target groups?**

- Use `elb_application_lb` module, loop over regions, configure listeners, rules, and target groups dynamically.

**1682. How do you automate multi-region CloudWatch log metric filters and alarms for high-volume applications?**

- Use `cloudwatch_log_metric_filter` and `cloudwatch_alarm` modules, loop over regions, create metrics and alarms dynamically.

**1683. How do you orchestrate multi-region CloudFront distributions with multiple origins, caching policies, and Lambda@Edge functions?**

- Use `cloudfront_distribution` module, loop over regions, configure origin groups, caching behaviors, and Lambda associations.

**1684. How do you deploy multi-region WAFv2 web ACLs with dynamic rule groups and rate-based rules?**

- Use `wafv2_web_acl` and `wafv2_rule_group` modules, loop over regions, attach rules dynamically.

**1685. How do you automate multi-region DynamoDB global tables with TTL, streams, and backups enabled?**

- Use `dynamodb_table` module, loop over regions, enable `replica_regions`, `ttl_specification`, and `stream_specification`.

**1686. How do you orchestrate multi-region AWS Backup vaults with cross-account selections and backup plans?**

- Use backup\_vault and backup\_selection modules, loop over regions and accounts, assign resources dynamically.

**1687. How do you deploy multi-region SSM parameters with hierarchical names, secure strings, and versioning?**

- Use ssm\_parameter\_store module, loop over regions, configure type, tier, and version dynamically.

**1688. How do you automate multi-region Kinesis Data Streams with dynamic shard scaling and encryption?**

- Use kinesis\_stream module, loop over regions, configure shard\_count, stream\_mode, and kms\_key\_id.

**1689. How do you orchestrate multi-region CloudFormation stack updates with change sets, drift detection, and rollback?**

- Use cloudformation module, loop over regions, create change sets, detect drift, and execute updates with rollback.

**1690. How do you deploy multi-region Elastic Beanstalk worker environments with dynamic queue names and IAM roles?**

- Use elasticbeanstalk module, loop over regions, configure environment tier, SQS queues, and IAM roles dynamically.

**1691. How do you automate multi-region S3 bucket encryption with KMS keys and access logging?**

- Use s3\_bucket module, loop over regions, configure server\_side\_encryption\_configuration and logging dynamically.

**1692. How do you orchestrate multi-region AWS Glue crawlers with dynamic S3 targets and IAM roles?**

- Use glue\_crawler module, loop over regions, configure database\_name, targets, and role dynamically.

**1693. How do you deploy multi-region ECS services with multiple task definitions, environment overrides, and secrets?**

- Use ecs\_taskdefinition and ecs\_service modules, loop over regions, configure environment variables, secrets, and volumes dynamically.

**1694. How do you automate multi-region API Gateway stages with throttling, logging, and stage variables?**

- Use apigateway\_stage module, loop over regions, configure throttling\_rate\_limit, logging\_level, and variables.

**1695. How do you orchestrate multi-region VPC peering with dynamic route table updates and CIDR blocks?**

- Use `ec2_vpc_peering` and `ec2_vpc_route_table` modules, loop over regions, configure CIDR routes dynamically.

**1696. How do you deploy multi-region Lambda function versions with alias routing and blue-green deployment?**

- Use `lambda` module, loop over regions, publish versions, create aliases, and route traffic dynamically.

**1697. How do you automate multi-region CloudTrail S3 bucket policy updates for cross-account logging?**

- Use `s3_bucket_policy` module, loop over regions, update bucket policies dynamically for CloudTrail logging.

**1698. How do you orchestrate multi-region ECS service discovery with Route53 private hosted zones and multiple services?**

- Use `ecs_service` and `route53_record` modules, loop over regions, configure service discovery dynamically.

**1699. How do you deploy multi-region AWS Batch jobs with dynamic job queues, compute environments, and dependencies?**

- Use `batch_job_queue` and `batch_compute_environment` modules, loop over regions, configure job dependencies dynamically.

**1700. How do you implement a fully automated multi-region disaster recovery architecture for AWS Lambda, S3, RDS, ECS, and CloudFront with Ansible?**

- Combine modules: `lambda`, `s3_bucket`, `rds_instance`, `ecs_service`, `cloudfront_distribution` dynamically, loop over regions, configure replication, failover, monitoring, and automated routing policies.
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**1701. How do you automate multi-region S3 bucket versioning and lifecycle policies dynamically?**

- Use `s3_bucket` module, enable versioning and configure `lifecycle_configuration` per region in loops.

**1702. How do you deploy multi-region Lambda functions with environment-specific VPC and subnet configurations?**

- Use `lambda` module, loop over regions, configure `vpc_config` dynamically with subnets and security groups per environment.

**1703. How do you orchestrate multi-region API Gateway with custom domains and certificate rotation?**

- Use `apigateway_domain_name` module, loop over regions, assign ACM certificates, and automate renewal.

**1704. How do you automate multi-region CloudFront distributions with multiple origins, caching policies, and WAF integration?**

- Use `cloudfront_distribution` module, loop over regions, configure origins, behaviors, and attach WAF ACLs.

**1705. How do you deploy multi-region RDS Aurora clusters with read/write splitting and automated failover?**

- Use `rds_instance` module with `multi_az`: yes, configure writer/reader endpoints, and cross-region replicas dynamically.

**1706. How do you orchestrate multi-region DynamoDB global tables with TTL, streams, and backups enabled?**

- Use `dynamodb_table` module, loop over regions, enable `replica_regions`, `ttl_specification`, and `stream_specification`.

**1707. How do you automate multi-region CloudWatch composite alarms for cross-region application monitoring?**

- Use `cloudwatch_composite_alarm` module, loop over regions, combine individual alarms dynamically.

**1708. How do you deploy multi-region Elastic Beanstalk applications with rolling updates and zero downtime?**

- Use `elasticbeanstalk` module, loop over regions, configure `DeploymentPolicy`: `RollingWithAdditionalBatch`.

**1709. How do you orchestrate multi-region Kinesis Data Streams with dynamic shard scaling based on CloudWatch metrics?**

- Use `kinesis_stream` module, loop over regions, adjust `shard_count` dynamically.

**1710. How do you automate multi-region SSM documents for patching, maintenance, and automation workflows?**

- Use `ssm_document` module, loop over regions, configure `document_type`: `Automation`, and assign targets dynamically.

**1711. How do you deploy multi-region ECS services with dynamic load balancer target groups and auto-scaling policies?**

- Use `ecs_service` module, loop over regions, assign `load_balancers`, configure `scaling_policies` dynamically.

**1712. How do you orchestrate multi-region EKS clusters with managed node groups using different instance types and scaling?**

- Use `eks_nodegroup` module, loop over regions, configure scaling and instance types per node group.

**1713. How do you automate multi-region CloudTrail with CloudWatch Logs integration for compliance auditing?**

- Use `cloudtrail` module, loop over regions, configure `cloud_watch_logs_group_arn` and multi-region logging.



**1714. How do you deploy multi-region AWS Batch jobs with dynamic compute environments and job dependencies?**

- Use `batch_compute_environment` and `batch_job_queue` modules, loop over regions, configure vCPUs, memory, and dependencies.

**1715. How do you orchestrate multi-region CodeBuild projects with cross-region CodePipeline integration dynamically?**

- Use `codebuild_project` module, loop over regions, configure source, environment, and artifacts dynamically.

**1716. How do you automate multi-region CloudFormation StackSets with parameter overrides, drift detection, and rollback?**

- Use `cloudformation` module, loop over regions, apply `parameter_overrides`, detect drift, and enable rollback.

**1717. How do you deploy multi-region ACM certificates with automatic DNS validation and renewal for multiple domains?**

- Use `acm_certificate` module, loop over regions, configure DNS validation and automatic renewal dynamically.

**1718. How do you orchestrate multi-region SNS topics with Lambda and SQS subscriptions across accounts?**

- Use `sns_topic` and `sns_subscription` modules, loop over regions, subscribe endpoints dynamically.

**1719. How do you automate multi-region AWS Config rules for compliance and remediation actions dynamically?**

- Use `aws_config_config_rule` module, loop over regions, assign rule parameters and remediation actions.

**1720. How do you deploy multi-region CloudWatch dashboards aggregating metrics from multiple regions dynamically?**

- Use `cloudwatch_dashboard` module, loop over regions, create widgets and dynamic metric queries per application.

**1721. How do you orchestrate multi-region S3 bucket notifications triggering multiple Lambda functions and SQS queues?**

- Use `s3_bucket_notification` module, loop over regions, assign targets dynamically per bucket.

**1722. How do you deploy multi-region AWS Glue jobs with dynamic job dependencies and connections?**

- Use `glue_job` module, loop over regions, configure connections, role, and `depends_on` dynamically.

**1723. How do you automate multi-region Redshift clusters with snapshot copy, security groups, and IAM roles?**

- Use `redshift_cluster` module, loop over regions, configure `snapshot_copy_grant_name`, `vpc_security_group_ids`, and IAM roles dynamically.

**1724. How do you orchestrate multi-region EC2 instances with dynamic IAM roles, tags, security groups, and monitoring agents?**

- Use `ec2_instance` module, loop over regions, assign roles, tags, security groups, and CloudWatch agent dynamically.

**1725. How do you deploy multi-region Lambda layers with versioning, compatible runtimes, and dependencies dynamically?**

- Use `lambda_layer` module, loop over regions, configure compatible runtimes, versions, and dependencies dynamically.

**1726. How do you automate multi-region ECR repositories with replication, lifecycle policies, and image scanning?**

- Use `ecr_repository` module, loop over regions, configure replication, `lifecycle_policy`, and `image_scanning_configuration`.

**1727. How do you orchestrate multi-region ElastiCache replication groups with automatic failover, backup retention, and maintenance windows?**

- Use `elasticache_replication_group` module, loop over regions, configure `automatic_failover`, backups, and maintenance.

**1728. How do you deploy multi-region Step Functions workflows with dynamic input parameters, Lambda integration, and error handling?**

- Use `stepfunctions_state_machine` module, loop over regions, pass input dynamically, configure tasks and retries.

**1729. How do you automate multi-region AWS Batch jobs with dynamic job definitions, queues, and container overrides?**

- Use `batch_job_definition` and `batch_job_queue` modules, loop over regions, configure containers and environment variables dynamically.

**1730. How do you orchestrate multi-region Route53 geolocation routing with health checks, weighted failover, and dynamic traffic policies?**

- Use `route53_record` module, loop over regions, configure `geo_location`, `health_check_id`, weight, and failover policies dynamically.

**1731. How do you deploy multi-region Application Load Balancers with listener rules, target groups, and path-based routing dynamically?**

- Use `elb_application_lb` module, loop over regions, configure listeners, rules, and target groups dynamically.

**1732. How do you automate multi-region CloudWatch log metric filters and alarms for high-volume multi-service applications?**

- Use `cloudwatch_log_metric_filter` and `cloudwatch_alarm` modules, loop over regions, configure filters and alarms dynamically.

**1733. How do you orchestrate multi-region CloudFront distributions with multiple origins, caching policies, Lambda@Edge functions, and WAF ACLs?**

- Use `cloudfront_distribution` module, loop over regions, configure origins, behaviors, Lambda associations, and attach WAF rules dynamically.

**1734. How do you deploy multi-region WAFv2 web ACLs with dynamic rule groups, rate-based rules, and logging configuration?**

- Use `wafv2_web_acl` and `wafv2_rule_group` modules, loop over regions, attach rules, configure rate limits, and logging.

**1735. How do you automate multi-region DynamoDB global tables with TTL, streams, backups, and auto-scaling enabled?**

- Use `dynamodb_table` module, loop over regions, enable `replica_regions`, `ttl_specification`, `stream_specification`, and auto-scaling policies.

**1736. How do you orchestrate multi-region AWS Backup vaults with cross-account selections, backup plans, and lifecycle rules?**

- Use `backup_vault` and `backup_selection` modules, loop over regions and accounts dynamically, configure lifecycle policies.

**1737. How do you deploy multi-region SSM parameters with hierarchical names, secure strings, versioning, and dynamic updates?**

- Use `ssm_parameter_store` module, loop over regions, configure type, tier, version, and automatic updates dynamically.

**1738. How do you automate multi-region Kinesis Data Streams with dynamic shard scaling, encryption, and retention periods?**

- Use `kinesis_stream` module, loop over regions, configure `shard_count`, `kms_key_id`, and `retention_period`.

**1739. How do you orchestrate multi-region CloudFormation stack updates with change sets, drift detection, and rollback configurations?**

- Use `cloudformation` module, loop over regions, create change sets, detect drift, and apply rollback policies.

**1740. How do you deploy multi-region Elastic Beanstalk worker environments with dynamic SQS queues, IAM roles, and environment variables?**

- Use `elasticbeanstalk` module, loop over regions, configure environment tier, queues, IAM roles, and variables dynamically.

**1741. How do you automate multi-region S3 bucket encryption with KMS keys, versioning, and access logging dynamically?**

- Use `s3_bucket` module, loop over regions, configure `server_side_encryption_configuration`, versioning, and logging.

**1742. How do you orchestrate multi-region AWS Glue crawlers with dynamic S3 targets, databases, and IAM roles?**

- Use glue\_crawler module, loop over regions, configure database\_name, targets, and roles dynamically.

**1743. How do you deploy multi-region ECS services with multiple task definitions, environment variables, secrets, and volumes dynamically?**

- Use ecs\_taskdefinition and ecs\_service modules, loop over regions, configure all parameters dynamically.

**1744. How do you automate multi-region API Gateway stages with throttling, logging, metrics, and stage variables dynamically?**

- Use apigateway\_stage module, loop over regions, configure throttling, logging, metrics, and variables.

**1745. How do you orchestrate multi-region VPC peering with route table updates, CIDR management, and security group rules dynamically?**

- Use ec2\_vpc\_peering and ec2\_vpc\_route\_table modules, loop over regions, configure routes and rules dynamically.

**1746. How do you deploy multi-region Lambda function versions with aliases, routing, and blue/green deployment dynamically?**

- Use lambda module, loop over regions, publish versions, create aliases, and route traffic dynamically.

**1747. How do you automate multi-region CloudTrail S3 bucket policies for cross-account logging and compliance?**

- Use s3\_bucket\_policy module, loop over regions, update policies dynamically.

**1748. How do you orchestrate multi-region ECS service discovery with Route53 private hosted zones, multiple services, and dynamic DNS records?**

- Use ecs\_service and route53\_record modules, loop over regions, configure service discovery and DNS dynamically.

**1749. How do you deploy multi-region AWS Batch jobs with dynamic job queues, compute environments, dependencies, and container overrides?**

- Use batch\_job\_queue and batch\_compute\_environment modules, loop over regions, configure all parameters dynamically.

**1750. How do you implement a fully automated multi-region disaster recovery architecture for AWS Lambda, S3, RDS, ECS, and CloudFront using Ansible playbooks?**

- Combine modules: lambda, s3\_bucket, rds\_instance, ecs\_service, cloudfront\_distribution, loop over regions, configure replication, failover, monitoring, and automated routing policies dynamically.
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**1751. How do you automate multi-region S3 bucket replication with cross-account IAM roles and dynamic prefix filters?**

- Use s3\_bucket module, configure replication\_configuration, loop over regions, and set role and prefix filters dynamically.

**1752. How do you deploy multi-region Lambda functions with dynamic environment variables, VPC subnets, and security groups?**

- Use lambda module, loop over regions, assign environment variables, configure vpc\_config dynamically for each environment.

**1753. How do you orchestrate multi-region API Gateway with custom domain names, base path mappings, and SSL certificates?**

- Use apigateway\_domain\_name and apigateway\_base\_path\_mapping modules, loop over regions, assign ACM certificates dynamically.

**1754. How do you automate multi-region CloudFront distributions with multiple origins, caching policies, and Lambda@Edge functions?**

- Use cloudfront\_distribution module, loop over regions, configure origins, behaviors, and attach Lambda@Edge dynamically.

**1755. How do you deploy multi-region RDS Aurora clusters with global databases and automated failover?**

- Use rds\_instance module, enable multi\_az, configure writer/readers, and global replication dynamically per region.

**1756. How do you orchestrate multi-region DynamoDB global tables with dynamic stream processing and TTL settings?**

- Use dynamodb\_table module, enable stream\_enabled, ttl\_specification, and configure replica\_regions in loops.

**1757. How do you automate multi-region CloudWatch composite alarms combining multiple services' metrics?**

- Use cloudwatch\_composite\_alarm module, loop over regions, combine metric alarms for cross-region applications.

**1758. How do you deploy multi-region Elastic Beanstalk applications with rolling and immutable update policies?**

- Use elasticbeanstalk module, loop over regions, configure DeploymentPolicy as Rolling or Immutable dynamically.

**1759. How do you orchestrate multi-region Kinesis Data Streams with auto-scaling shard count and encryption?**

- Use kinesis\_stream module, loop over regions, configure shard\_count and kms\_key\_id dynamically.

**1760. How do you automate multi-region SSM automation documents with patch baselines and maintenance windows?**

- Use `ssm_document` module, loop over regions, configure automation tasks and targets dynamically.

**1761. How do you deploy multi-region ECS services with multiple target groups, auto-scaling, and rolling updates?**

- Use `ecs_service` module, loop over regions, assign `load_balancers`, `scaling_policies`, and `deployment_configuration`.

**1762. How do you orchestrate multi-region EKS clusters with managed node groups and Fargate profiles dynamically?**

- Use `eks_nodegroup` and `eks_fargate_profile` modules, loop over regions, assign instance types, scaling, and selectors dynamically.

**1763. How do you automate multi-region CloudTrail logging with S3 bucket encryption and CloudWatch integration?**

- Use `cloudtrail` module, loop over regions, configure `s3_bucket_name`, encryption, and `cloud_watch_logs_group_arn`.

**1764. How do you deploy multi-region AWS Batch jobs with spot and on-demand instances, and compute environment priorities?**

- Use `batch_compute_environment` and `batch_job_queue` modules, loop over regions, configure allocation strategy dynamically.

**1765. How do you orchestrate multi-region CodeBuild projects with environment variables, IAM roles, and artifacts across regions?**

- Use `codebuild_project` module, loop over regions, configure environment, roles, and output artifacts dynamically.

**1766. How do you automate multi-region CloudFormation StackSets with parameter overrides, failure tolerance, and rollback configurations?**

- Use `cloudformation` module, loop over regions, configure `parameter_overrides`, `failure_tolerance_count`, and `rollback` options.

**1767. How do you deploy multi-region ACM certificates with automated DNS validation and cross-region usage?**

- Use `acm_certificate` module, loop over regions, configure DNS validation and automated renewal.

**1768. How do you orchestrate multi-region SNS topics with Lambda, SQS, and cross-account subscriptions?**

- Use `sns_topic` and `sns_subscription` modules, loop over regions, assign endpoints dynamically.

**1769. How do you automate multi-region AWS Config rules and remediation actions dynamically for compliance enforcement?**

- Use `aws_config_config_rule` module, loop over regions, assign rules and remediation actions per account.

**1770. How do you deploy multi-region CloudWatch dashboards aggregating logs and metrics dynamically?**

- Use `cloudwatch_dashboard` module, loop over regions, configure widgets and metrics for multiple services dynamically.

**1771. How do you orchestrate multi-region S3 bucket notifications to trigger multiple Lambda functions and SQS queues dynamically?**

- Use `s3_bucket_notification` module, loop over buckets and regions, configure multiple targets dynamically.

**1772. How do you deploy multi-region AWS Glue jobs with dynamic dependencies, connections, and IAM roles?**

- Use `glue_job` module, loop over regions, configure connections, role, and job dependencies dynamically.

**1773. How do you automate multi-region Redshift clusters with snapshot copy, cross-region replication, and security groups?**

- Use `redshift_cluster` module, loop over regions, configure snapshots, replication, and security groups dynamically.

**1774. How do you orchestrate multi-region EC2 instances with dynamic IAM roles, monitoring, tags, and security groups?**

- Use `ec2_instance` module, loop over regions, assign IAM roles, CloudWatch agents, tags, and security groups.

**1775. How do you deploy multi-region Lambda layers with versioning, compatible runtimes, and dependencies dynamically?**

- Use `lambda_layer` module, loop over regions, configure versions, runtimes, and dependencies dynamically.

**1776. How do you automate multi-region ECR repositories with replication, lifecycle policies, and image scanning configurations?**

- Use `ecr_repository` module, loop over regions, configure `replication_configuration`, `lifecycle_policy`, and scanning.

**1777. How do you orchestrate multi-region ElastiCache replication groups with automatic failover, backups, and maintenance windows?**

- Use `elasticache_replication_group` module, loop over regions, configure failover, backups, and maintenance dynamically.

**1778. How do you deploy multi-region Step Functions workflows with dynamic input, Lambda integrations, retries, and error handling?**

- Use `stepfunctions_state_machine` module, loop over regions, pass input dynamically, configure Lambda tasks and retries.

**1779. How do you automate multi-region AWS Batch jobs with dynamic job definitions, queues, container overrides, and dependencies?**

- Use `batch_job_definition` and `batch_job_queue` modules, loop over regions, configure all parameters dynamically.

**1780. How do you orchestrate multi-region Route53 geolocation and weighted routing with health checks and failover policies dynamically?**

- Use `route53_record` module, loop over regions, configure `geo_location`, `weight`, `health_check_id`, and failover policies dynamically.

**1781. How do you deploy multi-region Application Load Balancers with listener rules, target groups, path-based routing, and SSL certificates dynamically?**

- Use `elb_application_lb` module, loop over regions, configure listeners, target groups, routing rules, and SSL dynamically.

**1782. How do you automate multi-region CloudWatch log metric filters and alarms for high-volume multi-service applications dynamically?**

- Use `cloudwatch_log_metric_filter` and `cloudwatch_alarm` modules, loop over regions, configure filters and alarms dynamically.

**1783. How do you orchestrate multi-region CloudFront distributions with multiple origins, caching policies, Lambda@Edge, and WAF ACLs dynamically?**

- Use `cloudfront_distribution` module, loop over regions, configure origins, caching behaviors, Lambda@Edge, and attach WAF dynamically.

**1784. How do you deploy multi-region WAFv2 web ACLs with dynamic rules, rate-based rules, and logging configurations?**

- Use `wafv2_web_acl` and `wafv2_rule_group` modules, loop over regions, configure rules and logging dynamically.

**1785. How do you automate multi-region DynamoDB global tables with TTL, streams, backups, and auto-scaling dynamically?**

- Use `dynamodb_table` module, loop over regions, configure `replica_regions`, `TTL`, `streams`, and auto-scaling policies dynamically.

**1786. How do you orchestrate multi-region AWS Backup vaults with cross-account selections, backup plans, and lifecycle rules?**

- Use `backup_vault` and `backup_selection` modules, loop over regions and accounts, configure backup plans dynamically.

**1787. How do you deploy multi-region SSM parameters with hierarchical names, secure strings, versioning, and automatic updates?**

- Use `ssm_parameter_store` module, loop over regions, configure types, tiers, versions, and automatic updates dynamically.

**1788. How do you automate multi-region Kinesis Data Streams with dynamic shard scaling, encryption, and retention periods?**

- Use `kinesis_stream` module, loop over regions, configure `shard_count`, `kms_key_id`, and `retention_period` dynamically.



**1789. How do you orchestrate multi-region CloudFormation stack updates with change sets, drift detection, rollback, and notifications?**

- Use cloudformation module, loop over regions, create change sets, detect drift, apply rollback, and send notifications.

**1790. How do you deploy multi-region Elastic Beanstalk worker environments with dynamic SQS queues, IAM roles, and environment variables?**

- Use elasticbeanstalk module, loop over regions, configure environment tier, queues, IAM roles, and variables dynamically.

**1791. How do you automate multi-region S3 bucket encryption with KMS keys, versioning, and logging dynamically?**

- Use s3\_bucket module, loop over regions, configure encryption, versioning, and logging dynamically.

**1792. How do you orchestrate multi-region AWS Glue crawlers with dynamic targets, databases, and IAM roles?**

- Use glue\_crawler module, loop over regions, assign database\_name, targets, and roles dynamically.

**1793. How do you deploy multi-region ECS services with multiple task definitions, environment variables, secrets, and volumes dynamically?**

- Use ecs\_taskdefinition and ecs\_service modules, loop over regions, configure all parameters dynamically.

**1794. How do you automate multi-region API Gateway stages with throttling, logging, metrics, and stage variables?**

- Use apigateway\_stage module, loop over regions, configure throttling, logging, metrics, and variables dynamically.

**1795. How do you orchestrate multi-region VPC peering with route table updates, CIDR management, and security group rules dynamically?**

- Use ec2\_vpc\_peering and ec2\_vpc\_route\_table modules, loop over regions, configure routes and rules dynamically.

**\*\*1796. How do**

**you deploy multi-region Lambda function versions with aliases, routing, and blue/green deployment dynamically?\*\***

- Use lambda module, loop over regions, publish versions, create aliases, and route traffic dynamically.

**1797. How do you automate multi-region CloudTrail S3 bucket policies for cross-account logging and compliance dynamically?**

- Use s3\_bucket\_policy module, loop over regions, configure bucket policies dynamically.

**1798. How do you orchestrate multi-region ECS service discovery with Route53 private hosted zones, multiple services, and dynamic DNS records?**

- Use `ecs_service` and `route53_record` modules, loop over regions, configure service discovery and DNS records dynamically.

**1799. How do you deploy multi-region AWS Batch jobs with dynamic job queues, compute environments, container overrides, and dependencies?**

- Use `batch_job_queue` and `batch_compute_environment` modules, loop over regions, configure all parameters dynamically.

**1800. How do you implement a fully automated multi-region disaster recovery architecture for AWS Lambda, S3, RDS, ECS, and CloudFront using Ansible playbooks?**

- Combine modules: `lambda`, `s3_bucket`, `rds_instance`, `ecs_service`, `cloudfront_distribution`, loop over regions, configure replication, failover, monitoring, and automated routing policies dynamically.
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**1801. How do you automate multi-region AWS CodePipeline deployments with cross-region CodeBuild and CodeDeploy stages?**

- Use `codepipeline` module, loop over regions, configure stages for source, build, deploy, and assign dynamic cross-region CodeBuild/CodeDeploy actions.

**1802. How do you deploy multi-region Amazon MQ brokers with HA and cross-region replication dynamically?**

- Use `mq_broker` module, loop over regions, configure `deployment_mode`, `engine_version`, replication, and HA settings dynamically.

**1803. How do you orchestrate multi-region AWS IoT Core rules and topics with Lambda triggers and DynamoDB actions?**

- Use `iot_topic_rule` module, loop over regions, configure SQL statements, Lambda functions, and DynamoDB actions dynamically.

**1804. How do you automate multi-region Amazon MQ user management and access control dynamically?**

- Use `mq_user` module, loop over regions, configure username, password, permissions, and policies dynamically.

**1805. How do you deploy multi-region Elasticache Redis clusters with automatic failover, backup retention, and parameter groups?**

- Use `elasticache_replication_group` module, loop over regions, configure failover, backups, and parameter group settings dynamically.

**1806. How do you orchestrate multi-region AWS Secrets Manager secrets with rotation and cross-region replication?**

- Use `secretsmanager_secret` and `secretsmanager_secret_version` modules, loop over regions, enable automatic rotation and replication.

**1807. How do you automate multi-region AWS Organizations SCP policies using Ansible dynamically?**

- Use `organizations_policy` module, loop over accounts and regions, configure Service Control Policies dynamically.

**1808. How do you deploy multi-region AWS Step Functions with branching, retries, and cross-service integrations?**

- Use `stepfunctions_state_machine` module, loop over regions, configure branching, retries, and integrate Lambda, SQS, and SNS dynamically.

**1809. How do you orchestrate multi-region AWS CloudWatch synthetic canaries with dynamic scripts and monitoring alarms?**

- Use `cloudwatch_synthetics_canary` module, loop over regions, configure scripts, schedules, and CloudWatch alarms dynamically.

**1810. How do you automate multi-region AWS Transit Gateway attachments with VPCs and route table propagation?**

- Use `ec2_transit_gateway_vpc_attachment` module, loop over regions, attach VPCs, configure route propagation dynamically.

**1811. How do you deploy multi-region AWS App Mesh services with virtual nodes, virtual routers, and dynamic routing rules?**

- Use `appmesh_virtual_node` and `appmesh_virtual_router` modules, loop over regions, configure routing rules dynamically.

**1812. How do you orchestrate multi-region AWS CloudFront invalidations automatically after code deployments?**

- Use `cloudfront_distribution` module, loop over regions, trigger invalidations dynamically based on deployment events.

**1813. How do you automate multi-region AWS CodeArtifact repositories with package permissions, domains, and upstreams?**

- Use `codeartifact_repository` module, loop over regions, configure permissions, domains, and upstream repositories dynamically.

**1814. How do you deploy multi-region AWS Systems Manager Patch Baselines with rules for different OS types dynamically?**

- Use `ssm_patch_baseline` module, loop over regions, configure approval rules, OS types, and patch groups dynamically.

**1815. How do you orchestrate multi-region AWS Config conformance packs with dynamic remediation actions?**

- Use `aws_config_conformance_pack` module, loop over regions, assign rules and remediation actions dynamically.

**1816. How do you automate multi-region AWS KMS key rotation and cross-region replication?**

- Use kms\_key module, loop over regions, enable enable\_key\_rotation, and configure replica keys dynamically.

**1817. How do you deploy multi-region Amazon FSx file systems with automatic backups and security groups dynamically?**

- Use fsx\_file\_system module, loop over regions, configure backup, security groups, and deployment type dynamically.

**1818. How do you orchestrate multi-region AWS Elastic Beanstalk configuration updates with dynamic environment variables and scaling?**

- Use elasticbeanstalk module, loop over regions, configure environment variables, scaling, and deployment policies dynamically.

**1819. How do you automate multi-region Amazon MQ broker maintenance windows and engine version upgrades?**

- Use mq\_broker module, loop over regions, configure maintenance\_window\_start\_time and engine\_version dynamically.

**1820. How do you deploy multi-region AWS Lambda functions with provisioned concurrency and alias traffic shifting dynamically?**

- Use lambda module, loop over regions, configure provisioned\_concurrent\_executions, versions, aliases, and routing dynamically.

**1821. How do you orchestrate multi-region AWS WAFv2 rule updates with dynamic IP sets and rate limits?**

- Use wafv2\_web\_acl module, loop over regions, update rules, attach dynamic IP sets, and configure rate limits.

**1822. How do you automate multi-region Amazon Athena workgroups, named queries, and data catalog integration?**

- Use athena\_workgroup module, loop over regions, configure queries, workgroups, and catalog integration dynamically.

**1823. How do you deploy multi-region Amazon RDS Proxy with dynamic target groups and IAM authentication?**

- Use rds\_db\_proxy module, loop over regions, configure target\_groups, IAM authentication, and connection settings dynamically.

**1824. How do you orchestrate multi-region AWS CloudFormation custom resources with Lambda backends?**

- Use cloudformation module, loop over regions, configure custom\_resources pointing to Lambda functions dynamically.

**1825. How do you automate multi-region AWS Service Quotas updates and limit increases dynamically?**

- Use servicequotas\_service\_quota module, loop over regions, request quota updates, and monitor responses dynamically.

**1826. How do you deploy multi-region Amazon ElastiCache Redis clusters with cross-region backups and dynamic parameter groups?**

- Use `elasticache_replication_group` module, loop over regions, configure backup, failover, and parameter group dynamically.

**1827. How do you orchestrate multi-region AWS SNS FIFO topics with Lambda and SQS subscriptions dynamically?**

- Use `sns_topic` and `sns_subscription` modules, loop over regions, configure FIFO ordering and endpoints dynamically.

**1828. How do you automate multi-region AWS Lambda Layer deployments with dynamic dependencies and versioning?**

- Use `lambda_layer` module, loop over regions, configure compatible runtimes, dependencies, and versions dynamically.

**1829. How do you deploy multi-region AWS Backup plans with vaults, rules, and lifecycle policies dynamically?**

- Use `backup_plan` and `backup_selection` modules, loop over regions, configure rules, vaults, and retention dynamically.

**1830. How do you orchestrate multi-region AWS CloudWatch metric math alarms for complex multi-service monitoring?**

- Use `cloudwatch_metric_alarm` module, loop over regions, configure metric math expressions and thresholds dynamically.

**1831. How do you automate multi-region AWS Glue DataBrew projects with recipes and datasets dynamically?**

- Use `databrew_project` module, loop over regions, configure datasets, recipes, and output dynamically.

**1832. How do you deploy multi-region AWS Step Functions with Map states and dynamic parallel execution?**

- Use `stepfunctions_state_machine` module, loop over regions, configure Map states, iterations, and parallel tasks dynamically.

**1833. How do you orchestrate multi-region Amazon MQ ActiveMQ clusters with failover and replication dynamically?**

- Use `mq_broker` module, loop over regions, configure replication, failover, and broker settings dynamically.

**1834. How do you automate multi-region Amazon S3 bucket policies with cross-account access and dynamic object tagging rules?**

- Use `s3_bucket_policy` module, loop over regions, configure policies with dynamic tags and accounts.

**1835. How do you deploy multi-region AWS ECS Fargate tasks with dynamic CPU/memory, environment variables, and secrets?**

- Use `ecs_taskdefinition` and `ecs_service` modules, loop over regions, configure all parameters dynamically.

**1836. How do you orchestrate multi-region Amazon VPC endpoint services with private DNS and cross-account acceptance?**

- Use `ec2_vpc_endpoint_service` module, loop over regions, configure private DNS and cross-account permissions dynamically.

**1837. How do you automate multi-region Amazon API Gateway WebSocket APIs with Lambda integrations dynamically?**

- Use `apigatewayv2_api` and `apigatewayv2_integration` modules, loop over regions, configure routes and Lambda integrations.

**1838. How do you deploy multi-region Amazon Athena workgroups with encryption, output locations, and query limits dynamically?**

- Use `athena_workgroup` module, loop over regions, configure encryption, S3 output locations, and query limits dynamically.

**1839. How do you orchestrate multi-region Amazon RDS read replicas with cross-region replication and failover dynamically?**

- Use `rds_instance` module, loop over regions, configure `source_db_instance_identifier`, replication, and failover dynamically.

**1840. How do you automate multi-region AWS IAM role trust policies and permission boundaries dynamically?**

- Use `iam_role` module, loop over regions, configure `assume_role_policy` and permission boundaries dynamically.

**1841. How do you deploy multi-region AWS Secrets Manager rotation with Lambda functions and cross-region replication?**

- Use `secretsmanager_secret` module, loop over regions, enable rotation, assign Lambda, replicate secrets dynamically.

**1842. How do you orchestrate multi-region AWS Step Functions with error handling, retries, and fallback states?**

- Use `stepfunctions_state_machine` module, loop over regions, configure retries, catch, and fallback states dynamically.

**1843. How do you automate multi-region Amazon EC2 Auto Scaling groups with lifecycle hooks and dynamic scaling policies?**

- Use `ec2_asg` module, loop over regions, configure lifecycle hooks, min/max/desired capacities, and dynamic scaling policies.

**1844. How do you deploy multi-region Amazon RDS Proxy with IAM authentication and failover target groups dynamically?**

- Use `rds_db_proxy` module, loop over regions, configure IAM auth, target groups, and failover dynamically.

**1845. How do you orchestrate multi-region AWS CloudWatch Logs subscription filters to Lambda or Kinesis dynamically?**

- Use `cloudwatch_log_subscription_filter` module, loop over regions, configure destination dynamically.

**1846. How do you automate multi-region Amazon S3 replication metrics and notifications dynamically?**

- Use `s3_bucket_replication_configuration` module, loop over regions, configure metrics, events, and notifications dynamically.

**1847. How do you deploy multi-region Amazon ECS services with dynamic service discovery using private hosted zones?**

- Use `ecs_service` and `route53_record` modules, loop over regions, configure service discovery, private DNS dynamically.

**1848. How do you orchestrate multi-region Amazon VPC peering with dynamic route table updates and CIDR management?**

- Use `ec2_vpc_peering` and `ec2_vpc_route_table` modules, loop over regions, configure routes dynamically.

**1849. How do you automate multi-region AWS CloudFormation stack updates with change sets, drift detection, and rollback triggers dynamically?**

- Use `cloudformation` module, loop over regions, create change sets, detect drift, apply rollback triggers dynamically.

**1850. How do you deploy a fully automated multi-region disaster recovery environment for Lambda, S3, RDS, ECS, and CloudFront with Ansible playbooks?**

- Combine modules (`lambda`, `s3_bucket`, `rds_instance`, `ecs_service`, `cloudfront_distribution`), loop over regions, configure replication, failover, monitoring, and automated routing dynamically.

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**1851. How do you automate multi-region AWS CodeDeploy deployments with dynamic application revisions and deployment groups?**

- Use `codedeploy_app` and `codedeploy_deployment_group` modules, loop over regions, configure revision locations, deployment strategies, and hooks dynamically.

**1852. How do you deploy multi-region Amazon MQ brokers with dynamic user credentials and cross-region replication?**

- Use `mq_broker` and `mq_user` modules, loop over regions, configure engine type, replication, and credentials dynamically.

**1853. How do you orchestrate multi-region AWS Lambda function aliases with weighted traffic shifting and automatic rollbacks?**

- Use `lambda` module, loop over regions, create versions, configure aliases, route traffic, and rollbacks dynamically.

**1854. How do you automate multi-region Amazon S3 bucket lifecycle policies with versioning, archival, and cross-region replication?**

- Use `s3_bucket` module, loop over regions, configure versioning, lifecycle rules, and replication dynamically.

**1855. How do you deploy multi-region Amazon RDS Aurora global databases with automated failover and backup retention?**

- Use `rds_instance` module, loop over regions, configure global cluster settings, failover priorities, and backup retention.

**1856. How do you orchestrate multi-region AWS Step Functions with dynamic Map states and parallel Lambda invocations?**

- Use `stepfunctions_state_machine` module, loop over regions, configure Map states, parallel tasks, and error handling dynamically.

**1857. How do you automate multi-region Amazon ECS Fargate service deployments with environment variables, secrets, and dynamic task definitions?**

- Use `ecs_service` and `ecs_taskdefinition` modules, loop over regions, configure secrets, CPU/memory, and environment dynamically.

**1858. How do you deploy multi-region AWS Config conformance packs with dynamic remediation actions and compliance reporting?**

- Use `aws_config_conformance_pack` module, loop over regions, configure rules, remediation actions, and reporting dynamically.

**1859. How do you orchestrate multi-region Amazon CloudFront invalidations automatically after multi-region S3 deployments?**

- Use `cloudfront_distribution` module, loop over regions, trigger invalidations dynamically based on deployment events.

**1860. How do you automate multi-region AWS Secrets Manager secrets with rotation Lambda and cross-region replication?**

- Use `secretsmanager_secret` module, loop over regions, enable rotation, assign Lambda functions, replicate secrets dynamically.

**1861. How do you deploy multi-region AWS Backup plans with vaults, backup rules, and lifecycle policies dynamically?**

- Use `backup_plan` and `backup_selection` modules, loop over regions, configure vaults, rules, and retention dynamically.

**1862. How do you orchestrate multi-region Amazon DynamoDB global tables with TTL, streams, auto-scaling, and backup policies?**

- Use `dynamodb_table` module, loop over regions, configure replicas, TTL, streams, auto-scaling, and backup dynamically.

**1863. How do you automate multi-region AWS CloudWatch composite alarms aggregating metrics from multiple services dynamically?**



- Use `cloudwatch_composite_alarm` module, loop over regions, combine metrics from EC2, RDS, Lambda, and ECS dynamically.

**1864. How do you deploy multi-region Amazon Athena named queries, workgroups, and encryption settings dynamically?**

- Use `athena_workgroup` and `athena_named_query` modules, loop over regions, configure workgroups, queries, encryption, and output dynamically.

**1865. How do you orchestrate multi-region AWS IAM policies with dynamic permission boundaries and trust relationships?**

- Use `iam_role` and `iam_policy` modules, loop over regions, configure assume role policies, permission boundaries, and attachments dynamically.

**1866. How do you automate multi-region Amazon CloudWatch Logs subscription filters for Lambda and Kinesis destinations?**

- Use `cloudwatch_log_subscription_filter` module, loop over regions, configure destinations and filter patterns dynamically.

**1867. How do you deploy multi-region AWS Glue jobs with dynamic connections, scripts, and IAM roles?**

- Use `glue_job` module, loop over regions, assign connections, script locations, and IAM roles dynamically.

**1868. How do you orchestrate multi-region Amazon RDS read replicas with dynamic failover, backup retention, and multi-AZ deployment?**

- Use `rds_instance` module, loop over regions, configure read replicas, failover, backups, and multi-AZ dynamically.

**1869. How do you automate multi-region AWS Lambda layers with versioning, compatible runtimes, and dependencies?**

- Use `lambda_layer` module, loop over regions, configure versions, runtime compatibility, and dependencies dynamically.

**1870. How do you deploy multi-region AWS ECS service discovery using Route53 private hosted zones with dynamic DNS records?**

- Use `ecs_service` and `route53_record` modules, loop over regions, configure private DNS, service discovery, and records dynamically.

**1871. How do you orchestrate multi-region Amazon VPC endpoint services with cross-account acceptance and private DNS dynamically?**

- Use `ec2_vpc_endpoint_service` module, loop over regions, configure acceptance and DNS dynamically.

**1872. How do you automate multi-region AWS CloudFormation custom resources using Lambda backends for dynamic provisioning?**

- Use `cloudformation` module, loop over regions, configure custom resources with Lambda backends dynamically.

**1873. How do you deploy multi-region AWS Transit Gateway attachments with route propagation and dynamic VPC associations?**

- Use `ec2_transit_gateway_vpc_attachment` module, loop over regions, configure attachments and route tables dynamically.

**1874. How do you orchestrate multi-region AWS App Mesh virtual nodes and virtual routers with dynamic routing rules?**

- Use `appmesh_virtual_node` and `appmesh_virtual_router` modules, loop over regions, configure routes dynamically.

**1875. How do you automate multi-region Amazon S3 bucket policies with cross-account access, encryption, and lifecycle rules?**

- Use `s3_bucket_policy` and `s3_bucket` modules, loop over regions, configure policies, encryption, and lifecycle dynamically.

**1876. How do you deploy multi-region Amazon EC2 Auto Scaling groups with dynamic scaling policies, lifecycle hooks, and instance types?**

- Use `ec2_asg` module, loop over regions, configure scaling policies, hooks, and instance types dynamically.

**1877. How do you orchestrate multi-region AWS Lambda function aliases for blue/green deployments with weighted traffic?**

- Use `lambda` module, loop over regions, create versions, configure aliases, and traffic weights dynamically.

**1878. How do you automate multi-region AWS WAFv2 web ACL updates with dynamic IP sets and rate-based rules?**

- Use `wafv2_web_acl` and `wafv2_rule_group` modules, loop over regions, configure rules dynamically.

**1879. How do you deploy multi-region Amazon RDS Proxy with IAM authentication, target groups, and failover settings?**

- Use `rds_db_proxy` module, loop over regions, configure authentication, targets, and failover dynamically.

**1880. How do you orchestrate multi-region AWS CodeBuild projects with dynamic environment variables, buildspecs, and artifacts?**

- Use `codebuild_project` module, loop over regions, configure environment, buildspecs, and artifact locations dynamically.

**1881. How do you automate multi-region Amazon MQ ActiveMQ brokers with replication, failover, and dynamic user permissions?**

- Use `mq_broker` and `mq_user` modules, loop over regions, configure replication, HA, and permissions dynamically.

**1882. How do you deploy multi-region AWS Secrets Manager secrets with automatic rotation, cross-region replication, and versioning?**

- Use `secretsmanager_secret` module, loop over regions, enable rotation, replication, and version tracking dynamically.

**1883. How do you orchestrate multi-region AWS Batch job queues and compute environments with dynamic allocation strategies?**

- Use `batch_job_queue` and `batch_compute_environment` modules, loop over regions, configure allocation and priority dynamically.

**1884. How do you automate multi-region Amazon CloudWatch metric math alarms for multi-service monitoring dynamically?**

- Use `cloudwatch_metric_alarm` module, loop over regions, configure metric math expressions and thresholds dynamically.

**1885. How do you deploy multi-region AWS Step Functions workflows with dynamic error handling, retries, and fallback states?**

- Use `stepfunctions_state_machine` module, loop over regions, configure retries, catch, and fallback states dynamically.

**1886. How do you orchestrate multi-region Amazon ECS services with multiple task definitions, secrets, and environment variables?**

- Use `ecs_service` and `ecs_taskdefinition` modules, loop over regions, configure secrets, environment, and volumes dynamically.

**1887. How do you automate multi-region Amazon Athena named queries and workgroups with encryption and S3 output locations?**

- Use `athena_workgroup` and `athena_named_query` modules, loop over regions, configure encryption and output dynamically.

**1888. How do you deploy multi-region Amazon CloudFront distributions with multiple origins, caching policies, and Lambda@Edge integrations?**

- Use `cloudfront_distribution` module, loop over regions, configure origins, behaviors, and Lambda@Edge dynamically.

**1889. How do you orchestrate multi-region AWS Glue crawlers and jobs with dynamic databases, connections, and IAM roles?**

- Use `glue_crawler` and `glue_job` modules, loop over regions, configure targets, scripts, and roles dynamically.

**1890. How do you automate multi-region AWS Config rules with dynamic remediation actions and compliance reporting?**

- Use `aws_config_config_rule` module, loop over regions, assign rules, remediation actions, and reporting dynamically.

**1891. How do you deploy multi-region AWS Elastic Beanstalk environments with rolling and immutable deployment policies?**

- Use `elasticbeanstalk` module, loop over regions, configure deployment policies, environment variables, and scaling dynamically.

**1892. How do you orchestrate multi-region Amazon ECS service discovery using private hosted zones and dynamic DNS records?**

- Use `ecs_service` and `route53_record` modules, loop over regions, configure private DNS and service discovery dynamically.

**1893. How do you automate multi-region Amazon S3 replication with metrics, notifications, and lifecycle transitions dynamically?**

- Use `s3_bucket_replication_configuration` module, loop over regions, configure metrics, events, and lifecycle rules dynamically.

**1894. How do you deploy multi-region Amazon EC2 instances with dynamic IAM roles, monitoring agents, tags, and security groups?**

- Use `ec2_instance` module, loop over regions, configure IAM roles, CloudWatch agents, tags, and security groups dynamically.

**1895. How do you orchestrate multi-region Amazon VPC peering connections with dynamic route table updates and CIDR management?**

- Use `ec2_vpc_peering` and `ec2_vpc_route_table` modules, loop over regions, configure routes and CIDRs dynamically.

**1896. How do you automate multi-region AWS CloudFormation stack updates using change sets, drift detection, and rollback triggers?**

- Use `cloudformation` module, loop over regions, create change sets, detect drift, and apply rollback triggers dynamically.

**1897. How do you deploy multi-region Amazon Lambda provisioned concurrency with dynamic versioning and alias routing?**

- Use `lambda` module, loop over regions, configure provisioned concurrency, versions, and alias routing dynamically.

**1898. How do you orchestrate multi-region Amazon API Gateway stages with logging, throttling, and stage variables dynamically?**

- Use `apigateway_stage` module, loop over regions, configure throttling, logging, and stage variables dynamically.

**1899. How do you automate multi-region Amazon CloudWatch dashboards aggregating metrics from multiple services dynamically?**

- Use `cloudwatch_dashboard` module, loop over regions, configure widgets, metrics, and layout dynamically.

**1900. How do you deploy a fully automated multi-region disaster recovery solution for Lambda, ECS, RDS, S3, and CloudFront using Ansible playbooks?**

- Combine modules (`lambda`, `ecs_service`, `rds_instance`, `s3_bucket`, `cloudfront_distribution`), loop over regions, configure replication, failover, monitoring, and automated routing dynamically.
-

**1901. How do you automate multi-region AWS CodePipeline with dynamic cross-region CodeBuild and CodeDeploy actions?**

- Use codepipeline module, loop over regions, configure stages dynamically for cross-region builds and deployments.

**1902. How do you deploy multi-region Amazon MQ brokers with HA, replication, and dynamic user credentials?**

- Use mq\_broker and mq\_user modules, loop over regions, configure engine, replication, and credentials dynamically.

**1903. How do you orchestrate multi-region AWS Lambda aliases with weighted traffic shifting and rollback automation?**

- Use lambda module, loop over regions, create versions, assign aliases, configure weights, and automatic rollback.

**1904. How do you automate multi-region Amazon S3 bucket lifecycle policies with cross-region replication and versioning?**

- Use s3\_bucket module, loop over regions, configure lifecycle, versioning, and replication dynamically.

**1905. How do you deploy multi-region Amazon RDS Aurora global databases with failover and backup retention dynamically?**

- Use rds\_instance module, loop over regions, configure global cluster, failover priorities, and backup retention dynamically.

**1906. How do you orchestrate multi-region AWS Step Functions with dynamic Map states, parallel executions, and error handling?**

- Use stepfunctions\_state\_machine module, loop over regions, configure Map states, parallel tasks, and error handling dynamically.

**1907. How do you automate multi-region ECS Fargate service deployments with dynamic secrets, environment variables, and task definitions?**

- Use ecs\_service and ecs\_taskdefinition modules, loop over regions, configure secrets, environment variables, and CPU/memory dynamically.

**1908. How do you deploy multi-region AWS Config conformance packs with dynamic remediation actions and compliance reporting?**

- Use aws\_config\_conformance\_pack module, loop over regions, assign rules and remediation actions dynamically.

**1909. How do you orchestrate multi-region Amazon CloudFront invalidations after S3 deployments automatically?**

- Use cloudfront\_distribution module, loop over regions, trigger invalidations dynamically after deployments.

**1910. How do you automate multi-region AWS Secrets Manager secrets with rotation Lambda and cross-region replication?**

- Use `secretsmanager_secret` module, loop over regions, enable rotation, assign Lambda, and replicate secrets dynamically.

**1911. How do you deploy multi-region AWS Backup plans with vaults, rules, and lifecycle policies dynamically?**

- Use `backup_plan` and `backup_selection` modules, loop over regions, configure vaults, rules, and retention dynamically.

**1912. How do you orchestrate multi-region Amazon DynamoDB global tables with TTL, streams, auto-scaling, and backup policies?**

- Use `dynamodb_table` module, loop over regions, configure replicas, TTL, streams, auto-scaling, and backup dynamically.

**1913. How do you automate multi-region CloudWatch composite alarms aggregating multiple service metrics dynamically?**

- Use `cloudwatch_composite_alarm` module, loop over regions, combine metrics from EC2, RDS, Lambda, ECS dynamically.

**1914. How do you deploy multi-region Athena named queries, workgroups, and encryption dynamically?**

- Use `athena_workgroup` and `athena_named_query` modules, loop over regions, configure workgroups, queries, encryption dynamically.

**1915. How do you orchestrate multi-region AWS IAM policies with dynamic permission boundaries and trust relationships?**

- Use `iam_role` and `iam_policy` modules, loop over regions, configure assume role policies, boundaries, and attachments dynamically.

**1916. How do you automate multi-region CloudWatch Logs subscription filters for Lambda and Kinesis dynamically?**

- Use `cloudwatch_log_subscription_filter` module, loop over regions, configure destinations and filter patterns dynamically.

**1917. How do you deploy multi-region AWS Glue jobs with dynamic connections, scripts, and IAM roles?**

- Use `glue_job` module, loop over regions, assign connections, script locations, and roles dynamically.

**1918. How do you orchestrate multi-region RDS read replicas with dynamic failover, backup retention, and multi-AZ deployment?**

- Use `rds_instance` module, loop over regions, configure read replicas, failover, backups, multi-AZ dynamically.

**1919. How do you automate multi-region Lambda layers with versioning, compatible runtimes, and dependencies?**

- Use `lambda_layer` module, loop over regions, configure versions, runtime compatibility, and dependencies dynamically.

**1920. How do you deploy multi-region ECS service discovery using Route53 private hosted zones with dynamic DNS records?**

- Use `ecs_service` and `route53_record` modules, loop over regions, configure private DNS and service discovery dynamically.

**1921. How do you orchestrate multi-region VPC endpoint services with cross-account acceptance and private DNS dynamically?**

- Use `ec2_vpc_endpoint_service` module, loop over regions, configure acceptance and DNS dynamically.

**1922. How do you automate multi-region CloudFormation custom resources using Lambda backends dynamically?**

- Use `cloudformation` module, loop over regions, configure custom resources with Lambda backends.

**1923. How do you deploy multi-region Transit Gateway attachments with route propagation and dynamic VPC associations?**

- Use `ec2_transit_gateway_vpc_attachment` module, loop over regions, configure attachments and route tables dynamically.

**1924. How do you orchestrate multi-region App Mesh virtual nodes and routers with dynamic routing rules?**

- Use `appmesh_virtual_node` and `appmesh_virtual_router` modules, loop over regions, configure routes dynamically.

**1925. How do you automate multi-region S3 bucket policies with cross-account access, encryption, and lifecycle rules?**

- Use `s3_bucket_policy` and `s3_bucket` modules, loop over regions, configure policies, encryption, and lifecycle rules dynamically.

**1926. How do you deploy multi-region EC2 Auto Scaling groups with dynamic scaling policies, lifecycle hooks, and instance types?**

- Use `ec2_asg` module, loop over regions, configure scaling policies, hooks, and instance types dynamically.

**1927. How do you orchestrate multi-region Lambda aliases for blue/green deployments with weighted traffic?**

- Use `lambda` module, loop over regions, create versions, configure aliases and weights dynamically.

**1928. How do you automate multi-region WAFv2 Web ACL updates with dynamic IP sets and rate-based rules?**

- Use `wafv2_web_acl` and `wafv2_rule_group` modules, loop over regions, configure rules dynamically.

**1929. How do you deploy multi-region RDS Proxy with IAM authentication, target groups, and failover settings dynamically?**

- Use `rds_db_proxy` module, loop over regions, configure authentication, targets, and failover dynamically.

**1930. How do you orchestrate multi-region CodeBuild projects with dynamic environment variables, buildspecs, and artifact locations?**

- Use `codebuild_project` module, loop over regions, configure environment, buildspecs, and artifact locations dynamically.

**1931. How do you automate multi-region Amazon MQ ActiveMQ brokers with replication, failover, and dynamic user permissions?**

- Use `mq_broker` and `mq_user` modules, loop over regions, configure replication, HA, and permissions dynamically.

**1932. How do you deploy multi-region Secrets Manager secrets with automatic rotation, cross-region replication, and versioning?**

- Use `secretsmanager_secret` module, loop over regions, enable rotation, replication, and version tracking dynamically.

**1933. How do you orchestrate multi-region AWS Batch job queues and compute environments with dynamic allocation strategies?**

- Use `batch_job_queue` and `batch_compute_environment` modules, loop over regions, configure allocation strategies dynamically.

**1934. How do you automate multi-region CloudWatch metric math alarms for multi-service monitoring dynamically?**

- Use `cloudwatch_metric_alarm` module, loop over regions, configure metric math expressions and thresholds dynamically.

**1935. How do you deploy multi-region Step Functions workflows with dynamic error handling, retries, and fallback states?**

- Use `stepfunctions_state_machine` module, loop over regions, configure retries, catch, and fallback states dynamically.

**1936. How do you orchestrate multi-region ECS services with multiple task definitions, secrets, and environment variables dynamically?**

- Use `ecs_service` and `ecs_taskdefinition` modules, loop over regions, configure secrets, environment, and volumes dynamically.

**1937. How do you automate multi-region Athena named queries and workgroups with encryption and S3 output locations dynamically?**

- Use `athena_workgroup` and `athena_named_query` modules, loop over regions, configure encryption and output locations dynamically.

**1938. How do you deploy multi-region CloudFront distributions with multiple origins, caching policies, and Lambda@Edge integrations dynamically?**

- Use `cloudfront_distribution` module, loop over regions, configure origins, behaviors, and Lambda@Edge dynamically.



**1939. How do you orchestrate multi-region Glue crawlers and jobs with dynamic databases, connections, and IAM roles?**

- Use glue\_crawler and glue\_job modules, loop over regions, configure targets, scripts, and IAM roles dynamically.

**1940. How do you automate multi-region AWS Config rules with dynamic remediation actions and compliance reporting?**

- Use aws\_config\_config\_rule module, loop over regions, assign rules, remediation actions, and reporting dynamically.

**1941. How do you deploy multi-region Elastic Beanstalk environments with rolling and immutable deployment policies dynamically?**

- Use elasticbeanstalk module, loop over regions, configure deployment policies, environment variables, and scaling dynamically.

**1942. How do you orchestrate multi-region ECS service discovery using private hosted zones and dynamic DNS records?**

- Use ecs\_service and route53\_record modules, loop over regions, configure private DNS and service discovery dynamically.

**1943. How do you automate multi-region S3 replication with metrics, notifications, and lifecycle transitions dynamically?**

- Use s3\_bucket\_replication\_configuration module, loop over regions, configure metrics, events, and lifecycle rules dynamically.

**1944. How do you deploy multi-region EC2 instances with dynamic IAM roles, monitoring agents, tags, and security groups?**

- Use ec2\_instance module, loop over regions, configure IAM roles, CloudWatch agents, tags, and security groups dynamically.

**1945. How do you orchestrate multi-region VPC peering connections with dynamic route table updates and CIDR management?**

- Use ec2\_vpc\_peering and ec2\_vpc\_route\_table modules, loop over regions, configure routes and CIDRs dynamically.

**1946. How do you automate multi-region CloudFormation stack updates using change sets, drift detection, and rollback triggers dynamically?**

- Use cloudformation module, loop over regions, create change sets, detect drift, and apply rollback triggers dynamically.

**1947. How do you deploy multi-region Lambda provisioned concurrency with dynamic versioning and alias routing?**

- Use lambda module, loop over regions, configure provisioned concurrency, versions, and alias routing dynamically.

**1948. How do you orchestrate multi-region API Gateway stages with logging, throttling, and stage variables dynamically?**

- Use `apigateway_stage` module, loop over regions, configure throttling, logging, and stage variables dynamically.

**1949. How do you automate multi-region CloudWatch dashboards aggregating metrics from multiple services dynamically?**

- Use `cloudwatch_dashboard` module, loop over regions, configure widgets, metrics, and layout dynamically.

**1950. How do you deploy a fully automated multi-region disaster recovery environment for Lambda, ECS, RDS, S3, and CloudFront using Ansible playbooks dynamically?**

- Combine modules (`lambda`, `ecs_service`, `rds_instance`, `s3_bucket`, `cloudfront_distribution`), loop over regions, configure replication, failover, monitoring, and automated routing dynamically.

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**1951. How do you orchestrate multi-region AWS Organizations accounts creation and Service Control Policies using Ansible?**

Use `aws_organizations_account` and `aws_organizations_policy` modules, loop over regions/accounts, attach SCPs dynamically.

**1952. How do you automate multi-region AWS Service Catalog portfolios, products, and launch constraints dynamically?**

Use `servicecatalog_portfolio` and `servicecatalog_product` modules, loop over regions, configure launch constraints, and tags dynamically.

**1953. How do you deploy multi-region Amazon VPCs with dynamic subnets, route tables, NAT gateways, and security groups?**

Use `ec2_vpc`, `ec2_vpc_subnet`, `ec2_vpc_route_table`, and `ec2_security_group` modules, loop over regions, configure networking dynamically.

**1954. How do you orchestrate multi-region AWS EKS clusters with dynamic node groups, IAM roles, and addon configurations?**

Use `eks_cluster` and `eks_nodegroup` modules, loop over regions, configure nodes, IAM roles, and addons dynamically.

**1955. How do you automate multi-region AWS FSx for Lustre or Windows file systems with backups and security groups?**

Use `fsx_file_system` module, loop over regions, configure type, backup policies, and access dynamically.

**1956. How do you deploy multi-region AWS AppConfig applications and environments with dynamic deployment strategies?**

Use `appconfig_application` and `appconfig_environment` modules, loop over regions, configure strategies, and deployment dynamically.

**1957. How do you orchestrate multi-region Amazon ElastiCache clusters with replication, sharding, and automatic failover?**

Use `elasticache_replication_group` module, loop over regions, configure shards, replication, and failover dynamically.

**1958. How do you automate multi-region AWS CloudTrail trails with multi-account and multi-region logging dynamically?**

Use `cloudtrail` module, loop over regions, configure S3 logging, SNS notifications, and multi-account aggregation dynamically.

**1959. How do you deploy multi-region Amazon Kinesis Data Streams and Firehose delivery streams with dynamic processing and encryption?**

Use `kinesis_stream` and `kinesis_firehose` modules, loop over regions, configure shards, encryption, and destinations dynamically.

**1960. How do you orchestrate multi-region AWS DataSync tasks with dynamic source/destination locations and scheduling?**

Use `datasync_task` module, loop over regions, configure source/destination locations, scheduling, and filters dynamically.

**1961. How do you automate multi-region AWS Config aggregators for multi-account compliance reporting?**

Use `aws_config_configuration_aggregator` module, loop over regions/accounts, configure sources, aggregators, and notifications dynamically.

**1962. How do you deploy multi-region Amazon EventBridge rules with dynamic targets, retries, and dead-letter queues?**

Use `events_rule` module, loop over regions, configure targets, retry policies, and DLQs dynamically.

**1963. How do you orchestrate multi-region AWS Systems Manager Parameter Store hierarchies with dynamic naming, encryption, and permissions?**

Use `ssm_parameter` module, loop over regions, configure hierarchy, KMS encryption, and IAM permissions dynamically.

**1964. How do you automate multi-region AWS CloudWatch Contributor Insights rules and dashboards dynamically?**

Use `cloudwatch_contributor_insights` module, loop over regions, configure rules, metrics, and dashboards dynamically.

**1965. How do you deploy multi-region Amazon Redshift clusters with snapshots, encryption, and IAM roles dynamically?**

Use `redshift_cluster` module, loop over regions, configure snapshots, encryption, and IAM roles dynamically.

**1966. How do you orchestrate multi-region AWS OpsWorks stacks and layers with dynamic instance profiles, lifecycle events, and scaling?**

Use `opsworks_stack` and `opsworks_layer` modules, loop over regions, configure instances, profiles, events, and scaling dynamically.

**1967. How do you automate multi-region AWS Service Quotas requests and monitoring using Ansible?**

Use `service_quota_request` module, loop over regions, configure quotas, thresholds, and monitoring dynamically.

**1968. How do you deploy multi-region Amazon SageMaker endpoints with dynamic models, variants, and auto-scaling?**

Use `sagemaker_endpoint` and `sagemaker_model` modules, loop over regions, configure variants, scaling policies, and IAM roles dynamically.

**1969. How do you orchestrate multi-region AWS App Runner services with dynamic environment variables, secrets, and auto-scaling?**

Use `apprunner_service` module, loop over regions, configure environment, secrets, and scaling dynamically.

**1970. How do you automate multi-region AWS CodeArtifact repositories, domains, and upstreams dynamically?**

Use `codeartifact_repository` and `codeartifact_domain` modules, loop over regions, configure upstreams and policies dynamically.

**1971. How do you deploy multi-region Amazon MQ ActiveMQ brokers with encrypted storage, cross-region replication, and failover configurations?**

Use `mq_broker` module, loop over regions, configure encryption, replication, and failover dynamically.

**1972. How do you orchestrate multi-region AWS CloudWatch log groups with retention policies, metric filters, and subscription filters dynamically?**

Use `cloudwatch_log_group` module, loop over regions, configure retention, filters, and destinations dynamically.

**1973. How do you automate multi-region AWS Lambda function concurrency and versioning for high availability deployments?**

Use `lambda` module, loop over regions, configure reserved/provisioned concurrency, versions, and aliases dynamically.

**1974. How do you deploy multi-region Amazon ECS services with multiple load balancers, task sets, and dynamic scaling?**

Use `ecs_service` and `ecs_taskdefinition` modules, loop over regions, configure load balancers, task sets, and scaling dynamically.

**1975. How do you orchestrate multi-region AWS S3 access points with policies and cross-account access dynamically?**

Use `s3_access_point` module, loop over regions, configure policies, VPC restrictions, and cross-account access dynamically.

**1976. How do you automate multi-region AWS Step Functions Map states with dynamic parallel execution and retries?**

Use `stepfunctions_state_machine` module, loop over regions, configure Map states, parallel tasks, retries, and error handling dynamically.

**1977. How do you deploy multi-region Amazon ECR repositories with replication, image scanning, and lifecycle policies dynamically?**

Use `ecr_repository` module, loop over regions, configure replication rules, scan on push, and lifecycle policies dynamically.

**1978. How do you orchestrate multi-region AWS AppConfig feature flags with dynamic deployment strategies and environments?**

Use `appconfig_feature` module, loop over regions, configure strategies, environments, and versions dynamically.

**1979. How do you automate multi-region AWS KMS keys with rotation, alias management, and cross-account usage dynamically?**

Use `kms_key` module, loop over regions, configure rotation, aliases, and cross-account permissions dynamically.

**1980. How do you deploy multi-region Amazon CloudWatch anomaly detection and alarms dynamically for multi-service monitoring?**

Use `cloudwatch_metric_alarm` module, loop over regions, configure anomaly detection, metrics, and alarms dynamically.

**1981. How do you orchestrate multi-region Amazon API Gateway custom domain names, mappings, and TLS certificates dynamically?**

Use `apigateway_domain_name` module, loop over regions, configure domain mappings and TLS certificates dynamically.

**1982. How do you automate multi-region AWS Elastic Load Balancers with listeners, target groups, and cross-zone load balancing?**

Use `elb` and `elb_target_group` modules, loop over regions, configure listeners, targets, and cross-zone balancing dynamically.

**1983. How do you deploy multi-region AWS CloudFormation StackSets with dynamic accounts, regions, and parameter overrides?**

Use `cloudformation_stack_set` module, loop over regions/accounts, configure parameters and deployment targets dynamically.

**1984. How do you orchestrate multi-region Amazon SNS topics and subscriptions with dynamic endpoints, protocols, and filtering?**

Use `sns_topic` and `sns_subscription` modules, loop over regions, configure protocols, endpoints, and filters dynamically.

**1985. How do you automate multi-region Amazon SQS queues with encryption, dead-letter queues, and access policies dynamically?**

Use `sqs_queue` module, loop over regions, configure encryption, DLQs, and policies dynamically.

**1986. How do you deploy multi-region AWS CloudFront distributions with dynamic origins, cache behaviors, and Lambda@Edge functions?**

Use `cloudfront_distribution` module, loop over regions, configure origins, cache policies, and Lambda@Edge dynamically.

**1987. How do you orchestrate multi-region AWS ECS clusters with dynamic capacity providers, services, and task placement strategies?**

Use `ecs_cluster` and `ecs_service` modules, loop over regions, configure capacity providers, placement strategies, and scaling dynamically.

**1988. How do you automate multi-region AWS CloudTrail Insights events for anomalous activity detection dynamically?**

Use `cloudtrail` module, loop over regions, configure Insights selectors and logging dynamically.

**1989. How do you deploy multi-region AWS S3 Intelligent-Tiering with dynamic lifecycle policies and replication?**

Use `s3_bucket` module, loop over regions, configure Intelligent-Tiering, lifecycle, and replication dynamically.

**1990. How do you orchestrate multi-region AWS CodeDeploy blue/green deployments for ECS and Lambda services dynamically?**

Use `codedeploy_app`, `codedeploy_deployment_group` modules, loop over regions, configure ECS/Lambda blue/green deployments dynamically.

**1991. How do you automate multi-region Amazon RDS Proxy failover configurations with IAM authentication dynamically?**

Use `rds_db_proxy` module, loop over regions, configure failover, targets, and IAM authentication dynamically.

**1992. How do you deploy multi-region AWS Secrets Manager secrets with cross-region rotation and automatic failover?**

Use `secretsmanager_secret` module, loop over regions, enable rotation, replication, and failover dynamically.

**1993. How do you orchestrate multi-region AWS Step Functions with dynamic workflow versioning and deployment strategies?**

Use `stepfunctions_state_machine` module, loop over regions, configure versions, aliases, and deployment strategies dynamically.

**1994. How do you automate multi-region Amazon DynamoDB auto-scaling, backup, and global tables dynamically?**

Use `dynamodb_table` module, loop over regions, configure auto-scaling, backups, and global tables dynamically.

**1995. How do you deploy multi-region Amazon Athena queries with dynamic workgroups, encryption, and S3 outputs?**

Use `athena_named_query` and `athena_workgroup` modules, loop over regions, configure queries, encryption, and outputs dynamically.

**1996. How do you orchestrate multi-region AWS Lambda functions with environment variables, VPC configuration, and dead-letter queues?**

Use `lambda` module, loop over regions, configure environment, VPC, and DLQs dynamically.

**1997. How do you automate multi-region AWS Glue crawlers with dynamic connections, targets, and IAM roles?**

Use `glue_crawler` module, loop over regions, configure connections, databases, and roles dynamically.

**1998. How do you deploy multi-region AWS CloudWatch dashboards and alarms aggregating metrics from multiple accounts dynamically?**

Use `cloudwatch_dashboard` and `cloudwatch_metric_alarm` modules, loop over regions/accounts, configure metrics, widgets, and alarms dynamically.

**1999. How do you orchestrate multi-region AWS Elastic Beanstalk environments with dynamic configuration templates, rolling updates, and monitoring?**

Use `elasticbeanstalk` module, loop over regions, configure templates, deployment policies, and monitoring dynamically.

**2000. How do you implement a fully automated multi-region disaster recovery solution for Lambda, ECS, RDS, S3, CloudFront, and API Gateway using Ansible?**

Combine modules (`lambda`, `ecs_service`, `rds_instance`, `s3_bucket`, `cloudfront_distribution`, `apigateway_stage`), loop over regions, configure replication, failover, monitoring, and automated routing dynamically.

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Here's a comprehensive **Ansible interview tips, common mistakes, and scenario-based guidance** tailored for a **DevOps person with AWS experience**:

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## **1. Ansible Interview Tips for AWS DevOps**

### **A. Preparation**

#### **Understand Core Concepts**

Inventory (static vs dynamic)

Playbooks, Roles, Tasks

Variables, Facts, Templates (Jinja2)

Handlers, Conditionals, Loops

Modules (AWS-specific: `ec2`, `s3`, `rds`, `lambda`, `cloudformation`, etc.)

#### **AWS Integration**

Know how to use **Ansible AWS modules**.

Understand **boto3** credentials and roles.

Be ready to automate:

EC2 provisioning

RDS deployment & backups

Lambda & ECS deployments

S3 bucket policies, replication, versioning

CloudFormation stack automation

### **Practical Scenarios**

Automating CI/CD pipelines

Multi-region deployments

Infrastructure as Code (IaC) with Ansible + CloudFormation

Disaster recovery automation

Security hardening (IAM, Security Groups, KMS)

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## **B. Key Points to Mention in Interviews**

Explain **Idempotency**: Ansible ensures the same playbook produces the same results without causing unwanted changes.

Highlight **Dynamic Inventory** usage for AWS EC2 instances.

Showcase **Roles and Modular Playbooks** for reusable and maintainable code.

Emphasize **Error Handling**:

`ignore_errors`, `failed_when`, `block/rescue/always`

Stress **Integration with AWS Services**:

Using `cloudformation` module to deploy stacks

Using `ec2`, `s3_bucket`, `rds_instance`, `lambda` modules

Explain **Security Automation**: Secrets management via Ansible Vault or AWS Secrets Manager.

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## **2. Common Mistakes in Ansible Interviews**

**Not using roles:**

Writing huge playbooks without structure.

**Ignoring Idempotency:**



Running tasks that fail on repeated runs.

**Hardcoding credentials:**

Instead of using AWS IAM roles, environment variables, or Vault.

**Poor variable management:**

Mixing global, group, and host variables inconsistently.

**Not using dynamic inventory for AWS:**

Hardcoding EC2 host IPs instead of using ec2.py or AWS Inventory plugin.

**Ignoring error handling:**

Not using block/rescue/always or checks for failures.

**No testing or dry runs:**

Skipping ansible-playbook --check or --diff to validate changes.

**Overlooking module-specific options:**

E.g., not setting wait or state properly for EC2 or RDS modules.

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### 3. Scenario-Based Questions & Answers

#### Scenario 1: EC2 Deployment with Auto Scaling

**Q:** How would you deploy EC2 instances dynamically in multiple regions with auto-scaling and monitoring?

**A:**

Use ec2\_instance module with loop over regions.

Configure IAM roles for instance access.

Use ec2\_asg module to create Auto Scaling groups.

Integrate CloudWatch monitoring with cloudwatch\_alarm module.

Ensure idempotency by using tags and state: present.

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#### Scenario 2: Lambda & S3 Automation

**Q:** How to automate Lambda deployment triggered by S3 uploads in multiple regions?

**A:**

Use lambda module to deploy functions.

Configure s3\_bucket module for triggering events.

Use dynamic inventory for multiple regions.

Include versioning, aliases, and environment variables.

Add rollback logic using `lambda_version` and `aliases`.

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### Scenario 3: Disaster Recovery for Multi-Region Infrastructure

**Q:** How would you create an automated multi-region DR setup?

**A:**

Use `s3_bucket` for replication, `rds_instance` for read replicas.

Deploy `ecs_service` and `lambda` functions across regions.

Use `cloudfront_distribution` for multi-region endpoints.

Automate failover using Route53 with health checks.

Encapsulate all in reusable roles and include DR validation playbooks.

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### Scenario 4: Security Automation

**Q:** How would you enforce IAM policies and Secrets Manager access?

**A:**

Use `iam_role` and `iam_policy` modules.

Apply policies dynamically across accounts/regions.

Manage secrets with `secretsmanager_secret` module.

Use Ansible Vault for local sensitive information.

Implement audit and compliance reporting using CloudWatch & Config.

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## 4. Tips for AWS + Ansible Interview

**Know your modules:** `ec2`, `s3_bucket`, `rds_instance`, `lambda`, `cloudformation`, `ecs_service`, `cloudwatch_alarm`.

**Demonstrate hands-on experience:** Mention real projects using playbooks to automate infrastructure.

**Explain dynamic inventory:** Show how you can fetch EC2 instances automatically.

**Show understanding of DR & HA:** Multi-region deployment, replication, failover.

**Highlight security best practices:** IAM roles, key rotation, encrypted S3 buckets, Ansible Vault.

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